

GROUNDS
 OF
Natural Philosophy :
 Divided into
THIRTEEN PARTS:
 WITH AN
APPENDIX
 CONTAINING
FIVE PARTS.

The Second Edition, much altered from the *First*,
 which went under the Name of

PHILOSOPHICAL
 AND
PHYSICAL OPINIONS.

written by the
 Thrice Noble, Illustrious, and Excellent Princess,
 THE
DUCHESS of NEWCASTLE.

LONDON, C
 Printed by *A. Maxwell*, in the Year 1668.

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To all the UNIVERSITIES IN EUROPE.

Most Learned Societies,

All Books, without exception, being undoubtedly under your Jurisdiction, it is very strange that some Authors of good note, are not asham'd to repine at it; and the more forward they are in judging others, the less liberty they will allow to be judg'd themselves. But, if there was not a necessity, yet I would make it my choice, To submit, willingly, to your Censures, these *Grounds of Natural Philosophy*, in hopes that you will not condemn them, because they want *Art*, if they be found fraught with Sense and Reason. You are the *Starrs of the First Magnitude*, whose Influence governs the *World of Learning*; and it is my confidence, That you will be propitious to the Birth of this beloved Child of my Brain, whom I take the boldness to recommend to your Patronage; and as, if you vouchsafe to look on it favourably, I shall be extreamly obliged to your Goodness, for its everlasting Life: So, if you resolve to Frown upon it, I beg the favour, That it be not buried in the hard and Rocky Grave of your Displeasure; but be suffer'd, by your gentle silence, to lye still in the soft and easie Bed of Oblivion, which is incomparably the less Punishment of the Two. It is so commonly the error of indulgent Parents, to spoil their Children out of Fondness, that I may be forgiven for spoiling This, in never putting it to suck at the Breast of some Learned Nurse, whom I might have got from among your Students, to have assisted me; but would, obstinately, suckle it my self, and bring it up alone, without the help of any Scholar: Which having caused in the First Edition, (which was published under the name of *Philosophical and Physical*

Opinions) many Imperfections; I have endeavoured in this Second, by many Alterations and Additions, (which have forc'd me to give it another Name) to correct them; whereby, I fear, my Faults are rather *changed* and *encreased*, than *amended*. If you expect fair Proportions in the Parts, and a Beautiful Symmetry in the Whole, having never been taught at all, and having read but little; I acknowledg my self too illiterate to afford it, and too impatient to labour much for Method.

But, if you will be contented with *pure Wit*, and the Effects of *meer Contemplation*; I hope, that somewhat of that kind may be found in this Book, and in my other *Philosophical, Poetical, and Oratorical Works*: All which I leave, and this especially, to your kind Protection, and am,

Your most humble Servant, and Admirer,

MARGARET NEWCASTLE.

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FOUNDATIONS OF Natural Philosophy.

The First Part.

CHAP. I. OF MATTER.

Matter is that we name Body; which Matter cannot be less, or more, than Body: Yet some Learned Persons are of opinion, That there are Substances that are not Material Bodies. But how they can prove any sort of Substance to be no Body, I cannot tell: neither can any of Nature's Parts express it, because a Corporeal Part cannot have an Incorporeal Perception. But as for Matter, there may be degrees, as, *more pure*, or *less pure*; but there cannot be any Substances in Nature, that are between Body, and no Body: Also,

Matter cannot be figureless, neither can Matter be without Parts. Likewise, there cannot be Matter without Place, nor Place without Matter; so that Matter, Figure, or Place, is but one thing: for, it is as impossible for One Body to have Two Places, as for One Place to have Two Bodies; neither can there be Place, without Body.

CHAP. II. Of MOTION.

Though Matter might be without Motion, yet Motion cannot be without Matter; for it is impossible (in my opinion) that there should be an Immaterial Motion in Nature: and if Motion is corporeal, then Matter, Figure, Place, and Motion, is but one thing, viz. a corporeal figurative Motion. As for a First Motion, I cannot conceive how it can be, or what that First Motion should be: for, an Immaterial cannot have a Material Motion; or, so strong a Motion, as to set all the Material Parts in Nature, or this World, a-moving; but (in my opinion) every particular part moves by its own Motion: If so, then all the Actions in Nature are self-corporeal, figurative Motions. But this is to be noted, That as there is but one Matter, so there is but one Motion; and as there are several Parts of Matter, so there are several Changes of Motion: for, as Matter, of what degree soever it is, or can be, is but Matter; so Motion, although it make Infinite Changes, can be but Motion.

CHAP. III. Of the degrees of MATTER.

Though Matter can be neither more nor less than Matter; yet there may be degrees of Matter, as *more pure*, or *less pure*; and yet the purest Parts are as much material, in relation to the nature of Matter, as the grossest: Neither can there be more than two sorts of Matter, namely, that sort which is Self-moving, and that which is not Self-moving. Also, there can be but two sorts of the Self-moving Parts; as, that sort that moves intirely without Burdens,

and that sort that moves with the Burdens of those Parts that are not Self-moving: So that there can be but these three sorts; Those parts that are not moving, those that move free, and those that move with those parts that are not moving of themselves: Which degrees are (in my opinion) the Rational Parts, the Sensitive Parts, and the Inanimate Parts; which three sorts of Parts are so join'd, that they are but as one Body; for, it is impossible that those three sorts of Parts should subsist single, by reason Nature is but one united material Body.

CHAP. IV. Of VACUUM.

In my opinion, there cannot possibly be any *Vacuum*: for, though Nature, as being material, is divisible and compoundable; and, having Self-motion, is in perpetual action: yet Nature cannot divide or compose *from* her self, although she may move, divide, and compose in her self: But, were it possible Nature's Parts could wander and stray in, and out of *Vacuum*, there would be a Confusion; for, where Unity is not, Order cannot be: Wherefore, by the Order and Method of Nature's corporeal Actions, we may perceive, there is no *Vacuum*: For, what needs a *Vacuum*, when as Body and Place is but one thing; and as the Body alters, so doth the Place?

CHAP. V. The difference of the Two Self-moving Parts of Matter.

The Self-moving Parts of Nature seem to be of two sorts, or degrees; one being purer, and so more agil and free than the other; which (in my opinion) are the Rational Parts of Nature. The other sort is not so pure; and are the Architectonical Parts, which are the Labouring Parts, bearing the grosser Materials about them, which are the Inanimate Parts; and this sort (in my opinion) are the Sensitive Parts of Nature; which form, build, or compose themselves with the Inanimate Parts, into all kinds and sorts of Creatures, as Animals, Vegetables, Minerals, Elements, or what Creatures soever there

are in Nature: Whereas the Rational are so pure, that they cannot be so strong Labourers, as to move with Burdens of Inanimate Parts, but move freely without Burdens: for, though the Rational and Sensitive, with the Inanimate, move together as one Body; yet the Rational and Sensitive, do not move as one Part, as the sensitive doth with the Inanimate. But, pray mistake me not, when I say, the Inanimate Parts are grosser; as if I meant, they were like some densed Creature; for, those are but Effects, and not Causes: but, I mean gross, dull, heavy Parts, as, that they are not Self-moving; nor do I mean by Purity, Rarity; but Agility: for, Rare or Dense Parts, are Effects, and not Causes: And therefore, if any should ask, Whether the Rational and Sensitive Parts were Rare, or Dense; I answer, They may be Rare or Dense, according as they contract, or dilate their Parts; for there is no such thing as a Single Part in Nature: for Matter, or Body, cannot be so divided, but that it will remain Matter, which is divisible.

CHAP. VI. Of Dividing and Uniting of Parts.

Though every Self-moving Part, or Corporeal Motion, have free-will to move after what manner they please; yet, by reason there can be no Single Parts, several Parts unite in one Action, and so there must be united Actions: for, though every particular Part may divide from particular Parts; yet those that divide from some, are necessitated to join with other Parts, at the same point of time of division; and at that very same time, is their uniting or joining: so that Division, and Composition or Joining, is as one and the same act. Also, every altered Action, is an altered figurative Place, by reason Matter, Figure, Motion, and Place, is but one thing; and, by reason Nature is a perpetual motion, she must of necessity cause infinite Varieties.

CHAP. VII. Of Life and Knowledg.

All the Parts of Nature have Life and Knowledg; but, all the Parts have not Active Life, and a perceptive Knowledg, but onely the Rational and Sensitive: And this is to be noted, That the variousness, or variety of Actions, causes varieties of 7] Lives and Knowledges: For, as the Self-moving parts alter, or vary their Actions; so they alter and vary their Lives and Knowledges; but there cannot be an Infinite particular Knowledg, nor an Infinite particular Life; because Matter is divisible and compoundable.

CHAP. VIII. Of Nature's Knowledg and Perception.

If Nature were not Self-knowing, Self-living, and also Perceptive, she would run into Confusion: for, there could be neither Order, nor Method, in Ignorant motion; neither would there be distinct kinds or sorts of Creatures, nor such exact and methodical Varieties as there are: for, it is impossible to make orderly and methodical Distinctions, or distinct Orders, by Chances: Wherefore, Nature being so exact (as she is) must needs be Self-knowing and Perceptive: And though all her Parts, even the Inanimate Parts, are Self-knowing, and Self-living; yet, onely her Self-moving Parts have an active Life, and a perceptive Knowledg.

CHAP. IX. Of PERCEPTION in general.

Perception is a sort of Knowledg, that hath reference to Objects; that is, Some Parts to know other Parts: But yet Objects are not the cause of Perception; for the cause of Perception is Self-motion. But some would say, *If there were no Object, there could be no Perception.* I answer: It is true; for, that cannot be perceived, that is not: but yet, corporeal motions cannot be without Parts, and so not without Perception. But, put an impossible case, as, That there could be a single Corporeal Motion, and no more in Nature; that Corporeal Motion may make several Changes, somewhat like *Conceptions*, although not *Perceptions*: but, Nature being Corporeal, is

composed of Parts, and therefore there cannot be a want of Objects. But there are Infinite several manners and ways of Perception; which proves, That the Objects are not the Cause: for, every several kind and sort of Creatures, have several kinds and sorts of Perception, according to the nature and property of such a kind or sort of Composition, as makes such a kind or sort of Creature; as I shall treat of, more fully, in the following Parts of this Book.

CHAP. X. Of Double PERCEPTION.

There is a *Double Perception* in Nature, the Rational Perception, and the Sensitive: The Rational Perception is more subtil and penetrating than the Sensitive; also, it is more generally perceptive than the Sensitive; also, it is a more agil Perception than the Sensitive: All which is occasioned not onely through the *purity* of the Rational parts, but through the *liberty* of the Rational parts; whereas the Sensitive being incumbred with the Inanimate parts, is obstructed and retarded. Yet all Perceptions, both Sensitive and Rational, are in parts; but, by reason the Rational is freer, (being not a painful Labourer) can more easily make an united Perception, than the Sensitive; which is the reason the Rational parts can make a Whole Perception of a Whole Object: Whereas the Sensitive makes but Perceptions in part, of one and the same Object.

CHAP. XI. Whether the Triumphant Parts can be perceived distinctly from each other.

Some may make this Question, *Whether the Three sorts of Parts, the Rational, Sensitive, and Inanimate, may be singly perceived?* I answer, Not unless there were single Parts in Nature; but, though they cannot be singly perceived, yet they singly perceive; because, every Part hath its own motion, and so its own perception. And though those Parts, that have not self-motion, have not perception; yet, being joined, as one Body, to the Sensitive, they may by the Sensitive Motion, have some different sorts of Self-knowledg, caused by the different actions of the Sensitive parts; but that is not *Perception*. But, as I said, the *Triumphant Parts* cannot be

perceived distinctly asunder, though their Actions may be different: for, the joining, or intermixing of Parts, hinders not the several Actions; as for example, A Man is composed of several Parts, or, (as the Learned term them) *Corporeal Motions*; yet, not any of those different Parts, or Corporeal Motions, are a hindrance to each other: The same between the *Sensitive* and *Rational Parts*.

CHAP. XII. Whether Nature can know her self, or have an Absolute Power of her self, or have an exact Figure.

I was of an opinion, That Nature, because Infinite, could not know her Self; because Infinite hath no limit. Also, That Nature could not have an Absolute Power over her own Parts, because she had Infinite Parts; and, that the Infiniteness did hinder the Absoluteness: But since I have consider'd, That the Infinite Parts must of necessity be Self-knowing; and that those Infinite Self-knowing Parts are united in one Infinite Body, by which Nature must have both an United Knowledg, and an United Power. Also, I questioned, Whether Nature could have an Exact Figure, (but, mistake me not; for I do not mean the Figure of Matter, but a composed Figure of Parts) because Nature was composed of Infinite Variety of Figurative Parts: But considering, that those Infinite Varieties of Infinite Figurative Parts, were united into one Body; I did conclude, That she must needs have an Exact Figure, though she be Infinite: As for example, This World is composed of numerous and several Figurative parts, and yet the World hath an exact Form and Frame, the same which it would have if it were Infinite. But, as for Self-knowledg, and Power, certainly God hath given them to Nature, though her Power be limited: for, she cannot move beyond her Nature; nor hath she power to make her self any otherwise than what she is, since she cannot create, or annihilate any part, or particle: nor can she make any of her Parts, Immaterial; or any Immaterial, Corporeal: Nor can she give to one part, the Nature (*viz.* the Knowledg, Life, Motion, or Perception) of another part; which is the reason one Creature cannot have the properties, or faculties of another; they may have the like, but not the same.

CHAP. XIII. Nature cannot judg her self.

Although Nature knows her self, and hath a free power of her self; (I mean, a natural Knowledg and Power) yet, Nature cannot be an upright, and just Judg of her self, and so not of any of her Parts; because every particular part is a part of her self. Besides, as she is Self-moving, she is Self-changeing, and so she is alterable: Wherefore, nothing can be a perfect, and a just Judg, but something that is Individable, and Unalterable, which is the Infinite GOD, who is Unmoving, Immutable, and so Unalterable; who is the Judg of the Infinite Corporeal Actions of his Servant Nature. And this is the reason that all Nature's Parts appeal to God, as being the only Judg.

CHAP. XIV. Nature Poyses, or Balances her Actions.

Although Nature be Infinite, yet all her Actions seem to be *poysed*, or *balanced*, by Opposition; as for example, As Nature hath dividing, so composing actions: Also, as Nature hath regular, so irregular actions; as Nature hath dilating, so contracting actions: In short, we may perceive amongst the Creatures, or Parts of this World, slow, swift, thick, thin, heavy, leight, rare, dense, little, big, low, high, broad, narrow, light, dark, hot, cold, productions, dissolutions, peace, warr, mirth, sadness, and that we name *Life*, and *Death*; and infinite the like; as also, infinite varieties in every several kind and sort of actions: but, the infinite varieties are made by the Self-moving parts of Nature, which are the Corporeal Figurative Motions of Nature.

CHAP. XV. Whether there be Degrees of Corporeal Strength.

As I have declared, there are (in my Opinion) Two sorts of Self-moving Parts; the one Sensitive, the other Rational. The Rational parts of my Mind, moving in the manner of Conception, or Inspection, did occasion some Disputes, or Arguments, amongst those parts of my Mind. The Arguments were these: *Whether there were degrees of Strength, as there was of Purity, between their own sort, as, the Rational and the Sensitive?* The Major part of the Argument was, *That Self-motion could be but Self-motion: for, not any part of Nature could move beyond its power of Self-motion.* But the Minor part argued, *That the Self-motion of the Rational, might be stronger than the Self-motion of the Sensitive.* But the Major part was of the opinion, *That there could be no degrees of the Power of Nature, or the Nature of Nature: for Matter, which was Nature, could be but Self-moving, or not Self-moving; or partly Self-moving, or not Self-moving.* But the Minor argued, *That it was not against the nature of Matter to have degrees of Corporeal Strength, as well as degrees of Purity: for, though there could not be degrees of Purity amongst the Parts of the same sort, as amongst the Parts of the Rational, or amongst the Parts of the Sensitive; yet, if there were degrees of the Rational and Sensitive Parts, there might be degrees of Strength.* The Major part said, *That if there were degrees of Strength, it would make a Confusion, by reason there would be no Agreement; for, the Strongest would be Tyrants to the Weakest, in so much as they would never suffer those Parts to act methodically or regularly.* But the Minor part said, *that they had observed, That there was degrees of Strength amongst the Sensitive Parts.* The Major part argued, *That they had not degrees of Strength by Nature; but, that the greater Number of Parts were stronger than a less Number of Parts.* Also, *there were some sorts of Actions, that had advantage of other sorts. Also, some sorts of Compositions are stronger than other; not through the degrees of innate Strength, nor through the number of Parts; but, through the manner and form of their Compositions, or Productions.* Thus my Thoughts argued; but, after many Debates and Disputes, at last my Rational Parts agreed, *That, If there were degrees of Strength, it could not be between the Parts of the same degree, or sort; but, between the Rational and Sensitive; and if so, the Sensitive was Stronger, being less pure; and the Rational was more Agil, being more pure.*

CHAP. XVI. Of Effects, and Cause.

To treat of Infinite Effects, produced from an Infinite Cause, is an endless Work, and impossible to be performed, or effected; only this may be said, That the Effects, though Infinite, are so united to the material Cause, as that not any single effect can be, nor no Effect can be annihilated; by reason all Effects are in the power of the Cause. But this is to be noted, That some Effects producing other Effects, are, in some sort or manner, a Cause.

CHAP. XVII. Of INFLUENCE.

An *Influence* is this; When as the Corporeal Figurative Motions, in different kinds, and sorts of Creatures, or in one and the same sorts, or kinds, move sympathetically: And though there be antipathetical Motions, as well as sympathetical; yet, all the Infinite parts of Matter, are agreeable in their nature, as being all Material, and Self-moving; and by reason there is no *Vacuum*, there must of necessity be an Influence amongst all the Parts of Nature.

CHAP. XVIII. Of FORTUNE and CHANCE.

Fortune, is only various Corporeal Motions of several Creatures, design'd to one Creature, or more Creatures; either to *that* Creature, or *those* Creatures Advantage, or Disadvantage: If Advantage, Man names it *Good Fortune*; if Disadvantage, Man names it *Ill Fortune*. As for *Chance*, it is the visible Effects of some hidden Cause; and *Fortune*, a sufficient Cause to produce such Effects: for, the conjunction of sufficient Causes, doth produce such or such Effects; which Effects could not be produced, if any of those Causes were wanting: So that, *Chances* are but the Effects of Fortune.

CHAP. XIX. Of TIME and ETERNITY.

Time is not a Thing by it self; nor is *Time* Immaterial: for, *Time* is only the variations of Corporeal Motions; but Eternity depends not on Motion, but of a Being without Beginning, or Ending.

The Second Part.

CHAP. I. Of CREATURES.

All Creatures are Composed-Figures, by the consent of Associating Parts; by which Association, they joyn into such, or such a figured Creature: And though every Corporeal Motion, or Self-moving Part, hath its own motion; yet, by their Association, they all agree in proper actions, as actions proper to their Compositions: and, if every particular Part, hath not a perception of all the Parts of their Association; yet, every Part knows its own Work.

CHAP. II. Of Knowledg and Perception of different kinds and sorts of Creatures.

There is not any Creature in Nature, that is not composed of Self-moving Parts, (*viz.* both of Rational and Sensitive) as also of the Inanimate Parts, which are Self-knowing: so that all Creatures, being composed of these sorts of Parts, must have a Sensitive, and Rational Knowledg and Perception, as Animals, Vegetables, Minerals, Elements, or what else there is in Nature: But several kinds, and several sorts in these kinds of Creatures, being composed after different manners, and ways, must needs have different Lives, Knowledges, and Perceptions: and not only every several

kind, and sort, have such differences; but, every particular Creature, through the variations of their Self-moving Parts, have varieties of Lives, Knowledges, Perceptions, Conceptions, and the like; and not only so, but every particular part of one and the same Creature, have varieties of Knowledges, and Perceptions, because they have varieties of Actions. But, (as I have declared) there is not any different kind of Creature, that can have the like Life, Knowledg, and Perception; not only because they have different Productions, and different Forms; but, different Natures, as being of different kinds.

CHAP. III. Of Perception of Parts, and United Perception.

All the Self-moving Parts are perceptive; and, all Perception is in Parts, and is dividable, and compoundable, as being Material; also, Alterable, as being Self-moving: Wherefore, no Creature that is composed, or consists of many several sorts of Corporeal Figurative Motions, but must have many sorts of Perception; which is the reason that one Creature, as Man, cannot perceive another Man any otherwise but in Parts: for, the Rational, and Sensitive; nay, all the Parts of one and the same Creature, perceive their Adjoining Parts, as they perceive Foreign Parts; only, by their close conjunction and near relation, they unite in one and the same actions. I do not say, they always agree: for, when they move irregularly, they disagree: And some of those United Parts, will move after one manner, and some after another; but, when they move regularly, then they move to one and the same Design, or one and the same United Action. So, although a Creature is composed of several sorts of Corporeal Motions; yet, these several sorts, being properly united in one Creature, move all agreeably to the Property and Nature of the whole Creature; that is, the particular Parts move according to the property of the whole Creature; because the particular Parts, by conjunction, make the Whole: So that, the several Parts make one Whole; by which, a Whole Creature hath both a general Knowledg, and a Knowledg of Parts; whereas, the Perceptions of Foreign Objects, are but in the Parts: and this is the reason why one Creature perceives not the Whole of another Creature, but only some Parts. Yet this is to be noted, That not any Part hath another

Part's Nature, or Motion, nor therefore, their Knowledg, or Perception; but, by agreement, and unity of Parts, there is composed Perceptions.

CHAP. IV. Whether the Rational and Sensitive Parts have a Perception of each other.

Some may ask the Question, *Whether the Rational and Sensitive, have Perception of each other?* I answer: In my Opinion, they have. For, though the Rational and Sensitive Parts, be of two sorts; yet, both sorts have Self-motion; so that they are but as one, as, that they are both Corporeal Motions; and, had not the Sensitive Parts incumbrances, they would be, in a degree, as agil, and as free as the Rational. But, though each sort hath perception of each other, and some may have the like; yet they have not the same: for, not any Part can have another's Perception, or Knowledg; but, by reason the Rational and Sensitive, are both Corporeal Motions, there is a strong sympathy between those sorts, in one Conjunction, or Creature. Indeed, the Rational Parts are the Designing Parts; and the Sensitive, the Labouring Parts; and the Inanimate are as the Material Parts: not but all the three sorts are Material Parts; but the Inanimate, being not Self-moving, are the Burdensome Parts.

CHAP. V. Of Thoughts, and the whole Mind of a Creature.

AS for Thoughts, though they are several Corporeal Motions, or Self-moving Parts; yet, being united, by Conjunction in one Creature, into one whole Mind, cannot be perceived by some Parts of another Creature, nor by the same sort of Creature, as by another Man. But some may ask, *Whether the whole Mind of one Creature, as the whole Mind of one Man, may not perceive the whole Mind of another Man?* I answer, That if the Mind was not joyn'd and mix'd with the Sensitive and Inanimate Parts, and had not interior, as well as exterior Parts, the whole Mind of one Man, might

perceive the whole Mind of another Man; but, that being not possible, one whole Mind cannot perceive another whole Mind: By which Observation we may perceive, there are no *Platonick Lovers* in Nature. But some may ask, Whether the Sensitive Parts can perceive the Rational, in one and the same Creature? I answer, They do; for if they did not, it were impossible for the Sensitive Parts to execute the Rational Designs; so that, what the Mind designs, the Sensitive Body doth put in execution, as far as they have Power: But if, through Irregularities, the Body be sick, and weak, or hath some Infirmities, they cannot execute the Designs of the Mind.

CHAP. VI. Whether the Mind of one Creature, can perceive the Mind of another Creature.

Some may ask the reason, *Why one Creature, as Man, cannot perceive the Thoughts of another Man, as well as he perceives his exterior Sensitive Parts?* I answer, That the Rational Parts of one Man, perceive as much of the Rational Parts of another Man, as the Sensitive Parts of that Man doth of the Sensitive Parts of the other Man; that is, as much as is presented to his Perception: for, all Creatures, and every part and particle, have those three sorts of Matter; and therefore, every part of a Creature is perceiving, and perceived. But, by reason all Creatures are composed of Parts, (*viz.* both of the Rational and Sensitive) all Perceptions are in parts, as well the Rational, as the Sensitive Perception: yet, neither the Rational, nor the Sensitive, can perceive all the Interior Parts or Corporeal Motions, unless they were presented to their perception: Neither can one Part know the Knowledg and Perception of another Part: but, what Parts of one Creature are subject to the perception of another Creature, those are perceived.

CHAP. VII. Of Perception, and Conception.

Although the Exterior Parts of one Creature, can but perceive the Exterior Parts of another Creature; yet, the Rational can make Conceptions of the Interior Parts, but not Perception: for, neither the Sense, nor Reason, can perceive what is not present, but by rote, as after the manner of Conceptions, or Remembrances, as I shall in my following Chapters declare: So that, the Exterior Rational Parts, that are with the Exterior Sensitive Parts of an Object, are as much perceived, the one, as the other: but, those Exterior Parts of an Object, not moving in particular Parties, as in the whole Creature, is the cause that some Parts of one Creature, cannot perceive the whole Composition or Frame of another Creature: that is, some of the Rational Parts of one Creature, cannot perceive the whole Mind of another Creature. The like of the Sensitive Parts.

CHAP. VIII. Of Human Suppositions.

Although Nature hath an Infinite Knowledg and Perception; yet, being a Body, and therefore divisible and compoundable; and having, also, Self-motion, to divide and compound her Infinite Parts, after infinite several manners; is the reason that her finite Parts, or particular Creatures, cannot have a general or infinite Knowledg, being limited, by being finite, to finite Perceptions, or perceptive Knowledg; which is the cause of *Suppositions*, or Imaginations, concerning Forrein Objects: As for example, A Man can but perceive the Exterior Parts of another Man, or any other Creature, that is subject to Human Perception; yet, his Rational Parts may suppose, or presuppose, what another Man thinks, or what he will act: and for other Creatures, a Man may suppose or imagine what the innate nature of such a Vegetable, or Mineral, or Element is; and may imagine or suppose the Moon to be another World, and that all the fixed Starrs are Sunns; which Suppositions, Man names *Conjectures*.

CHAP. IX. Of Information between several Creatures.

No question but there is *Information* between all Creatures: but, several sorts of Creatures, having several sorts of Informations, it is impossible for any particular sort to know, or have perceptions of the Infinite, or Numberless Informations, between the Infinite and Numberless Parts, or Creatures of Nature: Nay, there are so many several Informations amongst one sort (as of Mankind) that it is impossible for one Man to perceive them all; no, nor can one Man generally perceive the particular Informations that are between the particular Parts of his Sensitive Body; or between the particular Informations of his Rational Body; or between the particular Rational and Sensitive Parts: much less can Man perceive, or know the several Informations of other Creatures.

CHAP. X. The Reason of several kinds and sorts of Creatures.

Some may ask, *Why there are such sorts of Creatures, as we perceive there are, and not other sorts?* I answer, That, 'tis probable, we do not perceive all the several kinds and sorts of Creatures in Nature: In truth, it is impossible (if Nature be Infinite) for a Finite to perceive the Infinite varieties of Nature. *Also they may ask, Why the Planets are of a Spherical Shape, and Human Creatures are of an Upright shape, and Beasts of a Bending and stooping shape? Also, Why Birds are made to flye, and not Beasts? And for what Cause, or Design, have Animals such and such sorts of shapes and properties? And Vegetables such and such sorts of shapes and properties? And so of Minerals and Elements?* I answer; That several sorts, kinds, and differences of Particulars, causes Order, by reason it causes Distinctions: for, if all Creatures were alike, it would cause a Confusion.

CHAP. XI. Of the several Properties of several Kinds and sorts of Creatures.

As I have said, There are several kinds, and several sorts, and several particular Creatures of several kinds and sorts; whereof there are some Creatures of a mixt kind, and some of a mixt sort, and some of a mixture of some particulars. Also, there are some kind of Creatures, and sorts of Creatures; as also Particulars of a Dense Nature, others of a Rate Nature; some of a Leight Nature, some of a Heavy Nature; some of a Bright Nature, some of a Dark Nature; some of an Ascending Nature, some of a Descending Nature; some of a Hard Nature, some of a Soft Nature; some of a Loose Nature, and some of a Fixt Nature; some of an Agil Nature, and some of a Slow Nature; some of a Consistent Nature, and some of a Dissolving Nature: All which is according to the Frame and Form of their Society, or Composition.

The Third Part.

CHAP. I. Of Productions in general.

The Self-moving Parts, or Corporeal Motions, are the Producers of all Composed Figures, such as we name *Creatures*: for, though all Matter hath Figure, by being Matter; for it were non-sense to say, *Figureless Matter*; since the most pure Parts of Matter, have Figure, as well as the grossest; the rarest, as well as the densed: But, such Composed Figures which we name *Creatures*, are produced by particular Associations of Self-moving Parts, into particular kinds, and sorts; and particular Creatures in every kind, or sort. The particular kinds, that are subject to Human Perceptions, are those we name Animals, Vegetables, Minerals, and Elements; of which kinds, there are numerous sorts; and of every sort, infinite particulars: And though there be Infinite Varieties in Nature, made by the Corporeal Motions, or Self-moving Parts, which might cause a Confusion: Yet, considering Nature is intire in her self, as being only Material, and as being but one United Body; also, poysing all her Actions by Opposites; 'tis impossible to be any ways in Extreame, or to have a Confusion.

CHAP. II. Of Productions in general.

The Sensitive Self-moving Parts, or Corporeal Motions, are the Labouring Parts of all Productions, or Fabricks of all Creatures; but yet, those Corporeal Motions, are parts of the Creature they produce: for, Production is only a Society of particular Parts, that joyn into particular Figures, or Creatures: but, as Parts produce Figures, by Association; so they dissolve those Figures by Division: for, Matter is a perpetual Motion, that is always dividing and composing; so that not any Creature can be eternally one and

the same: for, if there were no Dissolvings, and Alterings, there would be no varieties of Particulars; for, though the kinds and sorts may last, yet not the Particulars. But, mistake me not, I do not say those Figures are lost, or annihilated in Nature; but only, their Society is dissolved, or divided in Nature. But this is to be noted, That some Creatures are sooner produced and perfected, than others; and again, some Creatures are sooner decayed, or dissolved.

CHAP. III. Of Productions in general.

There are so many different composed Parts, and so much of variety of Action in every several Part of one Creature, as 'tis impossible for Human Perception to perceive them; nay, not every Corporeal Motion of one Creature, doth perceive all the varieties of the same Society; and, by the several actions, not only of several Parts, but of one and the same Parts, cause such obscurity, as not any Creature can tell, not only how they were produced, but, not how they consist: But, by reason every Part knows his own Work, there is Order and Method: For example, In a Human Creature, those Parts that produce, or nourish the Bones, those of the Sinews, those of the Veins, those of the Flesh, those of the Brains, and the like, know all their several Works, and consider not each several composed Part, but what belongs to themselves; the like, I believe, in Vegetables, Minerals, or Elements. But mistake me not; for, I do not say, those Corporeal Motions in those particulars, are bound to those particular Works, as, that they cannot change, or alter their actions if they will, and many times do: as some Creatures dissolve before they are perfect, or quite finished; and some as soon as finished; and some after some short time after they are finished; and some continue long, as we may perceive by many Creatures that dye, which I name Dissolving in several Ages; but, untimely Dissolutions, proceed rather from some particular Irregularities of some particular Parts, than by a general Agreement.

CHAP. IV. Of Productions in general.

The Reason that all Creatures are produced by the ways of Production, as one Creature to be composed out of other Creatures, is, That Nature is but one Matter, and that all her Parts are united as one Material Body, having no Additions, or Diminutions; no new Creations, or Annihilations: But, were not Nature one and the same, but that her Parts were of different natures; yet, Creatures must be produced by Creatures, that is, Composed Figures, as a Beast, a Tree, a Stone, Water, &c. must be composed of *Parts*, not a *single Part*: for, a single Part cannot produce composed Figures; nor can a single Part produce another single Part; for, Matter cannot create Matter; nor can one Part produce another Part out of it self: Wherefore, all Natural Creatures are produced by the consent and agreement of many Self-moving Parts, or Corporeal Motions, which work to a particular Design, as to associate into particular kinds and sorts of Creatures.

CHAP. V. Of Productions in general.

As I said in my former Chapter, That all Creatures are produced, or composed by the agreement and consent of particular Parts; yet some Creatures are composed of more, and some of fewer Parts: neither are all Creatures produced, or composed after one and the same manner; but some after one manner, and some after another manner: Indeed, there are divers manners of Productions, both of those we name Natural, and those we name *Artificial*; but I only treat of Natural Productions, which are so various, that it is a wonder if any two Creatures are just alike; by which we may perceive, that not only in several kinds and sorts, but in Particulars of every kind, or sort, there is some difference, so as to be distinguished from each other, and yet the species of some Creatures are like to their kind, and sort, but not all; and the reason that most Creatures are in *Species*, according to their sort, and kind, is not only, that Nature's Wisdom orders and regulates her Corporeal Figurative Motions, into kinds and sorts of Societies and Conjunctions; but, those Societies cause a perceptive Acquaintance, and an united Love, and good liking of the Compositions, or Productions: and not

only a love to their Figurative Compositions, but to all that are of the same sort, or kind; and especially, their being accustom'd to actions proper to their Figurative Compositions, is the cause that those Parts, that divide from the Producers, begin a new Society, and, by degrees, produce the like Creature; which is the cause that Animals and Vegetables produce according to their likeness. The same may be amongst Minerals and Elements, for all we can know. But yet, some Creatures of one and the same sort, are not produced after one and the same manner: As for example, One and the same sort of Vegetables, may be produced after several manners, and yet, in the effect, be the same, as when Vegetables are sowed, planted, engrafted; as also, Seeds, Roots, and the like, they are several manners, or ways of Productions, and yet will produce the same sort of Vegetable: but, there will be much alterations in replanting, which is occasioned by the change of associating Parts, and Parties; but as for the several Productions

of several kinds and sorts, they are very different; as for example, Animals are not produced as Vegetables, or Vegetables as Minerals, nor Minerals as any of the rest: Nor are all Animals produced alike, nor Minerals, or Vegetables; but after many different manners, or ways. Neither are all Productions like their Producers; for, some are so far from resembling their Figurative Society, that they produce another kind, or sort of Composed Figures; as for example, Maggots out of Cheese, other Worms out of Roots, Fruits, and the like: but these sorts of Creatures, Man names *Insects*; but yet they are Animal Creatures, as well as others.

CHAP. VI. Of Productions in general.

All Creatures are Produced, and Producers; and all these Productions partake more or less of the Producers; and are necessitated so to do, because there cannot be any thing New in Nature: for, whatsoever is produced, is of the same Matter; nay, every particular Creature hath its particular Parts: for, not any one Creature can be produced of any other Parts than what produced it; neither can the same Producer produce one and the same double, (as I may say to express my self:) for, though the same Producers may produce the like, yet not the same:

for, every thing produced, hath its own Corporeal Figurative Motions; but this might be, if Nature was not so full of variety: for, if all those Corporeal Motions, or Self-moving Parts, did associate in the like manner, and were the very same Parts, and move in the very same manner; the same Production, or Creature, might be produced after it was dissolved; but, by reason the Self-moving Parts of Nature are always dividing and composing *from*, and *to* Parts, it would be very difficult, if not impossible.

CHAP. VII. Of Productions in general.

As there are Productions, or Compositions, made by the Sensitive Corporeal Motions, so there are of the Rational Corporeal Motions, which are Composed Figures of the Mind: And the reason the Rational Productions are more various, as also more numerous, is, That the Rational is more loose, free, and so more agil than the Sensitive; which is also the reason that the Rational Productions require not such degrees of Time, as the Sensitive. But I shall treat more upon this Subject, when I treat of that Animal we name *MAN*.

CHAP. VII. *Lastly*, Of Productions in general.

Though all Creatures are made by the several Associations of Self-moving Parts, or (as the Learned name them) *Corporeal Motions*; yet, there are infinite varieties of Corporeal Figurative Motions, and so infinite several manners and ways of Productions; as also, infinite varieties of Figurative Motions in every produced Creature: Also, there is variety in the difference of Time, of several Productions, and of their Consistency and Dissolution: for, some Creatures are produced in few Hours, others not in many Years. Again, some continue not a Day; others, numbers of Years. But this is to be noted, That according to the Regularity, or Irregularity of the Associating Motions, their Productions are more or less perfect. Also, this is to be noted,

That there are Rational Productions, as well as Sensitive: for, though all Creatures are composed both of Sensitive and Rational Parts, yet the Rational Parts move after another manner.

CHAP. VIII. Productions must partake of some Parts of their Producers.

No Animal, or Vegetable, could be produced, but by such, or such particular Producers; neither could an Animal, or Vegetable, be produced without some Corporeal Motions of their Producers; that is, some of the Producers Self-moving Parts; otherwise the like Actions might produce, not only the like Creatures, but the same Creatures, which is impossible: Wherefore, the things produced, are part of the Producers; for, no particular Creature could be produced, but by such particular Producers. But this is to be noted, That all sorts of Creatures are produced by more, or fewer, Producers. Also, the first Producers are but the first Founders of the things produced, but not the only Builders: for, there are many several sorts of Corporeal Motions, that are the Builders; for, no Creature can subsist, or consist, by it self, but must assist, and be assisted: Yet, there are some differences in all Productions, although of the same Producers; otherwise all the Off-springs of one and the same Producer, would be alike: And though, sometimes, their several Off-springs may be so alike, as hardly to be distinguished; yet, that is so seldom, as it appears as a wonder; but there is a property in all Productions, as, for the Produced to belong as a Right and Property to the *Producer*.

CHAP. IX. Of Resemblances of several Off-springs, or Producers.

There are numerous kinds and sorts of Productions, and infinite manners and ways, in the actions of Productions; which is the cause that the Off-springs of the same Producers, are not so just alike, but that they are distinguishable; but yet there may not only be resemblances between

particular Off-springs of the same Producers, as also of the same sort; but, of different sorts of Creatures: but the Actions of all Productions that are according to their own *Species*, are Imitating Actions, but not Bare Imitations, as by an Incorporeal Motion; for if so, then a covetous Woman, that loves Gold, might produce a Wedg of Gold instead of a Child; also, *Virgins* might be as Fruitful as *Married Wives*.

CHAP. X. Of the Several Appearances of the Exterior Parts of One Creature.

Every altered Action of the Exterior Parts, causes an altered Appearance: As for example, A Man, or the like Creature, doth not appear when he is old, as when he was young; nor when he is sick, as when he is well in health; no, nor when he is cold, as when he is hot. Nor do they appear in several Passions alike: for, though Man can best perceive the Alteration of his own Kind, or Sort; yet, other Creatures have several Appearances, as well as Man; some of which, Man may perceive, though not all, being of a different sort. And not only Animals, but Vegetables, and Elements, have altered Appearances, and many that are subject to Man's perception.

The Fourth Part.

CHAP. I. Of Animal Productions; and of the Differences between Productions, and Transformations.

I understand Productions to be between Particulars; as, some particular Creatures to produce other particular Creatures; but not to transform from one sort of Creature, into another sort of Creature, as Cheese into Maggots, and Fruit into Worms, &c. which, in some manner, is like Metamorphosing. So by Transformation, the Intellectual Nature, as well as the Exterior Form, is transform'd: Whereas Production transforms only the Exterior Form, but not the Intellectual Nature; which is the cause that such Transformations cannot return into their former state; as a Worm to be a Fruit, or a Maggot a Cheese again, as formerly. Hence I perceive, that all sorts of Fowls are partly Produced, and partly Transformed: for, though an Egg be produced, yet a Chicken is but a Transformed Egg.

CHAP. II. Of different Figurative Motions in MAN's Production.

All Creatures are produced by Degrees; which proves, That not any Creature is produced, in perfection, by one Act, or Figurative Motion: for, though the Producers are the first Founders, yet not the Builders. But, as for Animal Creatures, there be some sorts that are composed of many different Figurative Motions; amongst which sorts, is Mankind, who has very different Figurative Parts, as Bones, Sinews, Nerves, Muscles, Veins, Flesh, Skin, and Marrow, Blood, Choler, Flegm, Melancholy, and the like; also, Head, Breast, Neck, Arms, Hands, Body, Belly, Thighs, Leggs, Feet, &c. also, Brains, Lungs, Stomack, Heart, Liver, Midriff, Kidnies, Bladder, Guts,

and the like; and all these have several actions, yet all agree as one, according to the property of that sort of Creature named MAN.

CHAP. III. Of the Quickning of a Child, or any other sort of Animal Creatures.

The Reason that a Woman, or such like Animal, doth not feel her Child so soon as it is produced, is, That the Child cannot have an Animal Motion, until it hath an Animal Nature, that is, until it be perfectly an Animal Creature; and as soon as it is a perfect Child, she feels it to move, according to its nature: but it is only the Sensitive Parts of the Child that are felt by the Mother, not the Rational; because those Parts are as the Designers, not the Builders; and therefore, being not the Labouring Parts, are not the Sensible Parts. But it is to be noted, That, according to the Regularity, or Irregularity of the Figurative Motions, the Child is *well shaped*, or **mishaped**.

CHAP. IV. Of the Birth of a Child.

The reason why a Child, or such like Animal Creature, stays no longer in the Mother's Body, than to such a certain Time, is, That a Child is not Perfect before that time, and would be too big after that time; and so big, that it would not have room enough; and therefore it strives and labours for liberty.

CHAP. V. Of Mischances, or Miscarriages of Breeding Creatures.

When a Mare, Doe, Hind, or the like Animal, cast their Young, or a Woman miscarries of her Child, the Mischance proceeds either through the Irregularities of the Corporeal Motions, or Parts of the Child; or through some Irregularity of the Parts of the Mother; or else of both Mother and Child. If the Irregularities be of the Parts of the Child, those Parts divide from the Mother, through their Irregularity: but, if the Irregularity be in the Parts of the Mother, then the Mother divides in some manner from the

Child; and if there be a distemper in both of them, the Child and Mother divide from each other: but, such Mischances are at different times, some sooner, and some later. As for false Conceptions, they are occasioned through the Irregularities of Conception.

CHAP. VI. Of the Encrease of Growth, and Strength of Mankind, or such like Creatures.

The reason most Animals, especially Human Creatures, are weak whilst they are Infants, and that their Strength and Growth encreases by degrees, is, That a Child hath not so many Parts, as when he is a Youth; nor so many Parts when he is a Youth, as when he is a Man: for, after the Child is parted from the Mother, it is nourished by other Creatures, as the Mother was, and the Child by the Mother; and according as the nourishing Parts be Regular, or Irregular, so is the Child, Youth, or Man, weaker, or stronger; healthful, or diseased; and when the Figurative Motions move (as I may say for expression sake) curiously, the Body is neatly shaped, and is, as we say, beautiful. But this is to be noted, That 'tis not Greatness, or Bulk of Body, makes a Body perfect; for, there are several sizes of every sort, or kind of Creatures; as also, in every particular kind, or sort; and every several size may be as perfect, one, as the other: But, I mean the Number of Parts, according to the proper size.

CHAP. VII. Of the several Properties of the several Exterior Shapes of several sorts of Animals.

The several Exterior Shapes of Creatures, cause several Properties, as Running, Jumping, Hopping, Leaping, Climbing, Galloping, Trotting, Ambling, Turning, Winding, and Rowling; also Creeping, Crawling, Flying, Soaring or Towing; Swimming, Diving, Digging, Stinging or Piercing; Pressing, Spinning, Weaving, Twisting, Printing, Carving, Breaking,

Drawing, Driving, Bearing, Carrying, Holding, Gripping or Grasping, Infolding, and Millions of the like. Also, the Exterior Shapes cause Defences, as Horns, Claws, Teeth, Bills, Talons, Finns, &c. Likewise, the Exterior Shapes cause Offences, and give Offences: As also, the different sorts of Exterior Shapes, cause different Exterior Perceptions.

CHAP. VIII. Of the Dividing and Uniting Parts of a particular Creature.

Those Parts (as I have said) that were the First Founders of an Animal, or other sort of Creature, may not be constant Inhabitants: for, though

the Society may remain, the particular Parts may remove: Also, all particular Societies of one kind, or sort, may not continue the like time; but some may dissolve sooner than others. Also, some alter by degrees, others of a sudden; but, of those Societies that continue, the particular Parts remove, and other particular Parts unite; so, as some Parts *were* of the Society, so some other Parts are of the Society, and *will be* of the Society: But, when the Form, Frame, and Order of the Society begins to alter, then that particular Creature begins to decay. But this is to be noted, That those particular Creatures that dye in their Childhood, or Youth, were never a full and regular Society; and the dissolving of a Society, whether it be a Full, or but a Forming Society, Man names *DEATH*. Also, this is to be noted, That the Nourishing Motion of Food, is the Uniting Motion; and the Cleansing, or Evacuating Motions, are the Dividing Corporeal Motions. Likewise it is to be noted, That a Society requires a longer time of uniting than of dividing; by reason uniting requires assistance of Foreign Parts, whereas dividings are only a dividing of home-Parts. Also, a particular Creature, or Society, is longer in dividing its Parts, than in altering its Actions; because a Dispersing Action is required in Division, but not in Alteration of Actions.

The Fifth Part.

CHAP. I. Of MAN.

Now I have discoursed, in the former Parts, after a general manner, of *Animals*: I will, in the following Chapters, speak more particularly of that sort we name *Mankind*; who believe (being ignorant of the Nature of other Creatures) that they are the most knowing of all Creatures; and yet a *whole Man* (as I may say for expression-sake) doth not know all the Figurative Motions belonging either to his Mind, or Body: for, he doth not generally know every particular Action of his Corporeal Motions, as, How he was framed, or formed, or perfected. Nor doth he know every particular Motion that occasions his present Consistence,

or Being: Nor every particular Digestive, or Nourishing Motion: Nor, when he is sick, the particular Irregular Motion that causes his Sickness. Nor do the Rational Motions in the Head, know always the Figurative Actions of those of the Heel. In short, (as I said) Man doth not generally know every particular Part, or Corporeal Motion, either of Mind, or Body: Which proves, Man's Natural Soul is not inalterable, or indivisible, and uncompoundable.

CHAP. II. Of the variety of Man's Natural Motions.

There is abundance of varieties of Figurative Motions in Man: As, first, There are several Figurative Motions of the Form and Frame of Man, as of his Innate, Interior, and Exterior Figurative Parts. Also, there are several Figures of his several Perceptions, Conceptions, Appetite, Digestions, Reparations, and the like. There are also several Figures of several Postures

of his several Parts; and a difference of his Figurative Motions, or Parts, from other Creatures; all which are Numberless: And yet all these different Actions are proper to the Nature of *MAN*.

CHAP. III. Of Man's Shape and Speech.

The Shape of Man's Sensitive Body, is, in some manner, of a mixt Form: but, he is singular in this, That he is of an upright and straight Shape; of which, no other Animal but Man is: which Shape makes him not only fit, proper, easie and free, for all exterior actions; but also for Speech: for being streight, as in a straight and direct Line from the Head to the Feet, so as his Nose, Mouth, Throat, Neck, Chest, Stomack, Belly, Thighs, and Leggs, are from a straight Line: also, his Organ-Pipes, Nerves, Sinews, and Joynts, are in a straight and equal posture to each other; which is the cause, Man's Tongue, and Organs, are more apt for Speech than those of any other Creature; which makes him more apt to imitate any other Creature's Voyces, or Sounds: Whereas other Animal Creatures, by reason of their bending Shapes, and crooked Organs, are not apt for Speech; neither (in my Opinion) have other Animals so melodious a Sound, or Voice, as Man: for, though some sorts of Birds Voices are sweet, yet they are weak, and faint; and Beasts Voices are harsh, and rude: but of all other Animals, besides Man, Birds are the most apt for Speech; by reason they are more of an upright shape, than Beasts, or any other sorts of Animal Creatures, as Fish, and the like; for, Birds are of a straight and upright shape, as from their Breasts, to their Heads; but, being not so straight as Man; causes Birds to speak uneasily, and constrainedly: Man's shape is so ingeniously contrived, that he is fit and proper for more several sorts of exterior actions, than any other Animal Creature; which is the cause he seems as Lord and Sovereign of other Animal Creatures.

CHAP. IV. Of the several Figurative Parts of Human Creatures.

The manner of Man's Composition, or Form, is of different Figurative Parts; whereof some of those Parts seem the Supreme, or (as I may say) Fundamental Parts; as the Head, Chest, Lungs, Stomack, Heart, Liver, Spleen, Bowels, Reins, Kidnies, Gaul, and many more: also, those Parts have other Figurative Parts belonging or adjoining to them, as the Head, Scull, Brains, *Pia-mater*, *Dura-mater*, Forehead, Nose, Eyes, Cheeks, Ears, Mouth, Tongue, and several Figurative Parts belonging to those; so of the rest of the Parts, as the Arms, Hands, Fingers, Leggs, Feet, Toes, and the like: all which different Parts, have different sorts of Perceptions; and yet (as I formerly said) their Perceptions are united: for, though all the Parts of the Human Body have different Perceptions; yet those different perceptions unite in a general Perception, both for the Subsistence, Consistence, and use of the Whole Man: but, concerning Particulars, not only the several composed Figurative Parts, have several sorts of Perceptions; but every Part hath variety of Perceptions, occasioned by variety of Objects.

CHAP. V. Of the several Perceptions amongst the several Parts of MAN.

There being infinite several Corporeal Figurative Motions, or Actions of Nature, there must of necessity be infinite several Self-knowledges and Perceptions: but I shall only, in this Part of my Book, treat of the Perception proper to Mankind: And first, of the several and different Perceptions, proper for the several and different Parts: for, though every Part and Particle of a Man's Body, is perceptive; yet, every particular Part of a Man, is not generally perceived; for, the Interior Parts do not generally perceive the Exterior; nor the Exterior, generally or perfectly, the Interior; and yet, both Interior and Exterior Corporeal Motions, agree as one Society; for, every Part, or Corporeal Motion, knows its own Office; like as Officers in a Common-wealth, although they may not be acquainted with each other, yet they know their Employments: So every particular Man in a Common-wealth, knows his own Employment, although he knows not every Man in the Common-wealth. The same do the Parts of a Man's Body, and Mind. But, if there be any Irregularity, or Disorder in a Common-wealth, every

Particular is disturbed, perceiving a Disorder in the Common-wealth. The same amongst the Parts of a Man's Body; and yet many of those Parts do not know the particular Cause of that general Disturbance. As for the Disorders, they may proceed from some Irregularities; but for Peace, there must be a general Agreement, that is, every Part must be Regular.

CHAP. VI. Of Divided and Composed Perceptions.

As I have formerly said, There is in Nature both Divided and Composed Perceptions; and for proof, I will mention Man's Exterior Perceptions; As for example, Man hath a Composed Perception of Seeing, Hearing, Smelling, Tasting, and Touching; whereof every several sort is composed, though after different manners, or ways; and yet are divided, being several sorts of Perceptions, and not all one Perception. Yet again, they are all Composed, being united as proper Perceptions of one Man; and not only so, but united to perceive the different Parts of one Object: for, as Perceptions are composed of Parts, so are Objects; and as there are different Objects, so there are different Perceptions; but it is not possible for a Man to know all the several sorts of Perceptions proper to every Composed Part of his Body or Mind, much less of others.

CHAP. VII. Of the Ignorances of the several Perceptive Organs.

As I said, That every several composed Perception, was united to the proper use of their whole Society, as one Man; yet, every several Perceptive Organ of Man is ignorant of each other; as the Perception of Sight is ignorant of that of Hearing; the Perception of Hearing, is ignorant of the Perception of Seeing; and the Perception of Smelling is ignorant of the Perceptions of the other two, and those of Scent, and the same of Tasting, and Touching: Also, every Perception of every particular Organ, is different; but some sorts of Human Perceptions require some distance between them and the Object: As

for example, The Perception of Sight requires certain Distances, as also Magnitudes; whereas the Perception of Touch requires a Joyning-Object, or Part. But this is to be noted, That although these several Organs are not perfectly, or throughly acquainted; yet in the Perception of the several parts of one Object, they do all agree to make their several Perceptions, as it were by one Act, at one point of time.

CHAP. VIII. Of the particular and general Perceptions of the Exterior Parts of Human Creatures.

There is amongst the Exterior Perceptions of Human Creatures, both particular sorts of Perceptions, and general Perceptions: For, though none of the Exterior Parts, or Organs, have the sense of Seeing, but the Eyes; of Hearing, but the Ears; of Smelling, but the Nose; of Tasting, but the Mouth: yet all the Exterior Parts have the Perception of Touching; and the reason is, That all the Exterior Parts are full of pores, or at least, of such composed Parts, that are the sensible Organs of Touching: yet, those several Parts have several Touches; not only because they have several Parts, but because those Organs of Touching, are differently composed. But this is to be noted, That every several part hath perception of the other parts of their Society, as they have of Foreign parts; and, as the Sensitive, so the Rational parts have such particular and general perceptions. But it is to be noted, That the Rational parts, are parts of the same Organs.

CHAP. IX. Of the Exterior Sensitive Organs of Human Creatures.

As for the manner, or ways, of all the several sorts, and particular perceptions, made by the different composed parts of Human Creatures; it is impossible, for a Human Creature, to know any otherwise, but in part: for, being composed of parts, into Parties, he can have but a parted knowledg, and a parted perception of himself: for, every different composed part of his

Body, have different sorts of Self-knowledg, as also, different sorts of Perceptions; but yet, the manner and way of some Human Perceptions, may probably be imagined, especially those of the exterior parts, Man names the *Sensitive Organs*; which Parts (in my opinion) have their perceptive actions, after the manner of patterning, or picturing the exterior Form, or Frame, of Foreign Objects: As for example, The present Object is a Candle; the Human Organ of Sight pictures the Flame, Light, Weck, or Snuff, the Tallow, the Colour, and the dimension of the Candle; the Ear patterns out the sparkling noise; the Nose patterns out the scent of the Candle; and the Tongue may pattern out the tast of the Candle: but, so soon as the Object is removed, the figure of the Candle is altered into the present Object, or as much of one present Object, as is subject to Human Perception. Thus the several parts or properties, may be patterned out by the several Organs. Also, every altered action, of one and the same Organ, are altered Perceptions; so as there may be numbers of several pictures or Patterns made by the Sensitive Actions of one Organ; I will not say, by one act; yet there may be much variety in one action. But this is to be noted, That the Object is not the *cause* of Perception, but is only the *occasion*: for, the Sensitive Organs can make such like figurative actions, were there no Object present; which proves, that the Object is not the Cause of the Perception. Also, when as the Sensitive parts of the Sensitive Organs, are Irregular, they will make false perceptions of present Objects; wherefore the Object is not the Cause. But one thing I desire, not to be mistaken in; for I do not say, that all the parts belonging to any of the particular Organs, move only in one sort or kind of perception; but I say, Some of the parts of the Organ, move to such, or such perception: for, all the actions of the Ears, are not only hearing; and all the actions of the Eye, seeing; and all the actions of the Nose, smelling; and all the actions of the Mouth, tasting; but, they have other sorts of actions: yet, all the sorts of every Organ, are according to the property of their figurative Composition.

CHAP. X. Of the Rational Parts of the Human Organs.

As for the Rational parts of the Human Organs, they move according to the Sensitive parts, which is, to move according to the Figures of Foreign Objects; and their actions are (if Regular) at the same point of time, with the Sensitive: but, though their Actions are alike, yet there is a difference in their Degree; for, the figure of an Object in the Mind, is far more pure than the figure in the Sense. But, to prove that the Rational (if Regular) moves with the Sense, is, That all the several Sensitive perceptions of the Sensitive Organs, (as all the several Sights, Sounds, Scents, Tastes, and Touches) are thoughts of the same.

CHAP. XI. Of the difference between the Human Conception, and Perception.

There are some differences between Perception, and Conception: for, Perception doth properly belong to present Objects; whereas Conceptions have no such strict dependency: But, Conceptions are not proper to the Sensitive Organs, or parts of a Human Creature; wherefore, the Sensitive never move in the manner of Conception, but after an irregular manner; as when a Human Creature is in some violent Passion, Mad, Weak, or the like Distempers. But this is to be noted, That all sorts of Fancies, Imaginations, &c. whether Sensitive, or Rational, are after the manner of Conceptions, that is, do move by Rote, and not by Example. Also, it is to be noted, That the Rational parts can move in more various Figurative Actions than the Sensitive; which is the cause that a Human Creature hath more Conceptions than Perceptions; so that the Mind can please it self with more variety of Thoughts than the Sensitive with variety of Objects: for variety of Objects consists of Foreign Parts; whereas variety of Conceptions consists only of their own Parts: Also, the Sensitive Parts are sooner satisfied with the perception of particular Objects, than the Mind with particular Remembrances.

CHAP. XII. Of the Several Varieties of Actions of Human Creatures.

To speak of all the Several Actions of the Sensitive and Rational parts of one Creature, is not possible, being numberless: but, some of those that are most notable, I will mention, as, Respirations, Digestions, Nourishments, Appetites, Satiety, Aversions, Conceptions, Opinions, Fancies, Passions, Memory, Remembrance, Reasoning, Examining, Considering, Observing, Distinguishing, Contriving, Arguing, Approving, Disapproving, Discoveries, Arts, Sciences. The Exterior Actions are, Walking, Running, Dancing, Turning, Tumbling, Bearing, Carrying, Holding, Striking, Trembling, Sighing, Groaning, Weeping, Frowning, Laughing, Speaking, Singing and Whistling: As for Postures, they cannot be well described; only, Standing, Sitting, and Lying.

CHAP. XIII. Of the manner of Information between the Rational and Sensitive Parts.

The manner of Information amongst the Self-moving Parts of a Human Creature, is after divers and several manners, or ways, amongst the several parts: but, the manner of Information between the Sensitive and Rational parts, is, for the most part, by Imitation; as, imitating each other's actions: As for example, The Rational parts invent some Sciences; the Sensitive endeavour to put those Sciences into an Art. If the Rational perceive the Sensitive actions are not just, according to that Science, they inform the Sensitive; then the Sensitive Parts endeavour to work, according to the directions of the Rational: but, if there be some obstruction or hindrance, then the Rational and Sensitive agree to declare their Design, and to require assistance of other Associates, which are other Men; as also, other Creatures. As for the several Manners and Informations between Man and Man, they are so ordinary, I shall not need to mention them.

CHAP. XIV. Of Irregularities and Regularities of the Self-moving Parts of Human Creatures.

Nature being poised, there must of necessity be Irregularities, as well as Regularities, both of the Rational and Sensitive parts; but when the Rational are Irregular, and the Sensitive Regular, the Sensitive endeavour to rectifie the Errors of the Rational. And if the Sensitive be Irregular, and the Rational Regular, the Rational do endeavour to rectifie the Errors of the Sensitive: for, the particular parts of a Society, are very much assistant to each other; as we may observe by the Exterior parts of Human Bodies; the Hands endeavour to assist any part in distress; the Leggs will run, the Eyes will watch, the Ears will listen, for any advantage to the Society; but when there is a general Irregularity, then the Society falls to ruine.

CHAP. XV. Of the Agreeing, or Disagreeing, of the Sensitive and Rational Parts of Human Creatures.

There is, for the most part, a general agreement between the Rational and Sensitive Parts of Human Creatures; not only in their particular, but general actions; only the Rational are the Designing-parts; and the Sensitive, the Labouring parts: As for proof, The Mind designs to go to such, or such Foreign Parts, or Places; upon which design the Sensitive Parts will labour to execute the Mind's intention, so as the whole Sensitive Body labours to go to the designed place, without the Mind's further Concern: for, the Mind takes no notice of every action of the Sensitive parts; neither of those of the Eyes, Ears; or of the Leggs, or feet; nor of their perceptions: for, many times, the Mind is busied in some Conception, Imagination, Fancy, or the like; and yet the Sensitive Parts execute the Mind's Design exactly. But, for better proof, When as the Sensitive parts are sick, weak, or defective, through some irregularities, the Sensitive parts cannot execute the Mind's Design: also, when the Sensitive parts are careless, they oft mistake their way; or when they are irregularly opposed, or busied about some Appetite, they will not obey the Mind's desire; all which

are different degrees of Parts. But, as it is amongst the particular parts of a Society; so, many times, between several Societies; for, sometimes, the Sensitive parts of two Men will take no notice of each other: As for example, When two men speak together, one man regards not what the

other says; so many times, the Sensitive parts regard not the Propositions of the Rational; but then the Sensitive is not perfectly Regular.

CHAP. XVI. Of the Power of the Rational; or rather, of the Indulgency of the Sensitive.

The Rational Corporeal Motions, being the purest, most free, and so most active, have great power over the Sensitive; as to perswade, or command them to obedience: As for example, When a man is studying about some Inventions of Poetical Fancies, or the like; though the Sensitive Corporeal Motions, in the Sensitive Organs, desire to desist from patterning of Objects, and would move towards sleep; yet the Rational will not suffer them, but causes them to work, viz. to write, or to read, or do some other Labour: Also, when the Rational Mind is merry, it will cause the Leggs to dance, the Organs of the Voice to sing, the Mouth to speak, to eat, to drink, and the like: If the Mind moves to sadness, it causes the Eyes to weep, the Lungs to sigh, the Mouth to speak words of Complaint. Thus the Rational Corporeal Motions of the Mind, will occasion the Senses to watch, to work, or to sport and play. But mistake me not; for I do not mean, the Senses are bound to obey the Rational Designs; for, the Sensitive Corporeal Motions, have as much freedom of Self-moving, as the Rational: for, the Command of the Rational, and the Obedience of the Sensitive, is rather an Agreement, than a Constraint: for, in many cases, the Sensitive will not agree, and so not obey: also, in many cases, the Rational submits to the Sensitive: also, the Rational sometimes will be irregular; and, on the other side, sometimes the Sensitive will be irregular, and the Rational regular; and sometimes both irregular.

CHAP. XVII. Of Human Appetites and Passions.

The Sensitive Appetites, and the Rational Passions do so resemble each other, as they would puzzle the most wise Philosopher to distinguish them; and there is not only a Resemblance, but, for the most part, a sympathetical Agreement between the Appetites, and the Passions; which strong conjunction, doth often occasion disturbances to the whole life of Man; with endless Desires, unsatiable Appetites, violent

Passions, unquiet Humors, Grief, Pain, Sadness, Sickness, and the like; through which, Man seems to be more restless, than any other Creature: but, whether the cause be in the Manner, or Form of Man's Composition, or occasioned by some Irregularities; I will leave to those who are wiser than I, to judg. But this is to be noted, That the more Changes and Alterations the Rational and Sensitive Motions make, the more variety of Passions and Appetites the Man hath: also, the quicker the Motions are, the sharper Appetite, and the quicker Wit, Man hath. But, as all the Human Senses are not bound to one Organ; so all Knowledges are not bound to one Sense, no more than all the Parts of Matter to the composition of one particular Creature: but, by some of the Rational and Sensitive actions, we may perceive the difference of some of the Sensitive and Rational actions; as, Sensitive Pain, Rational Grief; Sensitive Pleasure, Rational delight; Sensitive Appetite, Rational Desire; which are sympathetical actions of the Rational and Sensitive Parts: Also, through sympathy, Rational Passions will occasion Sensitive Appetites; and Appetites, the like Passions.

CHAP. XVIII. Of the Rational Actions of the Head and Heart of Human Creatures.

As I formerly said, In every Figurative Part of a Human Creature, the Actions are different, according to the Property of their different Composers; so that the Motions of the Heart are different to the Motions of the Head, and of the other several Parts: but, as for the Motions of the Head, they are (in my Opinion) more after the manner of Emboss'd Figures; and those of the Heart, more after the manner of Flat Figures; like Painting, Printing, Engraving, &c. For, if we observe, the Thoughts in our Heads are different from the Thoughts in our Hearts. I only name these two Parts, by

reason they seem to sympathize, or to agree, more particularly to each other's actions, than some of the other Parts of Human Creatures.

CHAP. XIX. Of Passions and Imaginations.

Some sorts of Passions seem to be in the Heart; as, Love, Hate, Grief, Joy, Fear, and the like; and all Imaginations, Fancies, Opinions, Inventions, &c. in the Head. But, mistake me not, I do not say, that none of the other Parts of a Man have not Passions and Conceptions: but, I say, they are not after the same manner, or way, as in the Heart, or Head: as for example, Every Part of a Man's Body is sensible, yet not after one and the same manner: for, every Part of a Man's Body hath different perceptions, as I have formerly declared, and yet may agree in general actions: but, unless the several composed Parts of a Human Creature, had not several perceptive actions, it were impossible to make a general perception, either amongst the several Parts of their own Society, or of Foreign Objects. But, it is impossible for me to describe the different manners and ways of the particular Parts, or the different actions of any one Part: for, what Man can describe the different perceptive actions of that composed Part, the Eye, and so of the rest of the Parts.

CHAP. XX. That Associations, Divisions, and Alterations, cause several Effects.

The Rational and Sensitive Corporeal Motions, are the perceptive Parts of Nature; and that which causes acquaintance amongst some parts, is their Uniting and Association: That which loses acquaintance of other Parts, is their Divisions and Alterations: for, as Self-compositions cause particular Knowledges, or Acquaintances: So Self-divisions cause particular Ignorances, or Forgetfulnesses: for, as all kinds and sorts of Creatures are produced, nourished, and encreased by the Association of Parts; so are all

kinds and sorts of Perceptions; and according as their Associations, or their Compositions do last, so doth their Acquaintance; which is the cause, that the Observations and Experiences of several and particular Creatures, such as Men, in several and particular Ages, joyned as into one Man or Age, causes strong and long-liv'd Opinions, subtile and ingenious Inventions, happy and profitable Advantages; as also, probable Conjectures, and many Truths, of many Causes and Effects: Whereas, the Divisions of particular Societies, causes what we name Death, Ignorance, Forgetfulness, Obscurity of particular Creatures, and of perceptive Knowledges; so that as particular perceptive Knowledges do alter and change, so do particular Creatures: for, though the Kinds and Sorts last, yet the Particulars do not.

CHAP. XXI. Of the Differences between Self-Love, and Passionate Love.

Self-love, is like Self-knowledg, which is an innate Nature; and therefore is not that Love Man names Passionate Love: for, *Passionate Love* belongs to several Parts; so that the several parts of one Society, as one Creature, have both Passionate Love, and Self-love, as being sympathetically united in one Society: Also, not only the Parts of one and the same Society, may have Passionate Love to each other; but, between several Societies; and not only several Societies of one Sort, but of different Sorts.

The Sixth Part.

CHAP. I. Of the Motions of some parts of the Mind; and of Forrein Objects.

Notions, Imaginations, Conceptions, and the like, are such Actions of the Mind, as concern not Forrein Objects: and some Notions, Imaginations, or Conceptions of one man, may be like to another man, or many men. Also, the Mind of one man may move in the like Figurative Actions, as the Sensitive Actions of other sorts of Creatures; and that, Man names *Understanding*: and if those Conceptions be afterwards produced, Man names them *Prudence*, or *Fore-sight*; but if those Parts move in such Inventions as are capable to be put into Arts, Man names that, *Ingenuity*: but, if not capable to be put into the practice of Arts, Man names it, *Sciences*: if those Motions be so subtile, that the Sensitive cannot imitate them, Man names them, *Fancies*: but, when those Rational Parts move promiscuously, as partly after their own inventions, and partly after the manner of Forrein or outward Objects; Man names them, *Conjectures*, or *Probabilities*: and when there are very many several Figurative, Rational Motions, then Man says, *The Mind is full of Thoughts*: when those Rational Figurative Motions, are of many and different Objects, Man names them, *Experiences*, or *Learning*: but, when there are but few different sorts of such Figurative Motions, Man names them *Ignorances*.

CHAP. II. Of the Motions of some Parts of the Mind.

When the Rational Figurative Corporeal Motions of an Human Creature, take no notice of Forrein Objects, Man nameth that, *Musing*, or *Contemplating*. And, when the Rational Parts repeat some former Actions,

Man names that, *Remembrances*. But, when those Parts alter those Repetitions, Man names that, *Forgetfulness*. And, when those Rational Parts move, according to a present Object, Man names it, *Memory*. And when those Parts divide in divers sorts of Actions, Man names it, *Arguing*, or *Disputing in the Mind*. And when those divers sorts of Actions are at some strife, Man names it, *A contradicting of himself*. And if there be a weak strife, Man names it, *Consideration*. But, when those different Figurative Motions move of one accord, and sympathetically, this Man names, *Discretion*. But, when those different sorts of Actions move sympathetically, and continue in that manner of action, without any alteration, Man names it, *Belief*, *Faith*, or *Obstinacy*. And when those Parts make often changes, as altering their Motions, Man names it *Inconstancy*. When their Rational Parts move slowly, orderly, equally, and sympathetically, Man names it *Sobriety*. When all the Parts of the Mind move regularly, and sympathetically, Man names it, *Wisdom*. When some Parts move partly regularly, and partly irregularly, Man names that, *Foolishness*, and *Simplicity*. When they move generally irregularly, Man names it *Madness*.

CHAP. III. Of the Motions of Human Passions, and Appetites; as also, of the Motions of the Rational and Sensitive Parts, towards Forrein Objects.

When some of the Rational Parts move sympathetically, to some of the Sensitive Perceptions; and those Sensitive Parts sympathize to the Object, it is *Love*. If they move antipathetically to the Object, it is *Hate*. When those Rational and Sensitive Motions, make many and quick repetitions of those sympathetical actions, it is *Desire* and *Appetite*. When those Parts move variously, (as concerning the Object) but yet sympathetically (concerning their own Parts) it is *Inconstancy*. When those Motions move cross towards the Object, and are perturbed, it is *Anger*. But when those perturbed Motions are in confusion, it is *Fear*. When the Rational Motions are partly sympathetical, and partly antipathetical, it is *Hope*, and *Doubt*. And if there be more sympathetical Motions than antipathetical, there is more *Hope* than

Doubt. If more antipathetical than sympathetical, then more *Doubt* than *Hope*. If those Rational Motions move after a dilating manner, it is *Joy*. If after a contracting manner, it is *Grief*. When those Parts move partly after a contracting, and partly after an attracting manner, as attracting from the Object, it is *Covetousness*. But, if those Motions are sympathetical to the Object, and move after a dilating manner towards the Object, it is *Generosity*. If those Motions are sympathetical to the Object, and move after the manner of a Contraction, it is *Pity* or *Compassion*. If those Motions move antipathetically towards the Object, yet after a dilating manner, it is *Pride*. When those Motions move sympathetically towards the Object, after a dilating manner, it is *Admiration*. If the dilating Action is not extream, it is only *Approving*. If those Motions are antipathetical towards the Object, and are after the manner of an extream contraction, it is *Horror*. But, if those Actions are not so extraordinary as to be extream, it is only *Disapproving*, *Despising*, *Rejecting*, or *Scorning*. If the Rational Parts move carelessly towards Forrein Objects, as also partly antipathetically, Man nameth it, *Ill-nature*. But, if sympathetically and industriously, Man nameth it, *Good-nature*. But this is to be noted, That there are many sorts of Motions of one and the same kind; and many several particular Motions, of one sort of Motion; which causes some difference in the Effects: but, they are so nearly related, that it requires a more subtile Observation than I have, to distinguish them.

CHAP. IV. Of the Repetitions of the Sensitive and Rational Actions.

Both the Rational and Sensitive Corporeal Motions, make often Repetitions of one and the same Actions: The Sensitive Repetitions, Man nameth, *Custom*. The Rational Repetitions, Man nameth, *Remembrances*: for, Repetitions cause a facility amongst the Sensitive Parts; but yet, in some Repeating Actions, the Senses seem to be tired, being

naturally delighted in variety. Also, by the Rational Repetitions, the Mind is either delighted, or displeased; and sometimes, partly pleased, and partly displeased: for, the Mind is as much pleased, or displeased in the absence of an Object, as in the presence; only the Pleasure, and Displeasure of the

Senses, is not joyned with the Rational: for, the Sense, if Regular, makes the most perfect Copies when the Object is present: but, the Rational can make as perfect Copies in the absence, as in the presence of the Object; which is the cause that the Mind is as much delighted, or grieved, in the absence of an Object, as with the presence: As for example, A Man is as much grieved when he knows his Friend is wounded, or dead, as if he had seen his Wounds, or had seen him dead: for, the Picture of the dead Friend, is in the mind of the living Friend; and if the dead Friend was before his Eyes, he could but have his Picture in his mind; which is the same for an absent Friend alive; only, as I said, there is wanting the Sensitive Perception of the absent Object: And certainly, the Parts of the Mind have greater advantage than the Sensitive Parts; for, the Mind can enjoy that which is not subject to the Sense; as those things Man names, *Castles in the Air*, or *Poetical Fancies*; which is the reason Man can enjoy Worlds of its own making, without the assistance of the Sensitive Parts; and can govern and command those Worlds; as also, dissolve and compose several Worlds, as he pleases: but certainly, as the pleasures of the Rational Parts are beyond those of the Sensitive, so are their Troubles.

CHAP. V. Of the Passionate Love, and Sympathetical Endeavours, amongst the Associate Parts of a Human Creature.

In every Regular Human Society, there is a Passionate Love amongst the Associated Parts, like fellow-Students of one Colledg, or fellow-Servants in one House, or Brethren in one Family, or Subjects in one Nation, or Communicants in one Church: So the Self-moving Parts of a Human Creature, being associated, love one another, and therefore do endeavour to keep their Society from dissolving. But perceiving, by the example of the lives of the same sort of Creatures, that the property of their Nature is such, that they must dissolve in a short time, this causes these Human sorts of Creatures, (being very ingenuous) to endeavour an after-life: but, perceiving again, that their after-life cannot be the same as the present life is, they endeavour (since they cannot keep their own Society from dissolving) that their Society may remain in remembrance amongst the particular and

general Societies of the same sort of Creatures, which we name Mankind: And this Design causes all the Sensitive and Rational Parts, in one Society, to be industrious, to leave some Mark for a lasting Remembrance, amongst their fellow-Creatures: which general remembrance, Man calls *Fame*; for which *Fame*, the Rational Parts are industrious to design the manner and way, and the Sensitive Parts are industrious to put those Designs in execution; as, their Inventions, into Arts or Sciences; or to cause their Heroick or Prudent, Generous or Pious Actions; their Learning, or witty Fancies, or subtile Conceptions, or their industrious Observations, or their ingenious Inventions, to be set in Print; or their Exterior Effigies to be cast, cut, or engraven in Brass, or Stone, or to be painted; or they endeavour to build Houses, or cut Rivers, to bear their Names; and millions of other Marks, for remembrance, they are industrious to leave to the perception of after-Ages: And many men are so desirous of this after-life, that they would willingly quit their present life, by reason of its shortness, to gain this after-life, because of the probability of a long continuance; and not only to live so in many several Ages, but in many several Nations. And amongst the number of those that prefer a long after-life, before a short present life, I am one. But, some men dispute against these Desires, saying, That *it doth a man no good to be remembred when he is dead*. I answer: It is very pleasing, whilst as man lives, to have in his Mind, or in his Sense, the Effigies of the Person, and of the good Actions of his Friend, although he cannot have his present company. Also, it is very pleasant to any body to believe, that the Effigies either of his own Person, or Actions, or both, are in the Mind of his Friend, when he is absent from him; and, in this case, Absence and Death are much alike. But, in short, God lives no other ways amongst his Creatures, but in their Rational Thoughts, and Sensitive Worship.

CHAP. VI. Of ACQUAINTANCE.

As there are Perceptive Acquaintances amongst the Parts of a Human Creature; so there is a Perceptive Acquaintance between, or amongst the Human sorts of Creatures. But, mistake me not; for I do not say, Men only

are acquainted with each other; for, there is not only an Acquaintance amongst every particular sort, as between one and the same sort of Creatures, but there are some Acquaintances between some sorts of different kinds: as for example, Between some sorts of Beasts, and Men; as also, some sorts of Birds, and Men, which understand each other, I will not say, so well as Man and Man; but so well, as to understand each other's Passions: but certainly, every particular sort of Creatures, of one and the same kind, understand each other, as well as Men understand one another; and yet, for all that, they may be unacquainted: for, Acquaintance proceeds from Association; so that, some Men, and some Beasts, by Association, may be acquainted with each other; when as some Men, not associating, are meer strangers. The truth is, Acquaintance belongs rather to Particularities, than Generalities.

CHAP. VII. Of the Effects of Forrein Objects of the Sensitive Body; and of the Rational Mind of a Human Creature.

According as the Rational Parts are affected, or disaffected with Forrein Objects, the Sensitive is apt to express the like affections, or disaffections: for, most Forrein Objects occasion either pleasure and delight, or displeasure and dislike: but, the effects of Forrein Objects are very many, and, many times very different; as, some Objects of Devotion, occasion a Fear, or Superstition, and Repentance in the Mind; and the Mind occasions the Sensitive Parts to several actions, as, Praying, Acknowledging Faults, Begging pardon, making Vows, imploring Mercy, and the like, in words: also, the Body bows, the Knees bend, the Eyes weep, the hands hold up, and many the like devout actions. Other sorts of Objects occasion pity and compassion in the Mind, which occasions the Sensitive Parts to attend the sick, relieve the poor, help the distressed, and many more actions of Compassion. Other sorts of Forrein Objects, occasion the Rational Mind to be dull and melancholy; and then the Sensitive Parts are dull, making no variety of Appetites, or regard Forrein Objects. Other sorts of Objects occasion the Mind to be vain and ambitious, and often to be proud; and those occasion the Sensitive Actions to be adventurous and bold; the

Countenance of the face, scornful; the Garb of the Body, stately; the Words, vaunting, boasting, or bragging. Other Objects occasion the Mind to be furious; and then the Sensitive Actions are, Cursing Words, Frowning Countenances, the Leggs stamping, the Hands and Arms fighting, and the whole Body in a furious posture. Other sorts of Objects occasion the Mind to a passionate Love; and then the Sensitive Actions are, Flattering, Professing, Protesting in words, the Countenance smiling, the Eyes glancing; also, the Body bows, the Leggs scrape, the Mouth kisses: also, the Hands mend their Garments, and do many of the like amorous actions. Other Objects occasion the Mind to Valour; and then the Sensitive Actions are, Daring, Encouraging, or Animating. Other Objects occasion the Mind to Mirth, or Cheerfulness; and they occasion the Sensitive Actions of the Voice, to Sing, or Laugh; the Words to be jesting, the Hands to be toying, the Leggs to be dancing. Other sorts of Objects occasion the Mind to be Prudent; and then the Sensitive Actions, are Sparing or Frugal. Other sorts of Objects occasion the Mind to be Envious, or Malicious; and then the Sensitive Actions are Mischievous. There are great numbers of Occasional Actions, but these are sufficient to prove, *That Sense and Reason understand each other's Actions or Designs.*

CHAP. VIII. Of the Advantage and Disadvantage of the Encounters of several Creatures.

There is a strong Sympathy between the Rational and Sensitive Parts, in one and the same Society, or Creature: not only for their Consistency, Subsistency, Use, Ease, Pleasure, and Delight; but, for their Safety, Guard, and Defence: as for example, When one Creature assaults another, then all the Powers, Faculties, Properties, Ingenuities, Agilities, Proportions, and Shape, of the Parts of the Assaulted, unite against the Assaulter, in the defence of every particular Part of their whole Society; in which Encounter, the Rational advises, and the Sensitive labours. But this is to be noted concerning advantage and disadvantage in such Encounters, That some sorts of Creatures have their advantage in the Exterior Shape, others meerly in the Number of Parts; others in the agility of their Parts, and some by the

ingenuity of their parts: but, for the most part, the greater Number have advantage over the less, if the greater number of Parts be as regular, and as ingenious as the less number: but, if the less number be more regular, and more ingenious than the greater, then 'tis a hundred to one but the less number of Parts have the advantage.

CHAP. IX. That All Human Creatures have the like Kinds and Sorts of Properties.

All Human Creatures have the like Kinds and Sorts of Properties, Faculties, Respirations, and Perceptions; unless some Irregularities in the Production, occasion some Imperfections, or some Misfortunes, in some time of his Age: yet, no Man knows what another Man perceives, but by guess, or information of the Party: but, as I said, if they have have no Imperfections, all Human Creatures have like Properties, Faculties, and Perceptions: As for example, All Human Eyes may see one and the same Object alike; or hear the same Tune, or Sound; and so of the rest of the Senses. They have also the like Respirations, Digestions, Appetites; and the like may be said of all the Properties belonging to a Human Creature. But, as one Human Creature doth not know what another Human Creature knows, but by Confederacy; so, no Part of the Body, or Mind of a Man, knows each Part's perceptive knowledg, but by Confederacy: so that, there is as much Ignorance amongst the Parts of Nature, as Knowledg. But this is to be noted, That there are several manners and ways of Intelligences, not only between several sorts of Creatures, or amongst particulars of one sort of Creatures; but, amongst the several Parts of one and the same Creature.

CHAP. X. Of the Irregularity of the Sensitive, and of the Rational Corporeal Motions.

As I have often mentioned, and do here again repeat, That the Rational and Sensitive Parts of one Society, or Creature, do understand, as perceiving each other's Self-moving Parts; and the proof is, That, sometimes, the Human Sense is regular, and the Human Reason irregular; and sometimes the Reason regular, and the Sense irregular: but, in these differences, the Regular Parts endeavour to reform the Irregular; which causes, many times, repetitions of one and the same Actions, and Examinations; as, sometimes the Reason examines the Sense; and sometimes the Sense, the Reason: and sometimes the Sense and Reason do examine the Object; for, sometimes an Object will delude both the Sense and Reason; and sometimes the Sense and Reason are but partly mistaken: As for example, A fired end of a Stick, by a swift exterior Circular Motion, appears a Circle of fire, in which they are not deceived: for, by the Exterior Motion, the fired end is a Circle; but they are mistaken, to conceive the Exterior Figurative Action to be the proper natural Figure: but when one man mistakes another, that is some small Error, both of the Sense and Reason. Also, when one man cannot readily remember another man, with whom he had formerly been acquainted, it is an Error; and such small Errors, the Sense and Reason do soon rectifie: but in causes of high Irregularities, as in Madness, Sickness, and the like, there is a great Bustle amongst the Parts of a Human Creature; so as those Disturbances cause unnecessary Fears, Grief, Anger, and strange Imaginations.

CHAP. XI. Of the Knowledg between the Sensitive Organs of a Human Creature.

The Sensitive Organs are only ignorant of each other, as they are of Forrein Objects: for, as all the Parts of Forrein Objects, are not subject to one Sensitive Organ; so all the Sensitive Organs are not subject to each Sensitive Organ of a Human Creature: yet, in the perceptive Actions of Forrein Objects, they do so agree, that they make an united Knowledg: Thus we may be particularly ignorant one way, and yet have a general Knowledg another way.

CHAP. XII. Of Human Perception, or Defects of a Human Creature.

It is not the great quantity of Brain, that makes a Man wise; nor a little quantity, that makes a Man foolish: but, the irregular, or regular Rational Corporeal Motions of the Head, Heart, and the rest of the Parts, that causes dull Understandings, short Memories, weak Judgments, violent Passions, extravagant Imaginations, wild Fancies, and the like. The same must be said of the Sensitive Irregular Corporeal Motions, which make Weakness, Pain, Sickness, disordered Appetites, and perturbed Perceptions, and the like: for, Nature poysing her Actions by Opposites, there must needs be Irregularities, as well as Regularities; which is the cause that seldom any Creature is so exact, but there is some Exception. But, when the Sensitive and Rational Corporeal Motions are regular, and move sympathetically, then the Body is healthful and strong, the Mind in peace and quiet, understands well, and is judicious: and, in short, there are perfect Perceptions, proper Digestions, easie Respirations, regular Passions, temperate Appetites. But when the Rational Corporeal Motions are curious in their change of Actions, there are subtile Conceptions, and elevated Fancies: and when the Sensitive Corporeal Motions move with curiosity, (as I may say) then there are perfect Senses, exact Proportions, equal Temperaments; and that, Man calls *Beauty*.

CHAP. XIII. Of Natural FOOLS.

There is great difference between a Natural Fool, and a Mad Man: for, Madness is a Disease, but a Natural Fool is a Defect; which Defect was some Error in his Production, that is, in the form and frame either of the Mind, or Sense, or both; for, the Sense may be a Natural Fool as well as the Reason; as we may observe in those sorts of Fools whom we name *Changelings*, whose Body is not only deformed, but all the Postures of the Body are defective, and appear as so many fools: but sometimes, only some Parts are fools; as for example; If a Man be born Blind, then only his Eyes

are Fools; if Deaf, then only his Ears are Fools, which occasions his dumbness; Ears being the informing Parts, to speak; and wanting those informations, he cannot speak a Language. Also, if a Man is born lame, his Leggs are Fools; that is, those Parts have no knowledg of such Properties that belong to such Parts; but the Sensitive Parts may be wise, as being knowing; and the Rational Parts may be defective; which Defects, Man names *Irrational*. But this is to be noted, That there may be Natural and Accidental Fools, by some extraordinary Frights, or by extraordinary Sickness, or through the defects of Old Age. As for the Errors of Production, they are incurable; as also, those of Old Age; the First being an Error in the very Foundation, and the other a Decay of the whole Frame of the Building: for, after a Human Creature is brought to that perfection, as to be, as we may say, at full growth and strength, at the prime of his age; the Human Motions, and the very Nature of Man, after that time, begins to decay; for then the Human Motions begin to move rather to the dissolution, than to the continuance; although some Men last to very old Age, by reason the unity of their Society is regular and orderly, and moves so Sympathetically, as to commit few or no Disorders, or Irregularities; and such old Men are, for the most part, Healthful, and very wise, through long Experience; and their Society having got a habit of Regularity, is not apt to be disturbed by Forrein Parts. But this is to be noted, That sometimes the Sensitive Body decays, before the Rational Mind; and sometimes the Rational Mind, before the Sensitive Body. Also, this is to be noted, That when the Body is defective, but not the Mind;

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then the Mind is very industrious to find out Inventions of Art, to help the Defects that are natural. But pray mistake me not; for I do not say, That *all* Deformities, or Defects, but only *some* particular sorts of Deformity, or Defects, are Foolish.

The Seventh Part.

CHAP. I. Of the Sensitive Actions of Sleeping and Waking.

The Sensitive and Rational Corporeal Figurative Motions, are the cause of infinite varieties: for, though Repetitions make no varieties; yet, every altered action is a variety: Also, different Actions, make different Effects; opposite Actions, opposite Effects; not only of the actions of the several Self-moving Parts, or Corporeal Motions, but of the same Parts: As for example, The same Parts, or Corporeal Motions, may move from that, Man names *Life*, to that which Man names *Death*; or, from Health to Sickness, from Ease to Pain, from Memory to Forgetfulness, from Forgetfulness to Remembrance, from Love to Hate, from Grief to Joy, from Irregularity to Regularity; or, from Regularity to Irregularity, and the like; and from one Perception to another: for, though all actions are perceptive, yet there are several kinds, several sorts, and several particular perceptions: But, amongst the several Corporeal Motions of Animal, or Human kind, there are the opposite Motions of what we name *Waking*, and *Sleeping*; the difference is, That Waking-actions are, most commonly, actions of Imitation, especially of the Sensitive Parts; and are more the Exterior, than the Interior actions of a Human Creature. But, the actions of Sleep, are the alterations of the Exterior Corporeal Motions, moving more interiorly, as it were inwardly, and voluntarily: As for example, The Optick Corporeal Motions, in Waking-actions, work, or move, according to the outward Object: but, in Sleeping-actions, they move by rote, or without Examples; also, as I said, they move, as it were, inwardly; like as a Man should turn himself inward, or outward, of a door, without removing from the door, or out of the place he stood in.

CHAP. II. Of SLEEPING.

Although the Rational and Sensitive Corporeal Motions, can never be tired, or weary of moving or acting, by reason it is their nature to be a perpetual Corporeal Motion; yet they may be weary, or tired with particular actions. Also, it is easier and more delightful, to move by Rote, than to take Copies, or Patterns; which is the reason that Sleep is easie and gentle, if the Corporeal Motions be regular; but if they be irregular, Sleep is perturbed. But this is to be noted, That the Corporeal Motions delight in varieties so well, that, many times, many and various Objects will cause the Sensitive and Rational Corporeal Motions in a Man, to retard their actions of Sleep; and, oft-times, want of variety of Forrein or outward Objects, will occasion the action of Sleep; or else Musing and Contemplating actions. Also, it is to be noted, That if some Parts of the Body, or Mind, be distempered with Irregularities, it occasions such disturbances to the Whole, as hinders that repose; but if the Regular Parts endeavour not to be disturbed with the Irregular; and the Irregulars do disturb the Regular; then it occasions that which Man names, *Half-sleeps*, or *Slumbers*, or *Drowsiness*. And if the Regular Corporeal Motions get

the better, (as many times they do) then we say, Sleep hath been the occasion of the Cure; and it oft proves so. And it is a common saying, *That a good Sleep will settle the Spirits*, or ease the Pains; that is, when the Regular Corporeal Motions have had the better of the Irregular.

CHAP. III. Of Human DREAMS.

There are several kinds, sorts, and particulars of Corporeal Irregularities, as well as of Regularities; and amongst the infinite kinds, sorts, and particulars, there is that of Human Dreams; for, the Exterior Corporeal Motions in Waking-actions, do copy or pattern outward Objects; whereas, in actions of Sleep, they act by rote, which, for the most part, is erroneous, making mixt Figures of several Objects; as, partly like a Beast; and partly, like a Bird, or Fish; nay, sometimes, partly like an Animal, and partly like a Vegetable; and millions of the like Extravagancies; yet, many times,

Dreams will be as exact as if a Man was awake, and the Objects before him; but, those actions by rote, are more often false than true: but, if the Self-moving Parts move after their own inventions, and not after the manner of Copying; or, if they move not after the manner of Human Perception, then a Man is as ignorant of

his Dreams, or any Human Perception, as if he was in a Swound; and then he says, he did not dream; and, that such Sleeps are like Death.

CHAP. IV. Of the Actions of DREAMS.

When the Figures of those Friends and Acquaintants that have been dead a long time, are made in our Sleep, we never, or seldom question the truth of their being alive, though we often question them how they came to be alive: And the reason that we make no doubt of their being alive, is, That those Corporeal Motions of Sleep, make the same pattern of that Object in Sleep, as when that Object was present, and patterned awake; so as the Picture in Sleep seems to be the Original awake: and until such times that the Corporeal Motions alter their Sleeping-Actions to Waking-Actions, the truth is not known. Though Sleeping and Dreaming, is somewhat after the manner of Forgetfulness and Remembrance; yet, perfect Dreams are as perceptive as Waking-patterns of present Objects; which proves, That both the Sensitive and Rational Motions, have Sleeping Actions; but both the Sensitive and Rational Corporeal Actions in Sleep, moving partly by rote, and partly voluntarily, or by invention, make Walking-Woods, or Wooden Men; or make Warrs and Battels, where some Figures of Men are kill'd, or wounded, others have victory: They also make Thieves, Murderers, falling Houses, great Fires, Floods, Tempests, high Mountains, great Precipices; and sometimes pleasant Dreams of Lovers, Marriage, Dancing, Banquetting, and the like: And the Passions in Dreams are as real, as in waking actions.

CHAP. V. Whether the Interior Parts of a Human Creature, do sleep.

The Parts of my Mind were in dispute, Whether the Interior Parts of a Human Creature, had sleeping and waking actions? The Major Part was of opinion, That Sleep was not proper to those Human Parts, because the Interior Motions were not like the Exterior. The Opinion of the Minor Part was, That change of Action, is like Ease after Labour; and therefore it was probable, the Interior Parts had sleeping and waking actions. The Opinion of the Major Parts, was, That if those Parts, as also the Food received into the Body, had sleeping actions, the Body could not be nourished; for, the Meat would not be digested into the like Parts of the Body, by reason sleeping actions were not such sorts of actions. The Opinion of the Minor Parts was, That the sleeping

actions were nourishing actions, and therefore were most proper for the Interior Parts; and, for proof, the whole Human Body becomes faint and weak, when they are hindred, either by some Interior Irregularity, or through some Exterior Occasion, from their sleeping actions. The Opinion of the Major Part, was, That sleeping actions are actions of rote, and not such altering actions as digesting actions, and nourishing actions, which are uniting actions. Besides, that the reason why the Interior actions are not sleeping actions, was, That when the Exterior Parts move in the actions of Sleep, the Interior Parts move when the Exterior are awake; as may be observed by the Human Pulse, and Human Respiration; and by many other Observations which may be brought.

CHAP. VI. Whether all the Creatures in Nature, have Sleeping and Waking Actions.

Some may ask this Question, *Whether all Creatures have sleeping Actions?* I answer, That though sleeping actions are proper to Human Creatures, as also, to most Animal Creatures; yet, such actions may not any ways be proper to other kinds and sorts of Creatures: and if (as in all probability it is) that the Exterior Parts of a Human Creature have no such sleeping actions, it is probable that other kinds and sorts of Creatures move not at

any time, in such sorts of actions. But some may say, *That if Nature is poysed, all Creatures must have sleeping actions, as well as waking actions.* I answer, That though Nature's actions are poysed, yet that doth not hinder the variety of Nature's actions, so as to tye Nature to particular actions: As for example, The Exterior Parts of Animals have both sleeping and waking actions; yet that doth not prove, that therefore all the Parts or Creatures in Nature, must have sleeping and waking actions. The same may be said of all the actions of an Animal Creature, or of a Human Creature; nay, of all the Creatures of the World: for, several kinds and sorts of Creatures, have several kinds and sorts of Properties: Wherefore, if there be other kinds and sorts of Worlds besides this, 'tis probable that those Worlds, and all the Parts, or several kinds and sorts of Creatures there, have different properties and actions, from those of this World; so that though Nature's actions are poysed and balanced, yet they are poysed and balanced after different manners and ways.

CHAP. VII. Of Human Death.

Death is not only a general Alteration of the Sensitive and Rational Motions, but a general Dissolution of their Society. And as there are degrees of Time in Productions, so in Dissolutions. And as there are degrees to Perfection, as from Infancy to Manhood; so there are degrees from Manhood to Old Age. But, as I said, *Death* is a general Dissolution, which makes a Human Creature to be no more: yet, some Parts do not dissolve so soon as others; as for example, Human Bones; but, though the Form or Frame of Bones is not dissolved; yet the Properties: of those Bones are altered. The same when a Human Creature is kept by Art from dissolving, so as the Form, or Frame, or Shape may continue; but all the Properties are quite altered; though the Exterior Shape of such Bodies doth appear somewhat like a Man, yet that Shape is not a Man.

CHAP. VIII. Of the Heat of Human Life, and the Cold of Human Death.

There are not only several sorts of Properties belonging to several sorts of Creatures, but several sorts of Properties belonging to one and the same sort of Creature; and amongst the several sorts of Human Properties, Human Heat is one, which Man names *Natural Heat*: but, when there is a general alteration of the Human Properties, there is that alteration of the Property as well of his Natural, as Human Heat: but, Natural Heat is not the cause of Human Life, though Human Life is the cause of that Natural Heat: so that, when Human Life is altered or dissolved, Human Heat is altered or dissolved: And as Death is opposite Actions to that Man names *Life*; so Cold is opposite Actions to that Man names *Heat*.

CHAP. IX. Of the Last Act of Human Life.

The reason some Human Creatures dye in more pain than others, is, That the Motions of some Human Creatures are in strife, because some would continue their accustomed Actions, others would alter their accustomed Actions; which Strife causes Irregularities, and those Irregularities cause Differences, or Difficulties, which causes Pain: but certainly, the last Act of Human Life is easie; not only that the Expulsive Actions of Human Respirations, are more easie than the Attracting Actions; but, that in the last act of Human Life, all the Motions do generally agree in one Action.

CHAP. X. Whether a Human Creature hath Knowledg in Death, or not?

Some may ask the Question, *Whether a dead Man hath any Knowledg or Perception?* I answer, That a dead Man hath not a Human Knowledg or Perception; yet all, and every Part, hath Knowledg and Perception: But, by reason there is a general alteration of the actions of the Parts of a Human Creature, there cannot possibly be a Human Knowledg or Perception. But some may say, That a Man in a Swound hath a general alteration of Human actions; and yet those Parts of a Human Creature do often repeat those former actions, and then a Man is as he was before he was in that Swound. I

answer, That the reason why a Man in a Swound hath not the same Knowledg as when he is not in a Swound, is, That the Human Motions are not generally altered, but only are generally irregular; which makes such a disturbance, that no Part can move so regularly, as to make proper Perceptions; as in some sorts of Distempers, a Man may be like a Natural Fool; in others, he may be Mad; and is subject to many several Distempers, which cause several Effects: but a Human Swound is somewhat like Sleeping without Dreaming; that is, the Exterior Senses do not move to Human Exterior Perception.

CHAP. XI. Whether a Creature may be new Formed, after a general Dissolution.

Some may ask the Question, *Whether a Human Creature, or any other Creature, after their Natural Properties are quite altered, can be repeated, and rechanged, to those Properties that formerly were?*

I answer, Yes, in case none of the Fundamental Figurative Parts be dissolved.

But some may ask, *That if those dissolved Parts were so inclosed in other Bodies, that none of them could easily disperse or wander; whether they might not joyn into the same Form and Figure again, and have the same Properties?*

I answer, I cannot tell well how to judg; but I am of the opinion, they cannot: for, it is the property of all such Productions, to be performed by degrees, and that there should be a dividing and uniting of Parts, as an intercourse of Home and Forrein Parts; and so there is requir'd all the same Parts, and every Part of the same Society, or that had any adjoining actions with that particular Creature; as all those Parts, or Corporeal Motions, that had been from the first time of Production, to the last of the Dissolving; and that could not be done without a Confusion in Nature.

But some may say, *That although the same Creature could not be produced after the same manner, nor return to the degree of his Infancy, and pass the*

degrees from his Infancy, to some degree of Age; yet, those parts that are together, might so joyn, and move, in the same manner, as to be the same Creature it was before its dissolution?

I answer, It may not be impossible: but yet, It is very improbable, that such numerous sorts of Motions, after so general an Alteration, should so generally agree in an unnatural action.

CHAP. XII. Of FOREKNOWLEDG.

I have had some Disputes amongst the Parts of my Mind, *Whether Nature hath Foreknowledg?* The Opinion of the Minor Parts was, That Nature had Foreknowledg, by reason all that was Material, was part of her self; and those Self-parts having Self-motion, she might foreknow what she would act, and so what they should know. The Opinion of the Major Parts was, That by reason every Part had Self-motion, and natural Free-will, Nature could not foreknow how they would move, although she might know how they have moved, or how they do move.

After this Dispute was ended, then there was a Dispute, *Whether the particular Parts had a Foreknowledg of Self-knowledg?* The Opinion of the Minor Parts was, That since every Part in Nature had Self-motion, and natural Free-will, every Part could know how they should move, and so what they should know. The Opinion of the Major Parts was, That first, the Self-knowledg did alter according to Self-action, amongst the Self-moving Parts: but, the Self-knowledg of the Inanimate Parts, did alter according to the actions of the Sensitive Self-moving Parts; and the Perceptive actions of the Self-moving Parts, were according to the form and actions of the Objects: so that Foreknowledg of Forein Parts, or Creatures, could not be: And for Foreknowledg of Self-knowledg of the Self-moving Parts, there were so many occasional actions, that it was impossible the Self-moving Parts could know how they should move, by reason that no Part had an Absolute Power, although they were Self-moving, and had a natural Free-will: which proves, That Prophecies are somewhat of the nature of Dreams,

whereof some may prove true by chance; but, for the most part, they are false.



The Eighth Part.

CHAP. I. Of the Irregularity of Nature's Parts.

Some may make this Question, that, *If Nature were Self-moving, and had Free-will, it is probable that she would never move her Parts so irregularly, as to put her self to pain.*

I answer, first, That Nature's Parts move themselves, and are not moved by any Agent. Secondly, Though Nature's Parts are Self-moving, and Self-knowing, yet they have not an infinite or uncontrollable Power; for, several Parts, and Parties, oppose, and oft-times obstruct each other; so that many times they are forced to move, and they may not when they would. Thirdly, Some Parts may occasion other Parts to be irregular, and keep themselves in a regular posture. Lastly, Nature's Fundamental actions are so poysed, that Irregular actions are as natural as Regular.

CHAP. II. Of the Human Parts of a Human Creature.

The Form of Man's Exterior and Interior Parts, are so different, and so numerous; that I cannot describe them, by reason I am not so learned to know them: But, some Parts of a Human Creature, Man names *Vital*; because, the least disturbance of any of those Parts, endangers the Human Life: and if any of those Vital Parts are diminished, I doubt whether they can be restored; but if some of those Parts can be restored, I doubt all cannot. The Vital Parts are, the Heart, Liver, Lungs, Stomack, Kidneys, Bladder, Gaul, Guts, Brains, Radical Humours, or Vital Spirits; and others which I know not of. But this is to be noted, That Man is composed of Rare and Solid Parts, of which there are more and less Solid, more and less Rare;

as also, different sorts of Solid, and different sorts of Rare: also, different sorts of Soft and Hard Parts; likewise, of Fixt and Loose Parts; also, of Swift and Slow Parts. I mean by Fixt, those that are more firmly united.

CHAP. III. Of Human Humours.

Humours are such Parts, that some of them may be divided from the whole Body, without danger to the whole Body; so that they are somewhat like Excremental parts, which Excremental parts, are the superfluous parts: for, though the Humours be so necessary, that the Body could not well subsist without them; yet, a Superfluity of them is as dangerous, (if not more) as a Scarcity. But there are many sorts of Humours belonging to a Human Creature, although Man names but Four, according to the Four Elements, viz. *Flegm*, *Choler*, *Melancholy*, and *Blood*: but, in my opinion, there are not only several sorts of *Choler*, *Flegm*, *Melancholy*, and *Blood*; but other sorts that are none of these Four.

CHAP. IV. Of BLOOD.

I have heard, that the Opinions of the most Learned Men, are, That all Animal Creatures have Blood, or at least, such Juyces that are in lieu of Blood; which Blood, or Juyces, move circularly: for my part, I am too ignorant to dispute with Learned Men; but yet I am confident, a *Moth* (which is a sort of Worm, or Fly, that eats Cloth) hath no Blood, no, nor any Juyce; for, so soon as it is touched, it dissolves straight to a dry dust, or like ashes. And there are many other Animals, or Insects, that have no appearance of Blood; therefore the life of an Animal doth not consist of Blood: And as for the Circulation of Blood, there are many Animal Creatures that have not proper Vessels, as Veins and Arteries, or any such Gutters, for their Blood, or Juyce, to circulate through. But, say the Blood of Man, or of such like Animal, doth circulate; then it is to be studied,

Whether the several parts of the Blood do intermix with each other, as it flows; or, whether it flows as Water seems to do; where the following parts may be as great strangers to the Leading parts, as in a Crowd of People, where some of those behind, do not know those that are before: but, if the Blood doth not intermix as it flows, then it will be very difficult for a Chyrurgion, or Physician, to find where the ill Blood runs: besides, if the Blood be continually flowing, when a sick Man is to be let blood, before the Vein is opened, the bad Blood may be past that Part, or Vein, and so only the good Blood will be let out; and then the Man may become worse than if he had not been let blood.

CHAP. V. Of the Radical Humours, or Parts.

There are many Parts in a Human Body, that are as the Foundation of a House; and being the Foundation, if any of those Parts be removed or decayed, the House immediately falls to ruine. These Fundamental Parts, are those we name the *Vital Parts*; amongst which are those Parts we name the *Vital* and *Radical Spirits*, which are the Oyl and Flame of a Human Creature, causing the Body to have that we name a *Natural Heat*, and a *Radical Moisture*. But it is to be noted, That these Parts, or Corporeal Motions, are not like gross Oyl, or Flame: for, I believe, there are more differences between those Flames, and ordinary Flames, than between the Light of the Sun, and the Flame of a Tallow Candle; and as much difference between this Oyl, and the greasie Oyl, as between the purest Essence, and Lamp-Oyl. But, these Vital Parts are as necessary to the Human Life, as the solid Vital Parts, viz. the Heart, Liver, Lungs, Brains, and the like.

CHAP. VI. Of Expelling Malignant Disorders in a Human Creature.

Expelling of Poyson, or any Malignity in the Body, is, when that Malignity hath not got, or is not settled into the Vital Parts; so that the Regular Motions

of the Vital Parts, and other Parts of the Body, endeavour to defend themselves from the Forrein Malignancies; which if they do, then the Malignant Motions do dilate to the Exterior Parts, and issue out of those Exterior Passages, at least, through some; as, either by the way of Purging, Vomiting, Sweating, or Transpiration, which is a breathing through the Pores, or other passages. After the same manner is the expelling of Surfeits, or Superfluities of Natural Humours: but, if the Malignity or Surfeit, Superfluity or superfluous Humours, have the better, (as I may say) then those Irregular Motions, by their Disturbances, cause the Regular Motions to be Irregular, and to follow the Mode; which is, to imitate Strangers, or the most Powerful; the most Fantastical, or the most debauch'd: for it is, many times, amongst the Interior Motions of the Body, as with the Exterior Actions of Men.

CHAP. VII. Of Human Digestions and Evacuations.

To treat of the several particular Digestive Actions of a Human Creature, is impossible: for, not only every part of Food hath a several manner of Digestive Action; but, every action in Transpiration, is a sort of Digestion and Evacuation: so that, though every sort of Digestion and Evacuation, may be ghest at; yet, every Particular is not so known, that it can be described. But this is to be noted, That there is no Creature that hath Digestive Motions, but hath Evacuating Motions; which Actions, although they are but Dividing, and Uniting; yet they are such different manners and ways of uniting and dividing, that the most observing Man cannot particularly know them, and so not express them: but, the Uniting actions, if regular, are the Nourishing actions; the Dividing actions, if regular, are the Cleansing actions: but if irregular, the Uniting actions are the Obstructive actions; and the Dividing actions, the Destructive actions.

CHAP. VIII. Of DISEASES in general.

There are many sorts of Human Diseases; yet, all sorts of Diseases are Irregular Corporeal Motions; but, every sort of Motion is of a different Figure: so that, several Diseases are different Irregular Figurative Motions; and according as the Figurative Motions vary, so do the Diseases: but, as there are Human Diseases, so there are Human Defects; which Defects (if they be those which Man names *Natural*) cannot be rectified by any Human Means. Also, there are Human Decays, and Old Age; which, although they cannot be prevented, or avoided; yet, they may, by good Order, and wise Observations, be retarded: but there are not only numerous sorts of Diseases, but every particular it self, and every particular sort, are more or less different; insomuch, that seldom a Disease of one and the same sort, is just alike, but there are some differences; as in Men, who though they be all of one sort of Animal-kind, yet seldom any two Men are just alike: and the same may be said of Diseases both of Body and Mind; as for example, concerning Irregular Minds, as in Mad-Men; Although all Mad-Men are mad, yet not mad alike; though they all have the Disease either of Sensitive or Rational Madness, or are both Sensitive and Rationally mad. Also, this is to be noted, That as several Diseases may be produced from several Causes, so several Diseases from one: Cause, and one Disease from several Causes; which is the cause that a Physician ought to be a long and subtile Observer and Practiser, before he can arrive to that Experience which belongs to a good Physician.

CHAP. IX. Of the Fundamental Diseases.

There are numerous sorts of Diseases, to which Human Creatures are subject; and yet there are but few Fundamental Maladies; which are these as follow; Pain, Sickness, Weakness, Dizziness, Numbness, Deadness, Madness, Fainting and Swoounding; of which one is particular, the rest are general: The particular is Sickness, to which no parts of the Body are subject, but the Stomack: for, though any parts of the Body may have Pain, Numbness, Dizziness, Weakness, or Madness; yet in no part can be that which we name Sickness, but the Stomack. As for Dizziness, the Effects are general, as may be observed in some drunken Men: for, many times, the

Head will be in good temper, when the Leggs (I cannot say, are dizzie, yet) will be so drunk, as neither to go or stand; and many times the Tongue will be so drunk, as not to speak plain, when all the rest of the body is well temper'd; at least so well, as not to be any ways perceived, but by the tripping of their Speech: but, as I said, no Part is subject to be sick, but the Stomack: And though there are numerous sorts of Pains to which every Part is subject, and every several Part hath a several Pain; yet they are still Pain. But some may say, *That there are also several sorts of Sickneses*. I grant it; but yet those several sorts of Sickneses, belong only to the Stomack, and to no other Part of the Body.

The Ninth Part.

CHAP. I. Of SICKNESS.

To go on as orderly as I can, I will treat of the Fundamental Diseases, and first of *Sickness*, by reason it is the most particular Disease: for though, as I have said, no part of a Human Creature is subject to that Disease, (namely, *Sickness*) but the Stomack; yet, there are different sorts of Sicknesses of the Stomack; as for example, Some sorts of Sickness is like the flowing and ebbing of the Sea: for, the Humours of the Stomack agitate in that manner, as, if the flowing motions flow upwards, it occasions Vomiting; if downwards, Purging: if the Humours divide, as, partly to flow upwards, and partly downwards, it occasions both Vomiting and Purging.

But the Question is, *Whether it is the motion of the Humours, that occasions the Stomack to be sick; or the sickness of the Stomack, that occasions the Humours to flow?*

I answer: That 'tis probable, that sometimes the flowing of the Humours causes the Stomack to be sick; and sometimes the sickness of the Stomack occasions the Humours to flow; and sometimes the Stomack will be sick without the flowing of Humours, as when the Stomack is empty; and sometimes the Humours will flow, without any disturbance to the Stomack; and sometimes both the Humours and the Stomack do jointly agree in Irregularities: but, as I said, there are several sorts of sicknesses of the Stomack, or at least, that sickness doth produce several sorts of Effects; as, for example, some sorts of sickness will occasion faint and cold Sweats; which sick Motion is not flowing up or down of the Humours; but it is a cold dilatation, or rarifying, after a breathing manner; also expelling of those rarified parts through the pores: Other sorts of Motions of the Humours, are like Boiling motions, viz. Bubbling motions; which occasion steaming or watry vapours, to ascend to the Head; which vapours are apt to cloud the perception of Sight. Other sorts of sick Motions, are Circular, and

those cause a swimming, or a dizzie motion in the Head, and sometimes a staggering motion in the Leggs. Other sorts of sick Motions are occasioned through

tough and clammy Humours, the motion of which Humours, is a winding or turning in such a manner, that it removes not from its Center; and until such time as that Turning or Winding Motions alter, or the Humour is cast out of the Stomack, the Patient finds little or no ease.

CHAP. II. Of PAIN.

As I said, No Part is subject to be sick, but the Stomack; but every several Part of a Human Creature, is subject to Pain; and not only so, but every particular Part is subject to several sorts of Pain; and every several sort of Pain, hath a several Figurative Motion: but to know the different Figurative Motions, will require a subtile Observation: for, though those painful Parts, know their own Figurative Motions; yet, the whole Creature (suppose *Man*) doth not know them. But it may be observed, Whether they are caused by Irregular Contractions or Attractions, Dilatations or Retentions, Expulsions or Irregular Pressures and Re-actions, or Irregular Transformations, or the like; and by those Observations, one may apply, or endeavour to apply proper Remedies: but all Pain proceeds from Irregular and perturbed Motions.

CHAP. III. Of DIZZINESS.

I Cannot say, *Dizziness* belongs only to the Head of an Animal Creature, because we may observe, by irregular Drinkers, that sometimes the Leggs will seem more drunk than their Heads; and sometimes all the Parts of their Body will seem to be temperate, as being Regular, but only the Tongue seems to be drunk: for, staggering of the Leggs, and a staggering of the Tongue, or the like, in a drunken Distemper, is a sort of Dizziness, although

not such a sort as that which belongs to the Head; so that, when a man is dead-drunk, we may say, that every part of the Body is *Dizzily drunk*. But mistake me not; for I do not mean, that all sorts of dizzinesses proceed from drinking; I only bring Drunkenness for an Example: but, the Effects of dizziness of the Head, and other parts of the Body, proceed from different Causes; for, some proceed from Wind, not Wine; others from Vapour; some from the perception of some Forrein Object; and numbers of the like Examples may be found. But this is to be noted, That all such sorts of Swimming and Dizziness in the Head, are produced from Circular Figurative Motions. Also it is to be noted, That many times the Rational Corporeal Motions are Irregular with the Sensitive, but not always: for, sometimes in these and the like Distempers, the Sensitive will be Irregular, and the Rational Regular; but, for the most part, the Rational is so compliant with the Sensitive, as to be Regular, or Irregular, as the Sensitive is.

CHAP. IV. Of the Brain seeming to turn round in the Head.

When the Human Brain seems to turn round, the cause is, that some Vapours do move in a Circular Figure, which causes the Head to be dizzy; as when a man turns round, not only his Head will be dizzy, but all the Exterior Parts of his Body; insomuch that some, by often turning round, will fall down; but if, before they fall, they turn the contrary way, they will be free from that dizziness: The reason of which is, That, by turning the contrary way, the Body is brought to the same posture it was before; as, when a man hath travell'd some way, and returns the same way back, he returns to the place where first he began his Journey.

CHAP. V. Of WEAKNESS.

There are many sorts of *Weakness*; some *Weakness* proceeds from Age; others, through want of Food; others are occasioned by Oppression; others, by Disorders and Irregularities; and so many other sorts, that it would be too tedious to repeat them, could I know them: But, such sorts of *Weakness*, as Human Creatures are subject to, after some Disease or Sickness, are somewhat like Weariness after a Laborious or over-hard Action; as, when a Man hath run fast, or laboured hard, he fetches his breath short and thick; and as most of the Sensitive Actions are by degrees, so is a Returning to Health after Sickness: but, all Irregularities are Laborious.

CHAP. VI. Of SWOUNDING.

The cause why a Man in *Swound*, is, for a time, as if he were dead; is, an Irregularity amongst some of the Interior Corporeal Motions, which causes an Irregularity of the Exterior Corporeal Motions, and so a general Irregularity; which is the cause that a Man appears as if he were dead.

But some may say, *A Man in a Swound is void of all Motion.*

I answer: That cannot be: for, if the Man was really dead, yet his Parts are moving, though they move not according to the property or nature of a living Man: but, if the Body had not consistent Motions, and the Parts did not hold together, it would be dissolved in a moment; and when the Parts do divide, they must divide by Self-motion: but, in a Man in a *Swound*, some of his Corporeal Motions are only altered from the property and nature of a living Man; I say, some of his Corporeal Motions, not all: Neither do those Motions quite alter from the nature of a living Man, so as the alterations of the Fundamental Motions do: but they are so alter'd, as Language may be alter'd, viz. From *Hebrew* to *Greek*, *Latin*, *French*, *Spanish*, *English*, and many others; and although they are all but Languages, yet they are several Languages or Speeches; so the alteration of the Corporeal Motions of a Man in a *Swound*, is but as the altering of one sort of Language to another; as put the case, *English* were the Natural Language or Speech, then all other Languages were unknown to him that knows no other than his Natural: So a Man in a *Swound* is ignorant of those Motions in the *Swound*: but, when

those Motions return to the Nature of a living Man, he hath the same knowledg he had before. Thus Human Ignorance, and Human Knowledg, may be occasioned by the alterations of the Corporeal Motions.

The truth is, that Swounding and Reviving, is like Forgetfulness and Remembrance, that is, Alteration and Repetition, or Exchange of the same Actions.

CHAP. VII. Of Numb and Dead Palsies, or Gangren's.

As for *Numb* and *Dead Palsies*, they proceed not only from disordered and Irregular Motions, but from such Figurative Motions as are quite different from the nature of the Creature: for, though it be natural for a Man to dye; yet the Figurative Motions of *Death* are quite different from the Figurative Motions of *Life*; so in respect to that which Man names *Life*, that which Man names *Death*, is unnatural: but, as there are several sorts of that Man names *Life*, or *Lives*; so there are several sorts of those Corporeal Motions, Man names *Death*: but, *Dead Palsies* of some Parts of a Man's Body, are not like those of a Man when he is, as we say, *quite dead*; for, those are not only such sorts of Motions that are quite, or absolutely different from the life of the Man, or such like Creature; but such as dissolve the whole Frame, or Figure of the Creature: But, the Motions of a *Dead Palsie*, are not dissolving Motions, although they are different from the natural living Motions of a Man. The same, in some manner, are *Numb Palsies*; only the Motions of *Numb Palsies* are not so absolutely different from the Natural living Motions; but have more Irregularities, than perfect Alterations. As for that sort of Numbness we name *Sleepy Numbness*, it is occasioned through some obstruction that hinders and stops the Exterior Sensitive Perception. As, when the Eyes are shut, or blinded, or the Ears stopt, or the Nostrils; the Sensitive Figurative Motions of those Sensitive Organs, cannot make Perceptions of Forrein Objects: so, when the Pores of the Flesh, which are the perceptive Organs of Forrein Touches, are stopt, either by too heavy burthens or pressings, or tying some Parts so hard, as to close the Exterior Organs, (*viz.* the Pores) they cannot make such Perceptions as belong to

Touch: but, when those hinderances are removed, then the Sensitive Perception of Touch, is, in a short time, as perfect as before.

As for *Gangren's*, although they are somewhat like *Dead Palsies*, yet they are more like those sorts of dead Corporeal Motions, that dissolve the Frame and Form of a Creature: for, *Gangren's* dissolve the Frame and Form of the Diseased Part; and the like do all those Corporeal Motions that cause Rottenness, or Parts to divide and separate after a rotten manner.

CHAP. VIII. Of MADNESS.

There are several sorts of that Distemper named *Madness*; but they all proceed through the Irregularities, either of the Rational, or the Sensitive Parts; and sometimes from the Irregularities both of Sense and Reason: but these Irregularities are not such as are quite different from the Nature or Property of a Human Creature, but are only such Irregularities as make false Perceptions of Forrein Objects, or else make strange Conceptions; or move after the manner of Dreams in waking-actions; which is not according to the Perception of present Objects: As for example, The Sensitive Motions of the Exterior Parts, make several Pictures on the outside of the Organs; when as no such Object is present; and that is the reason Mad-men see strange and unusual Sight, hear strange and unusual Sounds, have strange and unusual Tasts and Touch: but, when the Irregularities are only amongst the Rational Parts, then those that are so diseased, have violent Passions, strange Conceptions, wild Fancies, various Opinions, dangerous Designs, strong Resolutions, broken Memories, imperfect Remembrances, and the like. But, when both the Sensitive and Rational are sympathetically disorderly; then the Mad-men will talk extravagantly, or laugh, sing, sigh, weep, tremble, complain, &c. without cause.

CHAP. IX. The Sensitive and Rational Parts may be distinctly Mad.

The Senses may be irregularly Mad, and not the Reason; and the Reason may be irregularly mad, and not the Sense; and, both Sense and Reason may be both sympathetically mad: And, an evident proof that there is a Rational and Sensitive Madness, is, That those whose Rational Parts are Regular, and only some of the Sensitive Irregular, will speak soberly, and declare to their Friends, how some of their Senses are distemper'd, and how they see strange and unusual Sight, hear unusual Sounds, smell unusual Sents, feel unusual Touches, and desire some Remedy for their Distempers. Also, it may be observed, That sometimes the Rational Parts are madly distemper'd, and not the Sensitive; as when the Sensitive Parts make no false Perceptions, but only the Rational; and then only the Mind is out of order, and is extravagant, and not the Senses: but, when the Senses and Reason are madly Irregular, then the diseased Man is that we name, *Outragiously Mad*.

CHAP. X. The Parts of the Head are not only subject to Madness, but also the other Parts of the Body.

Madness is not only in the Head, but in other Parts of the Body: As for example, Some will feel unusual Touches in their Hands, and several other parts of their Body. We may also observe by the several and strange Postures of Mad-men, that the several Parts of the Body are madly distemper'd. And it is to be noted, That sometimes some Parts of the Body are mad, and not the other; as, sometimes only the Eyes, sometimes only the Ears; and so of the rest of the Organs, and of the rest of the Parts of the Body; one Part only being mad, and the rest in good order. Moreover, it is to be noted, That some are not continually mad, but only mad by fits, or at certain times; and those fits, or certain times of disorders, proceed from a custom or habit of the Rational or Sensitive Motions, to move Irregularly at such times; and a proof that all the Parts are subject to the Distemper of *Madness*, is, That every part of the Body of those sorts of Mad-men that believe their Bodies to be Glass, moves in a careful and wary motion, for fear of breaking in pieces: Neither are the Exterior Parts only subject to the Distemper of Madness, but the Interior Parts; as may be observed, when the

whole Body will tremble through a mad fear, and the Heart will beat disorderly, and the Stomack will many times be sick.

CHAP. XI. The Rational and Sensitive Parts of a Human Creature, are apt to disturb each other.

Although the Rational and Sensitive Corporeal Motions, may, and do sometimes disagree; yet, for the most part, there is such a sympathetic Agreement between the Sensitive and Rational Corporeal Motions of one Society, (viz. of one Creature) as they often disturb each other: As for example, If the Rational Motions are so irregular, as to make imaginary Fears, or fearful Imaginations, these fearful Imaginations cause the Sensitive Corporeal motions, to move according to the Irregularities of the Rational; which is the cause, in such fears, that a man seems to see strange and unusual Objects, to hear strange and unusual Sounds, to smell unusual Sents, to feel unusual Touches, and to be carried to unusual Places; not that there are such Objects, but the Irregular Senses make such Pictures in the Sensitive Organs; and the whole Body may, through the strength of the Irregular motions, move strangely to unusual places: As for example, A Mad-man, in a strong mad fit, will be as strong as Ten men; whereas, when the mad Fit is over, he seems weaker than usually, or regularly, he uses to be; not that the Self-moving Parts of Nature are capable of being weaker, or stronger, than naturally they are: but having liberty to move as they will, they may move stronger, or weaker, swifter or slower, regularly or irregularly, as they please; nor doth Nature commonly use Force. But this is to be noted, That there being a general Agreement amongst the particular Parts, they are more forcible than when those Parts are divided into Factions and Parties: so that in a general Irregular Commotion or Action, all the Sensitive Parts of the Body of a man, agree to move with an extraordinary force, after an unusual manner; provided it be not different from the property and nature of their Compositions; that is, not different from the Property and Nature of a Man. But this is likewise to be noted, That in a general Agreement, man may have other Properties, than when the whole Body is governed by Parts, as it is usual when the Body is Regular, and that

every Part moves in his proper Sphere, as I may say, (for example) the Head, Heart, Lungs, Stomack, Liver, and so the rest, where each Part doth move in several sorts of Actions. The like may also may be said of the Parts of the Leggs and Hands, which are different sorts of Actions; yet all move to the use and benefit of the whole Body: but, if the Corporeal motions in the Hands, and so in the Leggs, be irregular, they will not help the rest of the Parts; and so, in short, the same happens in all the Parts of the Body, whereof some Parts may be Regular, and others Irregular; and sometimes all may be Irregular. But, to conclude this Chapter, the Body may have unusual Force and Properties; as when a man says, He was carried and flung into a Ditch, or some place distant; and that he was pinch't, and did see strange sights, heard strange sounds, smelt strange scents; all which may very well be caused by the Irregular motions, either by a general Irregularity, or by some particular Irregularity; and the truth is, The particular Corporeal motions, know not the power of the general, until they unite by a general Agreement; and sometimes there may be such Commotions in the Body of a Man, as in a Common-wealth, where many times there is a general Uproar and Confusion, and none know the Cause, or who began it. But this is to be noted, That if the Sensitive motions begin the Disorder, then they cause the Rational to be so disordered, as they can neither advise wisely, or direct orderly, or perswade effectually.

CHAP. XII. Of Diseases produced by Conceit.

AS there are numerous sorts of *Diseases*, so there are numerous manners or ways of the production of Diseases; and those Diseases that are produced by *Conceit*, are first occasioned by the Rational Corporeal Figurative Motions: for, though every several Conceit, or Imagination, is a several Rational Corporeal Figurative Motion; yet, every Conceit or Imagination doth not produce a Sensitive Effect: but in those that do produce a Sensitive Effect, it is the Conceit or Imagination of some sorts of Diseases; but in most of those sorts that are dangerous to Life, or causes Deformity: The reason is, That as all the Parts of Nature are Self-knowing, so they are Self-loving: Also, Regular Societies beget an united Love, by Regular

Agreements, which cause a Rational Fear of a disuniting, or dissolving; and that is the reason, that upon the perception of such a Disease, the Rational, through some disorder, figures that Disease; and the Sensitive Corporeal Motions, take a pattern from the Rational, and so the Disease is produced.

The Tenth Part.

CHAP. I. Of FEVERS.

Some are of opinion, That all, or, at least, most Diseases, are accompanied, more or less, with a *Feverous Distemper*: If so, then we may say, A *Fever* is the *Fundamental Disease*: but, whether that Opinion is true, or no, I know not; but I observe, there are many sorts of Fevers, and so there are of all other Diseases or Distempers: for, every alteration, or difference, of one and the same kind of Disease, is a several sort. As for Fevers, I have observed, there are Fevers in the Blood, or Humours, and not in any of the Vital Parts; and those are ordinary Burning-Fevers: and there are other sorts of Fevers that are in the Vital Parts, and all other Parts of the Body, and those are *Malignant Fevers*; and there are some sorts of Fevers which are in the Radical Humours, and those are *Hectick Fevers*; and there are other sorts of Fevers that are in those Parts, which we name the *Spiritous Parts*. Also, all *Consumptions* are accompanied with a Feverish Distemper: but, what the several Figurative Motions are of these several sorts of Fevers, I cannot tell.

CHAP. II. Of the PLAGUE.

There are Two visible sorts of the Disease named the *Plague*: The weaker sort is that which produces Swellings, or inflamed or corrupted Sores, which are accompanied with a Fever. The other sort is that which is named the *Spotted Plague*. The First sort is sometimes Curable; but the Second is Incurable; at least, no Remedy as yet hath been found. The truth is, the *Spotted Plague* is a *Gangrene*, but is somewhat different from other sorts of *Gangren's*; for this begins amongst the Vital Parts, and, by an Infection, spreads to the Extreame Parts; and not only so, but to Forrein Parts; which

makes not only a general Infection amongst all the several Parts of the Body, but the Infection spreads it self to other Bodies. And whereas other sorts of *Gangren's* begin outwardly, and pierce inwardly; the *Plaguy Gangrene* begins

inwardly, and pierces outwardly: so as the difference (as I said) is, That the ordinary sort of *Gangren's* infect the next adjoining Parts of the Body, by moderate degrees; whereas the *Plaguy Gangrene* infects not only the adjoining Parts of the same Body, and that suddenly, but infects Forreign Bodies. Also, the ordinary *Gangren's* may be stopped from their Infection, by taking off the Parts infected, or diseased. But the *Plaguy Gangrene* can no ways be stopped, because the Vital Parts cannot be separated from the rest of the Parts, without a total ruine: besides, it pierces and spreads more suddenly, than Remedies can be applyed. But, whether there are Applications of Preventions, I know not; for, those Studies belong more to the *Physicians*, than to a *Natural Philosopher*. As for the Diseases we name the *Purples*, and the *Spotted Fever*, they are of the same Kind, or Kindred, although not of the same sort, as *Measles*, and the *Small-Pox*. But this is to be noted, That Infection is an act of Imitation: for, one Part cannot give another Part a Disease, but only that some imitate the same sorts of Irregular Actions of other Parts; of which some are near adjoining Imitators, and some occasion a general Mode.

CHAP. III. Of the Small-Pox, and Measles.

The *Small-Pox* is somewhat like the *Sore-Plague*, not only by being Infectious, as both sorts of Plagues are; but, by being of a corrupt Nature, as the *Sore-Plague* is; only the *Small-Pox* is innumerable, or very many small Sores; whereas the *Sore-Plague* is but one or two great Sores. Also, the *Small-Pox* and *Sore-Plague*, are alike in this, That if they rise and break, or if they fall not flat, but remain until they be dry and scabbed, the Patient lives: but, if they fall flat, and neither break, nor are scabbed, the Patient is in danger to dye. Also, it is to be noted, That this Disease is sometimes accompanied with a Feverish Distemper; I say, Sometimes, not Always; and that is the cause that many dye, either with too hot, or too cooling

Applications: for, in a Feverish Distemper, hot Cordials are Poyson; and when there is no Fever, Cooling Remedies are *Opium*: The like for letting Blood; for if the Disease be accompanied with a Fever, and the Fever be not abated by letting Blood, 'tis probable the Fever, joyned with the Pox, will destroy the Patient: and if no Fever, and yet loose Blood, the Pox hath not sufficient Moisture to dilate, nor a sufficient natural Vapour to breathe, or respirate; so as the Life of the Patient is choaked or

stifled with the contracted Corruptions. As for Measles, though they are of the same kind, yet not of the same sort; for they are rather Small Risings, than Corrupted Sores, and so are less dangerous.

CHAP. IV. Of the Intermission of Fevers or Agues.

Agues have several sorts of Distempers, and those quite opposite to each other, as Cold and Shaking, Hot and Burning, besides Sweating: Also, there are several times of Intermissions; as some are Every-day Agues, some Third-day Agues, and some *Quartan* Agues; and some Patient may be thus distempered, many times, in the compass of Four and twenty hours: but those are rather of the Nature of Intermitting Fevers, than of perfect Agues. Also, in Agues, there is many times a difference of the Hot and Cold fits: for sometimes the Cold Fits will be long, and the Hot short; other times, the Hot Fits will be long, and the Cold Fits short; other times, much of an equal degree: but, most Intermitting Fevers and Agues, proceed either from ill-digestive Motions, or from a superfluity of Cold and Hot Motions, or an Irregularity of the Cold, Hot, Dry; or Moist Motions, where each sort strives and struggles with each other. But, to make a comparison, Agues are somewhat like several sorts of Weather, as Freezing and Thawing,

Cloudy or Rainy, or Fair and Sun-shining days: or like the Four Seasons of the Year, where the Cold Fits are like *Winter*, cold and windy; the Hot Fits like *Summer*, hot and dry; the Sweating Fits like *Autumn*, warm and moist; and, when the Fit is past, like the *Spring*. But, to conclude, the chief Cause of Agues, is, Irregular Digestions, that make half-concocted Humours; and according as these half-concocted Humours digest, the Patient hath his

Aguish Distempers, where some are every day, others every second day, some every third day, and some *Quartans*: but, by reason those half-concocted Humours, are of several sorts of Humors, some Cold, some Hot, some Cold and Dry, some Hot and Dry, or Hot and Moist; and those different sorts, raw, or but half-concocted Humours; they occasion such disorder, not only by an unnatural manner of Digestion, as not to be either timely, or regular, by degrees; but, those several sorts of Raw Humours, strive and struggle with each other for Power or Supremacy: but, according as those different Raw Humours concoct, the Fits are longer or shorter: also, according to the quantity of those Raw Humours, and according as those Humours are a gathering, or breeding, so are the times of those Fits and Intermissions. But here is to be noted, That some Agues may be occasioned from some Particular Irregular Digestions; others from a General Irregular Digestion, some from some obscure Parts, others from ordinary Humours.

CHAP. V. Of CONSUMPTIONS.

There are many sorts of *Consumptions*; as, some are Consumptions of the Vital Parts, as the Liver, Lungs, Kidneys, or the like Parts: Others, a Consumption of the Radical Parts: Others a Consumption of the Spiritous Parts: Other Consumptions are only of the Flesh; which, in my opinion, is the only Curable Consumption. But, all Consumptions, are not only an Alteration, but a Wasting and Dis-uniting of the Fundamental Parts; only those Consuming Parts do, as it were, steal away by degrees; and so, by degrees, the Society of a Human Creature is dissolved.

CHAP. VI. Of DROPSIES.

Dropsies proceed from several Causes; as, some from a decay of some of the Vital Parts; others through a superfluity of indigested Humours; some

from a supernatural Driness of some Parts; others through a superfluity of Nourishing Motions; some, through some Obstructions; others, through an excess of Moist Dyet: but, all Dropsies proceed not only from Irregular Motions, but from such a particular Irregularity, as all the Motions endeavour to be of one Mode, (as I may say) that is, To move after the manner of those sorts of Motions which are the innate Nature of Water, and are some sorts of Circular Dilatations: but, by these actions, the Human Society endeavours to make a Deluge, and to turn from the Nature of Blood and Flesh, to the Nature of Water.

CHAP. VII. Of SWEATING.

All *Sweating-Diseases* are somewhat of the nature of Dropsies; but they are (at least, seem to be) more Exterior, than Interior Dropsies: but, though there be Sweating-Diseases which are Irregular; yet, Regular Sweating is as proper as Regular Breathing; and so healthful, that Sweating extraordinary, in some Diseases, occasions a Cure: for, Sweating is a sort of Purging; so that the evacuation of Sweat, through the Pores, is as necessary as other sorts of evacuation, as Breathing, Urine, Siege, Spitting, Purging through the Nose, and the like. But, Excess of Sweating, is like other sorts of Fluxes, of which, some will scowr to death; others vomit to death; and others the like Fluxes will occasion

death; the like is of Sweating: so that the *Sweating-Sickness* is but like a *Fluxive-Sickness*. But, as I said, Regular Sweating is as necessary as other ordinary Evacuations: and as some are apt to be restraining, others laxative; and sometimes one and the same Man will be laxative, other times, costive; so are Men concerning Sweating: and as some Men take Medicines to purge by Stool, or Vomits, or Urine; so they take Medicines to purge by Sweating. And, as Man hath several sorts of Excremental Humours, so, several sorts of Sweats; as, Clammy Sweats, Cold Sweats, Hot Sweats, and Faint Sweats: and, as all Excess of other sorts of Purgings, causes a Man to be weak and faint; so doth Sweating.

CHAP. VIII. Of COUGHS.

There are many several sorts of *Coughs*, proceeding from several Causes; as, some Coughs proceed from a Superfluity of Moisture; others from an Unnatural Heat; others from a Corruption of Humors; others from a Decay of the Vital Parts; others from sudden Colds upon Hot Distempers: Some are caused by an Interior Wind; some Coughs proceed from Salt Humors, Bitter, Sharp, and Sweet: some Coughs proceed from Flegm, which Flegm ariseth like a Scum in a Pot, when Meat is boiling on a Fire: for

when the Stomack is distemperedly hot, the Humors in the Stomack boyl as Liquid Substances on the Fire; those boiling Motions bearing up the gross Humors beyond the Mouth of the Stomack, and, causing a Dispute between the Breath and Humors, produce the Effect of Straining, or Reaching upwards towards the Mouth, much like the Nature and Motions of Vomiting: but, by reason those Motions are not so strong in Coughing, as in Vomiting, the Coughing Motions bring up only pieces or parts of superfluous Flegm, or gross Spittle. The like for corrupt Humors. Other Coughs proceed from Unnatural or Distempered Heats; which Heats cause Unnecessary Vapours, and those Vapours ascending up from the Bowels, or Stomack, to the Head, and finding a Depression, are converted or changed into a Watry Substance; which Watry Substance falls down, like mizling or small Rain, or in bigger drops, through the passage of the Throat and Wind-pipe: which being opprest, and the Breath hindered, causes a Strife; which Striving, is a Straining; like as when Crumbs of Bread, or Drops of Drink, go not rightly through the Throat, but trouble and obstruct the Wind-pipe, or when any such Matter sticks in the passage of the Throat: for, when any Part of the Body is obstructed, it endeavours to release it self from those Obstructions: Also, when the Vapour that arises, arises in very Thin and Rarified Vapour,

that Rarified Vapour thickens or condenses not so suddenly, being farther from the degree of Water; but when condensed into Water, it falls down by drops; which drops trickling down the Throat, (like as Tears from the Eyes trickle down the Cheeks of the Face) the Cough is not so violent, but more frequent: but if the Rheum be salt or sharp, that trickles down the Throat, it

causes a gentle or soft smart, which is much like the touch of Tickling or Itching, which provokes a faint or weak Strain or Cough. Also, Wind will provoke to Strain or Cough: The Motion of Wind is like as if Hair should tickle the Nose. Or, Wind will cause a tickling in the Nose, which causes the Effect of Sneezing: for, Sneezing is nothing but a Cough through the Nose; I may say, It is a Nose-Cough. And Hickops are but Stomach-Coughs, Wind causing the Stomack to strain. Also, the Guts have Coughs, which are caused by the Wind, which makes a strife in the Guts and Bowels. Other Coughs are produced from Decayed Parts: for, when any Part is corrupted, it becomes less Solid than naturally it should be: As for example, The Flesh of the Body, when corrupted, becomes from Dense Flesh, to a Slimy Substance; thence, into a Watry Substance, which falls into Parts, or changes from Flesh, into a Mixt Corrupted Matter, which falls into Parts. The several Mixtures, or Distempered Substances,

and Irregular Motions, causes Division of the composed Parts; but in the time of dissolving, and divisions of any Part, there is a strife which causes Pain: and if the strife be in the Lungs, it causes Coughs, by obstructing the Breath: but, some Coughs proceed from Vapours and Winds, arising from the decayed Interior Parts, sending up Vapours from the Dissolving Substance, which causeth Coughs; and some Coughs cause Decays of the Prime Interior Parts: for, when there falls from the Head a constant Distillation, this Distillation is like dropping Water, which will penetrate or divide Stone; and more easily will dropping or drilling Water do it, as Rheum, will corrupt Spongy Matter as Flesh is: but, according as the Rheum is Fresh, Salt, or Sharp, the Parts are a longer or shorter time decaying: for, Salt and Sharp is Corroding; and, by the Corroding Motions, Ulcerates those Parts the Salt Rheums fall on, which destroys them soon. As for *Chin-Cough*, 'tis a Wind or Vapour arising from the Lungs, through the Wind-pipe; and as long as the Wind or Vapour ascends, the Patient cannot draw in Reviving Air or Breath, but Coughs violently and incessantly, until it faint away, or have no Strength left; and with straining, will be as if it were choaked or strangled, and become black in the face, and, after the Cough is past, recover again; but some dye of these sorts of Coughs.

CHAP. IX. Of GANGREN'S.

Gangren's are of the Nature of the *Plague*; and they are of Two sorts, as the *Plague* is; the one more sudden and deadly than the other: The only difference of their Infecting Qualities, is, That *Gangren's* spread by infecting still the next, or Neighbouring Parts; whereas *Plagues* infect Forrein, as much as Home-Parts. Also, the deadly sort of *Gangren's*, infect (as I may say) from the Circumference towards the Center: when as the deadly sorts of *Plague*, infect from the Center, towards the Circumference. But, that sort of *Gangrene* that is the weaker sort, infects only the next adjoining Parts, by degrees, and after a spreading manner, rather than after a piercing manner.

But some may object, That *Plagues* and *Gangren's* are produced from different Causes; as for example, Extream Cold will cause *Gangren's*; and Extream Heat causes *Plagues*.

I answer, That Two opposite Causes may produce like Effects, for which may be brought numerous Examples.

CHAP. X. Of Cancers and Fistula's.

Cancers and *Fistula's* are somewhat alike, in that they are both produced from Salt, or sharp corroding Motions: but in this they differ, that *Cancers* keep their Center, and spread in streams; whereas *Fistula's* will run from place to place: for if it be stopt in one place, it is apt to remove and break out in another. Yet *Cancers* are somewhat like *Gangren's*, in infecting adjoining Parts; so that unless a *Cancer* be in such a place as can be divided from the Sound Parts, it destroys the Human Life, by eating (as I may say) the Sound Parts of the Body, as all Corroding, and Sharp or Salt Diseases do.

CHAP. XI. Of the GOVT.

As for the Disease named the *Gout*, I never heard but of Two sorts; the *Fixt*, and the *Running Gout*: but, mistake me not, I mean *Fixt* for *Place*, not *Time*. The *Fixt* proceeds from Hot, Sharp, or Salt Motions: The *Running Gout* from Cold, Sharp Motions; but, both sorts are Intermitting Diseases, and very painful; and I have heard those that have had the *Fixt Gout*, say, That the pain of the *Fixt Gout*, is somewhat like the Tooth-ach: but, all Gouts are occasioned by Irregular Pressures and Re-actions. As for that sort that is named the *Windy Gout*, it is rather a *Sciatica*, than a *Gout*.

CHAP. XII. Of the STONE.

Of the Disease of the *Stone* in Human Creatures, there are many sorts: for, though the *Stone* of the *Bladder*, of the *Kidneys*, and in the *Gaul*, be all of one kind of Disease called the *Stone*, yet they are of different sorts: but, whether the Disease of the *Stone* be produced of Hot or Cold Motions, I cannot judg: but 'tis probable, some are produced of Hot Motions, others of Cold; and perchance, others of such sorts of Motions as are neither perfectly Hot, nor Cold: for, the *Stone* is produced, as all other Creatures, by such or such sorts of Figurative Motions. Here is to be noted, That some of the Humours of the Body may alter their Motion, and turn from being Flegm, Choler, or the like, to be *Stone*; and so from being a Rare, Moist, or Loose Body, to be a Dry, Densed, Hard, or *Fixt* Body. But certainly, the *Stone* of the *Bladder*, *Kidneys* and *Gaul*, are of several sorts, as being produced by several sorts of Figurative Motions; as also, according to the Properties and Forms of those several Parts of the Body they are produced in: for, as several sorts of Soyls, or Parts of the Earth, produce several sorts of Minerals; so several Parts of the Body, several sorts of the Disease of the *Stone*: And, as there are several sorts of Stones in the several Parts of the Earth; so, no doubt, there may not only be several sorts of *Stone* in several Parts, but several sorts in one and the same Part; at least, in the like Parts of several Men.

CHAP. XII. Of Apoplexies, and Lethargies.

Apoplexies, Lethargies, and the like Diseases, are produced by some decay of the Vital Spirits, or by Obstructions, as being obstructed by some Superfluities, or through the Irregularities of some sorts of Motions, which occasion some Passages to close, that should be open. But mistake me not, I do not mean empty Passages; for there is no such thing (in my opinion) in Nature: but, I mean an open passage for a frequent Course and Recourse of Parts. But an *Apoplexy* is somewhat of the Nature of a *Dead-Palsie*; and a *Lethargy*, of a Numb-Palsie; but I have heard, that the Opinion of Learned Men is, That some sorts of Vaporous Pains are the Fore-runners of *Apoplexies* and *Palsies*: but, in my opinion, though a Man may have two Diseases at once; yet surely, where Vapour can pass, there cannot be an absolute Stoppage.

CHAP. XIII. Of EPILEPSIES.

Epilepsies, or that we name the *Falling-Sickness*, is of the nature of Swounding or Fainting Fits: but there are two visible sorts; the one is, that only the Head is affected, and not the other Parts of the Body; and for proof, Those that are thus distempered only in the Head, all the other Parts will struggle and strive to help or assist the affected or afflicted Parts, and those Parts of the Head that are not Irregular, as may be observed by their Motions; but, by the means of some other Parts, there will also be striving and struggling, as may be observed by foaming through the Mouth. The other sort is like ordinary Swounding-Fits, where all the Parts of the Body seem, for a time, to be dead. But this is to be observed, That those that are thus diseased, have certain times of Intermissions, as if the Corporeal Motions did keep a Decorum in being Irregular. But some have had *Epilepsies* from their Birth; which proves, That their Productive Motions was Irregular.

CHAP. XIV. Of Convulsions, and Cramps.

Convulsions and *Cramps* are somewhat alike; and both, in my Opinion, proceed from Cold Contractions: but, *Cramps* are caused by the Contractions of the *Capillary* Veins, or small *Fibers*, rather than of the Nerves and Sinews: for, those Contractions, if violent, are *Convulsions*: so that Cramps are Contractions of the small Fibers; and Convulsions are Contractions of the Nerves and Sinews. But the reason (I believe) that these Diseases proceed from Cold Contractions, is, That Hot Remedies produce, for the most part, perfect Cures; but, they must be such sorts of Hot Remedies, that are of a dilating or extenuating nature; and not such whose Properties are Hot and Dry, or Contracting: also, the Applications must be according to the strength of the Disease.

CHAP. XV. Of CHOLICKS.

Cholicks are like *Cramps* or *Convulsions*; or *Convulsions* and *Cramps*, like *Cholicks*: for, as *Convulsions* are Contractions of the Nerves and Sinews; and *Cramps*, Contractions of the small *Fibers*: so *Cholicks*

are a Contracting of the Gutts: and, for proof, So soon as the Contracting Motions alter, and are turn'd to Dilating or Expelling Actions, the Patient is at ease. But, there are several Causes that produce the *Cholick*: for, some *Cholicks* are produced by Hot and Sharp Motions, as *Bilious Cholicks*; others from Cold and Sharp Motions, as *Splenetick Cholicks*; others from Crude and Raw Humours; some from Hot Winds; some from Cold Winds. The same some sorts of *Convulsions* and *Cramps* may be: but, though these several *Cholicks* may proceed from several Causes; yet, they all agree in this, To be Contractions: for, as I said, when those Corporeal Motions alter their Actions to Dilatation or Expulsion, the Patient is at ease. But, those *Cholicks* that proceed from Hot and Sharp Motions, are the most painful and dangerous, by reason they are, for the most part, more strong and stubborn. As for *Cholicks* in the Stomack, they are caused by the same sorts of Motions that cause some sorts of Contractions: but, those sorts of

Cholick Contractions, are after the manner of wreathing, or wringing Contractions. The same in Convulsive-Contractions.

CHAP. XVI. Of Shaking Palsies.

Shaking Palsies proceed from a Slackness of the Nerves, or Sinew strings, as may be observed by those that hold or lay any heavy weight upon the Arms, Hands or Leggs: for, when the Burdens are removed, those Limbs will be apt to tremble and shake so much, for a short time, (until they have recovered their former strength) that the Leggs cannot go, or stand steadily; nor the Arms, or Hands, do any thing without shaking. The reason of these sorts of Slackness, is, That heavy Burdens occasion the Nerves and Sinews to extend beyond their Order; and being stretched, they become more slack, and loose, by how much they were stretched, or extended; until such time as they contract again into their proper Posture: And the reason that Old Age is subject to *Shaking-Palsies*, is, That the Frame of their whole Body is looser and slacker, than when it was young: As in a decayed House, every Material is looser than when it was first built; but yet, sometimes an old shaking House will continue a great while, with some Repairs: so old shaking Men, with Care, and good Dyet, will continue a great time. But this is to be noted, That trembling is a kind of a *Shaking-Palsie*, although of another sort; and so is Weakness

after Sickness: but, these sorts are occasioned, as when a House shakes in a great Wind, or Storm; and not through any Fundamental Decay.

CHAP. XVII. Of the Muther, Spleen, and Scurvy.

As for those Diseases that are named the *Fits of the Muther*, the *Spleen*, the *Scurvy*, and the like; although they are the most general Diseases, especially amongst the Females; yet, each particular sort is so various, and hath such different Effects, that, I observe, they puzzle the most Learned Men to find

out their juggling, intricate, and uncertain Actions. But this is to be observed, That the Richest sorts of Persons are most apt to these sorts of Diseases; which proves, That Idleness and Luxury is the occasion.

CHAP. XVIII. Of Food, or Digestions.

As I have said, *Digestions* are so numerous, and so obscure, that the most Learned Men know not how Food is converted and distributed to all the Parts of the Body: Which Obscurity occasions many Arguments, and much Dispute amongst the Learned; but, in my opinion, it is not the Parts of the Human Body, that do digest the Food, although they may be an occasion (through their own Regularities, or Irregularities) to cause good or bad digestions: but, the Parts of the Food, do digest themselves; that is, alter their actions to the Property and Nature of a Human Body: so that Digestive Parts are only Additional Parts; and, if those Nourishing Motions be Regular, they distribute their several Parts, and joyn their several Parts, to those several Parts of the Body that require Addition. Also, the Digestive Motions are according to the Nature or Property of each several Part of the Human Body, As for example, Those Digestive Parts alter into Blood, Flesh, Fat, Marrow, Brains, Humors, and so into any other Figurative Parts of the Sensitive Body. The same may be said of the Rational Parts of the Mind: but, if those Digestive Parts be Irregular, they will cause a Disorder in a well-ordered Body: and, if the Parts of the Body be Irregular, they will occasion a Disorder amongst the Digestive Parts: but, according to the Regularities and Irregularities of the Digestive Parts, is the Body more or less nourished. But this is to be noted, That according to the Superfluity or Scarcity of those Digestive Parts, the Body is oppress'd, or starved.

CHAP. XIX. Of SURFEITS.

Surfeits are occasioned after different manners: for, though many *Surfeits* proceed from those Parts that are received into the Body; yet, some are occasioned through often repetitions of one and the same actions: As for example, The Eyes may surfeit with too often viewing one Object; the Ears, with often hearing one Sound; the Nose, with smelling one Sent; the Tongue, with one Tast. The same is to be said of the Rational Actions; which *Surfeits*, occasion an aversion to such or such Particulars: but, for those *Surfeits* that proceed from the Parts that are received into the Body, they are either through the *quantity* that oppresses the Nature of the Body; or, through the *quality* of those Parts, being not agreeable to the Nature of the Body; or, through their Irregularities, that occasion the like Irregularities in the Body: and sometimes, the fault is through the Irregularities of the Body, that hinder those received Parts, or obstruct their Regular Digestions; and sometimes, the fault is both of the Parts of the Body, and those of the Food: but, the *Surfeits* of those Parts that receive not Food, are caused through the often repetition of one and the same Action.

CHAP. XX. Of Natural Evacuations, or Purgings.

There are many sorts, and several ways or means of Purging actions; whereof some we name *Natural*, which purge the Excremental Parts; and such Natural Purgings, are only of such Parts as are no ways useful to the Body; or of those that are not willing to convert themselves into the Nature and Property of the Substantial Parts. There must of necessity be Purging actions, as well as Digestive actions; because, no Creature can subsist singly of it self, but all Creatures subsist each by other; so that, there must be Dividing actions, as well as Uniting actions; only, several sorts of Creatures, have several sorts of Nourishments and Evacuations. But this is to be noted, in the Human Nourishments and Evacuations, that, through their Irregularities, some Men may nourish too much, and others purge too much; and some may nourish too little, and some may purge too little. The Irregularities concerning Nourishments, are amongst the adjoining Parts; the Errors concerning Purging, are amongst the Dividing Parts.

CHAP. XXI. Of PURGING DRUGGS.

There are many sorts of *Druggs*, whereof some are beneficial, by assisting those particular Parts of the Body that are oppressed and offended, either by Superfluous Humours, or Malignant Humours: but, there are some sorts of Druggs that are as malicious to the Human Life, as the Assistant Druggs are friendly. Several sorts of Druggs, have several sorts of Actions, which causes several Effects; as, some Druggs work by Siege; others, by Urine; some, by Vomit; others, by Spitting; others, by Sweating; some cause sleep; some are hot, others are cold; some dry, others moist. But this is to be noted, That 'tis not the Motions of the Druggs, but the Motion of the Humours, which the Druggs occasion to flow; and not only to flow, but to flow after such or such a manner and way. The Actions of Druggs, are like the Actions of Hounds, or Hawks, that flye at a particular Bird, or run after a particular beast of their own kind, although of a different sort: The only difference is, That Druggs are not only of a different sort, but of a different Kind from Animal Kind; at least, from Human Sort.

CHAP. XXII. Of the Various Humours of Druggs.

The reason, one and the same Quantity or Dose of one and the same sort of Purging-Druggs or Medicine, will often work differently in several Human Bodies; as also, differently in one and the same Body, at several times of taking the same sorts of Medicines; is, That several Parts of one and the same sort, may be differently humoured: as, some to be duller and slower than others; and some to be more active than others. Also, some Parts may be ill-natured, and cause Factions amongst the Parts of the Body; whereas others will endeavour to rectifie Disorders, or Factions. And sometimes both the Druggs, and the Body, falls out; and then there is a dangerous strife; the Body striving to expel the Physick, and the Physick endeavouring to stay in the Body, to do the Body some mischief. Also, some Parts of one and the same sort, may be so Irregular, as to hunt not only the superfluous

Humours, or the Malignant Humors, but all sorts of flowing Parts; which may cause so great and general Disorder, as may endanger Human Life.

CHAP. XXIII. Of CORDIALS.

There are many sorts of *Cordials*: for, I take every Beneficial Remedy to be a Cordial: but, many of the Vulgar believe, That there is no Cordial but Brandy, or such like Strong-waters; at least, they believe all such Remedies that are virtually Hot, to be Cordials: but, when they take too much of such Cordials, either in Sickness, or Health, they will, in some time, find them as bad as Poyson. But, all such Applications as are named *Cordials*, are not hot: for, some are cool, at least, of a temperate degree. And as there are Regular and Irregular Corporeal Motions; so there are Sympathetical, and Antipathetical Motions; and yet both sorts may be Regular. Also, there is a Neutral sort, that has neither Sympathy nor Antipathy, but is Indifferent. But in Disputes between Two different Parties, a Third may come in to the assistance of one Side, more out of hate to the Opposite, than love to the Assisted. The same may Cordials, or such like Applications, do, when the Corporeal Motions of Human Life are in disorder, and at variance: for, oftentimes there is as great a Mutiny and Disorder amongst the Corporeal Motions, both in the Mind and Body of a Man, as in a Publick State in time of Rebellion: but, all

Assistant Cordials, endeavour to assist the Regular Parts of the Body, and to perswade the Irregular Parts. As for Poysons, they are like Forrein Warr, that endeavours to destroy a Peaceable Government.

CHAP. XXIV. Of the different Actions of the several Sensitive Parts of a Human Creature.

Some Parts of a Human Creature will be Regular, and some Irregular: as, some of the Sensitive Parts will be Regular, and some Irregular; that is,

some Parts will be Painful, or Sick, others well: some Parts will make false Perceptions; others, true Perceptions: some Parts be Temperate; others, Intemperate: some Parts be Madd, other Parts Sober: some Parts be Wise; others, Foolish: and the same is to be said of the Rational Motions. But, in a Regular Society, every Part and Particle of the Body, is Regularly agreeable, and Sympathetical.

CHAP. XXV. Of the Antipathy of some Human Creatures, to some Forrein Objects.

As I have often said, There is often both Sympathy and Antipathy between the Parts of some particular Human and Forrein Objects; in so much, that some will occasion such a general Disturbance, as will cause a general Alteration, viz. cause a Man to swound, or at least, to be very faint, or sick: as for example, Some will Swound at some sorts of Sounds, some sorts of Scents, some sorts of Tast, some sorts of Touches, and some sorts of Sights. Again, on the other side, some Human Creatures will so sympathize with some sorts of Forrein Objects, as some will Long for that, another will Swound to have.

CHAP. XXVI. Of the Effects of Forrein Objects, on the Human Mind.

As there is often Antipathy of the Parts of a Human Creature, to Forrein Objects; so there are often Sympathetical Effects produced from Forrein Objects, with the Parts of a Human Creature. As for example, A timely, kind, and discreet Discourse from a Friend, will compose or quiet his troubled Mind: Likewise, an untimely, unkind, hasty, malicious, false, or sudden Discourse, will often disorder a well-temper'd, or Regular Mind, the Mind imitating the smooth or harsh strains of the Object: and the same Effects hath Musick, on the Minds of many Human Creatures.

CHAP. XXVII. Of CONTEMPLATION.

Human *Contemplation*, is a Conversation amongst some of the Rational Parts of the Human Mind; which Parts, not regarding present Objects, move either in devout Notions, or vain Fancies, Remembrances, Inventions, Contrivancies, Designs, or the like. But the question is, Whether the Sensitive Parts of a Human Society, do, at any time, Contemplate? I answer, That some of the Sensitive Parts are so sociable, that they are, for the most part, agreeable to the Rational: for, in deep Contemplations, some of the Sensitive Parts do not take notice of Forrein Objects, but of the Rational Actions. Also, if the Contemplations be in devout Notions, the Sensitive Parts express Devotion by their Actions, as I have formerly mentioned. Also, when the Rational Parts move in Actions of Desire, straight the Sensitive move in Sympathetical Appetites: Wherefore, if the Society be Regular, the Sensitive and Rational Parts are agreeable and sociable.

CHAP. XXVIII. Of Injecting of the Blood of one Animal, into the Veins of another Animal.

To put Blood of one Animal, into another Animal; as for example, Some Ounces of Blood taken, by some Art, out of a Dogg's Veins, and, by some Art, put into a Man's Veins, may very easily be done by *Injection*; and certainly, may as readily convert it self to the Nature of Human Blood, as Roots, Herbs, Fruit, and the like Food; and probably, will more aptly be transformed into Human Flesh, than Hogg's Blood, mixt with many Ingredients, and then put into Gutts, and boyled, (an ordinary Food amongst Country People;) but Blood being a loose Humourish Part, may encrease or diminish, as the other Humors, viz. *Flegm*, *Choler*, and *Melancholy*, are apt to do. But this is to be observed, That by reason Blood is the most flowing Humor, and of much more, or greater quantity than all the rest of the Humours, it is apt (if Regular) to cause, not only more frequent, but a more general Disturbance.

The Eleventh Part.

CHAP. I. Of the different Knowledges, in different Kinds and Sorts of Creatures.

If there be not Infinite Kinds, yet, it is probable, there are Infinite several Sorts; at least, Infinite particular Creatures, in every particular Kind and Sort; and the Corporeal Motions moving after a different manner, is the cause there are different Knowledges, in different Creatures; yet, none can be said to be *least knowing*, or *most knowing*: for, there is (in my opinion) no such thing as *least* and *most*, in Nature: for, several kinds and sorts of Knowledges, make not Knowledg to be more, or less; but only, they are different Knowledges proper to their kind, (as, Animal-kind, Vegetable-kind, Mineral-kind,

Elemental-kind) and are also different Knowledges in several sorts: As for example, Man may have a different Knowledg from Beasts, Birds, Fish, Flies, Worms, or the like; and yet be no wiser than those sorts of Animal-kinds. The same happens between the several Knowledges of Vegetables, Minerals, and Elements: but, because one Creature doth not know what another Creature knows, thence arises the Opinion of *Insensibility*, and *Irrationability*, that some Creatures have of others. But there is to be noted, That Nature is so Regular, or wise, in her Actions, that the *Species* and Knowledg of every particular Kind, is kept in an Even, or Equal Balance: For example, The Death or Birth of Animals, doth neither add or diminish from, or to the Knowledg of the Kind, or rather the Sort. Also, an Animal can have no Knowledg, but such as is proper to the *species* of his Figure: but, if there be a Creature of a mixt *Species*, or Figure, then their Knowledg is according to their mixt Form: for, the Corporeal Motions of every Creature, move according to the Form, Frame, or *Species* of their Society: but, there is not only different Knowledges, in different Kinds and Sorts of Creatures; but, there are different Knowledges in the different Parts of one

and the same; as, the different Senses of Seeing, Hearing, Smelling, Tasting, and Touching, have not only different Knowledges in different Sensitive Organs, but

in one Sense, they have several Perceptive Knowledges: and though the different Sensitive Organs of a Human Creature, are ignorant of each other; yet, each Sense is as knowing as another. The same (no question) is amongst all the Creatures in Nature.

CHAP. II. Of the Variety of Self-actions in particular Creatures.

There are numerous Varieties of Figurative Motions in some Creatures; and in others, very few, in comparison: but, the occasion of that, is the manner of the Frame and Form of a Creature: for, some Creatures that are but small, have much more variety of Figurative Motions, than others that are very bigg and large Creatures: so that, it is not only the Quantity of Matter, or Number of Parts, but the several Changes of Motion, by the Variety of their Active Parts, that is the cause of it: for, Nature is not only an Infinite Body, but, being Self-moving, causes Infinite Variety, by the altered Actions of her Parts; every altered Action, causing both an altered Self-knowledg, and an altered Perceptive Knowledg.

CHAP. III. Of the Variety of Corporeal Motion, of one and the same sort or kind of Motion.

There is Infinite Variety of Motion of the same sorts and kinds of Motions; as for example, Of Dilatations, or Extensions, Expulsions, Attractions, Contractions, Retentions, Digestions, Respirations: There is also Varieties of Densities, Rarities, Gravities, Levities, Measures, Sizes, Agilness, Slowness, Strength, Weakness, Times, Seasons, Growths, Decays, Lives, Deaths, Conceptions, Perceptions, Passions, Appetites, Sympathies, Antipathies, and Millions the like kinds, or sorts.

CHAP. IV. Of the Variety of particular Creatures.

Nature is so delighted with *Variety*, that seldom two Creatures (although of the same sort, nay, from the same Producers) are just alike; and yet Human Perception cannot perceive above four kinds of Creatures, viz. *Animals*, *Vegetables*, *Minerals*, and *Elements*: but, the several sorts seem to be very numerous; and the Varieties of the several Particulars, Infinite: but, Nature is necessitated to divide her Creatures into Kinds and Sorts, to keep Order and Method: for, there may be numerous Varieties of sorts; as for example, Many several Worlds, and infinite Varieties of Particulars in those *Worlds*: for, Worlds may differ from each other, as much as several sorts of Animals, Vegetables, Minerals, or Elements; and yet be all of that sort we name Worlds: but, as for the Infinite Varieties of Nature, we may say, That every Part of Nature is Infinite, in some sort; because every Part of Nature is a perpetual Motion, and makes Infinite Varieties, by change or alteration of Action: but, there is so much Variety of the several Shapes, Figures, Forms, and Sizes, as, Bigger, and Less; as also, several sorts of Heats, Colds, Droughts, Moistures, Fires, Airs, Waters, Earths, Animals, Vegetables, and Minerals, as are not to be expressed.

CHAP. V. Of Dividing, and Rejoyning, or Altering Exterior Figurative Motions.

The Interior and Exterior Figurative Motions of some sorts of Creatures, are so united by their Sympathetical Actions, as they cannot be separated without a Total Dissolution; and some cannot be altered without a Dissolution; and other Figurative Motions may separate, and unite again; and others, if separate, cannot unite again, as they were before: As for example, The Exterior Parts of a Human Creature, if once divided, cannot be rejoyned; when as some sorts of Worms may be divided, and if those divided Parts meet, can rejoyne, as before. Also, some Figurative Motions of different sorts, and so different, that they are opposite, may unite in agreement, in one Composition, or Creature; yet, when the very same sorts of Figurative Motions, are not so united, they are, as it were, deadly Enemies.

CHAP. VI. Of Different Figurative Motions in particular Creatures.

There are many Creatures that are composed of very opposite Figurative Motions; as for example, Some Parts of Fire and Water; also, all Cordials, Vitriols, and the like Waters; also, Iron and Stone, and Infinite the like: But, that which is composed of the most different Figurative Motions, is *Quick-silver*, which is exteriorly Cold, Soft, Fluid, Agil, and Heavy: also, Divisible, and Rejoynable; and yet so Retentive of its Innate Nature, that although it can be rarified, yet not easily dissolved; at least, not that Human Creatures can perceive; for, it hath puzzled the best *Chymists*.

CHAP. VII. Of the Alterations of Exterior and Innate Figurative Motions of several sorts of Creatures.

The Form of several Creatures, is after several manners and ways, which causes several Natures or Properties: As for example, The Exterior and Innate Corporeal Motions of some Creatures, depend so much on each other, That the least Alteration of the one, causes a Dissolution of the whole Creature; whereas the Exterior Corporeal Motions of other sorts of Creatures, can change and rechange their actions, without the least disturbance to the Innate Figurative Motions: In other sorts the Innate Motions shall be quite altered, but their Exterior Motions be in some manner consistent: As for proof, Fire is of that Nature, that both the Exterior and Innate Motions, are of one and the same sort; so that the Alteration of the one, causeth a Dissolution of the other; that is, Fire loses the Property of Fire, and is altered from being Fire. On the other side, the Exterior Figurative Motions of Water, can change and rechange, without any disturbance to the Innate Nature: but, though the Alteration of the Innate Figurative Motions of all Creatures, must of necessity alter the Life and Knowledg of that Creature; yet there may be such consistent Motions amongst the

Exterior Parts of some sorts of Creatures, that they will keep their Exterior Form: As for example, A Tree that is cut down, or into pieces, when those pieces are withered, and, as we say, dead; yet, they remain of the Figure of Wood. Also, a dead Beast doth not alter the Figure of Flesh or Bones, presently. Also, a dead Man doth not presently dissolve from the Figure of Man; and some, by the Art of embalming, will occasion the remaining Figurative Motions of the dead Man to continue, so that those sorts of Motions, that are the Frame and Form, are not quite altered: but yet, those Exterior Forms are so altered, that they are not such as those by which we name a *Living Man*. The same of Flyes, or the like, intomb'd in *Amber*: but by this we may perceive, That the Innate Figurative Motions may be quite altered, and yet the Exterior Figurative consistent Motions, do, in some manner, keep in the Figure, Form, or Frame of their Society. The truth is, (in my opinion) That all the Parts that remain undissolved, have quite altered their Animal actions; but only the Consistent actions, of the Form of their Society, remains, so as to have a resemblance of their Frame or Form.

CHAP. VIII. Of LOCAL MOTION.

All Corporeal Motion is *Local*; but only they are different Local Motions: and some sorts or kinds, have advantage of others, and some have power over others, as, in a manner, to inforce them to alter their Figurative motions; as for example, When one Creature doth destroy another, those that are the *Destroyers*, occasion those that we name the *Destroyed*, to dissolve their Unity, and to alter their actions: for, they cannot annihilate their actions; nor can they give or take away the Power of Self-motions; but, as I said, some Corporeal motions can occasion other Corporeal motions to move so, or so. But this is to be noted, That several sorts of Creatures have a mixture of several sorts of Figurative motions; as for example, There are Flying Fish, and Swimming Beasts; also, there are some Creatures that are partly Beasts, and partly Fish, as *Otters*, and many others; also, a *Mule* is partly a Horse, and an Ass; a *Batt* is partly a Mouse, and a Bird; an *Owle* is partly a Cat and a Bird; and numerous other Creatures there are, that are partly of one sort, and partly of another.

CHAP. IX. Of several manners, or ways of Advantages, or Disadvantages.

Not only the Manner, Form, Frame, or Shape of particular Creatures; but also, the Regularity or Irregularity of the Corporeal motions of particular Creatures, doth cause that which Man names *Strength* or *Weakness*, *Obedience* or *Disobedience*, *Advantages* or *Disadvantages* of Power and Authority, or the like: As for example, A greater Number will overpower a lesse: for, though there be no Differences (as being no Degrees) of Self-strength amongst the Self-moving Parts, or Corporeal motions; yet, there may be stronger and weaker Compositions, or Associations; and a greater Number of Corporeal motions, makes a stronger Party: but, if the greater Party be Irregular, and the lesser Party be Regular, a hundred to one, but the weaker Party is victorious. Also, the manner of the Corporeal motions; as, a Diving-motion may get the better of a Swimming-motion; and, in some cases, the Swimming, the better of the Diving. Jumping may have the advantage over Running; and, in other cases, Running, over Jumping. Also, Creeping may have the advantage over Flying; and, in other cases, Flying, over Creeping. A Cross Motion may have the advantage over a Straight; and, in other cases, a Straight, over a Cross. So it may be said, of Turning and Lifting, of Contracting and Dilating Motions. And many the like Examples may be had; but, as I have often said, There is much Advantage and Disadvantage in the manner and way of the Composed Form and Figure of Creatures.

CHAP. X. Of the Actions of some sorts of Creatures, over others.

Some sorts of Creatures are more Exteriorly active, than other sorts; and some more Interiorly active; some more rare, some more dense, and the like: also, some dense Creatures are more active than the rare; and some rare, are more active than other sorts that are dense. Also, some Creatures that are rare, have advantage of some that are dense; and some that are

dense, over some sorts that are rare; some leight Bodies, over some heavy Bodies; and some heavy Bodies, over some sorts of leight Bodies. Also, several sorts of Exterior Motions, of several sorts of Creatures, have advantage and disadvantage of each other; as for example, Springs of Water, and Air, will make Passages, and so divide hard strong Rocks. And, on the other side, a Straw will divide Parts of Water; and a small Flye, will divide Parts of the Air: but, mistake me not, I mean, that they occasion the Airy or Watry Parts, to divide.

CHAP. XI. Of GLASSIE BODIES.

Tis impossible, as I have said, to describe the Infinite Corporeal Figurative Motions: but, amongst those Creatures that are subject to Human Perception, there are some that resemble each other, and yet are of different Natures; as for example, *Black Ebony*, and *Black Marble*, they are both Glassie, smooth, and black; yet, one is Stone, the other Wood. Also, there be many light and shining Bodies, that are of different Natures; as for example, Metal is a bright shining Body; and divers sorts of Stones, are bright shining Bodies: also, clear Water is a bright shining Body; yet, the Metal and Stones are Minerals, and Water is an Element. Indeed, Most Bodies are of a Glassie Hue, or, as I may say, Complexion; as may be observed in most Vegetables; as also, Skins, Feathers, Scales, and the like.

But some may say, That *Glassiness is made by the Brightness of the Light that shines upon them.*

I answer: If so, then the ordinary Earth would have the like Glassiness: but, we perceive the Earth to appear dull in the clearest Sun-shining Day: wherefore, it is not the Light, but the nature of their own Bodies. Besides, every Body hath not one and the same sort of Glassiness, but some are very different: 'Tis true, some sorts of Bodies do not appear Glassie, or shining, until they be polished: but, as for such sorts of shining Bodies that appear in the dark, there is not many of them perceiv'd by us, besides the Moon and Starrs; but yet some there are, as Fire; but that is an Element. There are also

Glow-worms Tayles, Cats Eyes, Rotten Wood, and such like shining-Bodies.

CHAP. XII. Of Metamorphoses, or Transformations of Animals and Vegetables.

There are some Creatures that cannot be Metamorphosed: as for example, Animals and Vegetables, at least, most of those sorts, by reason they are composed of many several and different Figurative Motions; and I understand *Metamorphose*, to be a change and alteration of the Exterior Form, but not any change or alteration of the Interior or Intellectual Nature: and how can there be a general change of the Exterior Form or Shape of a Human Creature, or such like Animal, when the different Figurative Motions of his different Compositions, are, for the most part, ignorant of each others particular Actions? Besides, as Animals and Vegetables require degrees

of time for their Productions, as also, for their Perfections; so, some Time is requir'd for their Alterations: but, a sudden alteration amongst different Figurative Motions, would cause such a Confusion, that it would cause a Dissolution of the *whole* Creature, especially in actions that are not natural, as being improper to their kind, or sort: The same of Vegetables, which have many different Figurative Motions. This considered, I cannot chuse but wonder, that wise men should believe (as some do) the Change or Transformation of Witches, into many sorts of Creatures.

CHAP. XIII. Of the Life and Death of several Creatures.

That which Man names *Life*, and *Death*, (which are some sorts of Compositions and Divisions of Parts of Creatures) is very different, in different kinds and sorts of Creatures, as also, in one and the same sort: As for example, Some Vegetables are old and decrepit in a Day; others are not

in Perfection, or in their Prime, in less than a hundred years. The same may be said of Animal kinds. A *Silk-worm* is no sooner born, but dyes; when as other Animals may live a hundred years. As for Minerals, Tinn and Lead seem but of a short Life, to Gold; as a Worm to an Elephant, or a Tulip to an Oak for lasting; and 'tis probable, the several Productions of the Planets and Fixed Starrs, may be as far more lasting, than the parts of Gold more lasting than a Flye: for, if a Composed Creature were a Million of years producing, or Millions of years dissolving, it were nothing to Eternity: but, those produced Motions that make Vegetables, Minerals, Elements, and the like, the subtlest Philosopher, or Chymist, in Nature, can never perceive, or find out; because, Human Perception is not so subtle, as to perceive that which Man names Natural Productions: for, though all the Corporeal Motions in Nature are perceptive; yet, every Perceptive Part doth not perceive all the actions in Nature: for, though every different Corporeal Motion, is a different Perception; yet, there are more Objects than any one Creature can perceive: also, every particular kind or sort of Creatures, have different Perceptions, occasioned by the Frame and Form of their Compositions, or unities of their Parts: So as the Perceptions of Animals, are not like the Perceptions of Vegetables; nor Vegetables, like the Perceptions of Minerals; nor Minerals, like the Perceptions of Elements: For, though all these several kinds and sorts, be perceptive; yet, not after one and the same way, or manner of Perception: but, as there is infinite variety of Corporeal Motions, so there are infinite varieties of Perceptions: for, Infinite Self-moving Matter, hath infinite varieties of Actions. But, to return to the Discourse of the Productions and Dissolutions of Creatures; The reason, that some Creatures last longer than others, is, That some Forms or Frames of their Composition, are of a more lasting Figure. But this is to be observed, That the Figures that are most solid, are more lasting than those that are more slack and loose: but mistake me not; I say, *For the most part*, they are more lasting. Also, this is to be noted, That some Compositions require more labour; some, more curiosity; and some are more full of variety, than others.

CHAP. XIV. Of CIRCLES.

A *Circle* is a Round Figure, without End; which Figure can more easily and aptly alter the Exterior Form, than any other Figure. For example, A Circular Line may be drawn many several ways, into different and several sorts of Figures, without breaking the Circle: also, it may be contracted or extended into a less or wider compass; and drawn or formed into many several sorts of Figures, or Works; as, into a Square, or Triangle, or Oval, or Cylinder, or like several sorts of Flowers, and never dissolve the Circular Line. But this is to be noted, that there may be several sorts of Circular Lines; as, some Broad, some Narrow, some Round, some Flat, some Ragged or Twisted, some Smooth, some Pointed, some Edged, and numbers of the like; and yet the compass be exactly round.

But some may say, that, *When a Circle is drawn into several Works, it is not a Circle: As for example, When a Circle is squared, it is not a Circle, but a Square.*

I answer: It is a Circle squar'd, but not a Circle broken, or divided: for, the Interior Nature is not dissolved, although the Exterior Figure is altered: it is a Natural Circle, although it should be put into a Mathematical Square. But, to conclude this Chapter, I say, That all such sorts of Figures that are (like Circular Lines) of one piece, may change and rechange their Exterior Figures, or Shapes, without any alterations of their Interior Properties.

CHAP. XV. Human Creatures cannot so probably treat of other sorts of Creatures, as of their own.

To treat of the Productions of Vegetables, Minerals, and Elements, is not so easie a Task, as to treat of Animals; and, amongst Animals, the most easie Task is, to treat of Human Productions; by reason one Human Creature may more probably guess at the Nature of all Human Creatures (being of the same Nature) than he can of other kinds

of other kinds of Creatures, that are of another Nature. But, mistake me not, I mean not of another Nature, being not of the same kind of Creature, but concerning Vegetables, Minerals, and Elements. The Elements may more easily be treated of, than the other Two kinds: for, though there be

numerous sorts of them, at least, numerous several Particulars; yet, not so many several Sorts, as of Vegetables: and though Minerals are not, as to my knowledg, so numerous as Vegetables; yet, they are of more, or at least, of as many Sorts as Elements are. But, by reason I am unlearned, I shall only give my Opinion of the Productions of some sorts; in which, I fear, I shall rather discover my Ignorance, than the Truth of their Productions. But, I hope my *Readers* will not find fault with my Endeavour, though they may find fault with my little Experience, and want of Learning.

The Twelfth Part.

CHAP. I. Of the Equality of ELEMENTS.

As for the Four Elements, *Fire, Air, Water, and Earth*; they subsist, as all other Creatures, which subsist by each other: but, in my opinion, there should be an Equality of the Four Elements, to balance the World: for, if one sort should superabound, it would occasion such an Irregularity, that would cause a Dissolution of this World; as, when some particular Humour in Man's Body superabounds, or there is a scarcity of some Humours, it causes such Irregularities, that do, many times, occasion his Destruction. The same may be said of the Four Elements of the World: as for example, If there were not a sufficient quantity of Elemental Air, the Elemental Fire would go out; and if not a sufficient quantity of Elemental Fire, the Air would corrupt: also, if there were not a sufficient quantity of Elemental Water, the Elemental Fire would burn the Earth; and if there were not a sufficient quantity of Earth, there would not be a solid and firm Foundation for the Creatures of the Earth: for, if there were not Density, as well as Rarity; and Levity, as well as Gravity; Nature would run into Extreams.

CHAP. II. Of several TEMPERS.

Heat doth not make Drought: for, there is a *Temper* of Hot and Moist. Nor Cold doth not make Drought: for, there is a *Temper* of Cold and Moist. Neither doth Heat make Moisture: for, there is a *Temper* of Hot and Dry. Nor doth Cold make Moisture: for, there is a *Temper* of Cold and Dry. But, such or such sorts of Corporeal Figurative Motions, make Hot, Cold, Moist, Dry; Hot and Dry, Hot and Moist; Cold and Dry, Cold and Moist; and, as those Figurative Motions alter their Actions, those *Tempers* are altered: the

like happens in all Creatures. But this is to be observed, That there is some opposite or contrary *Temper*s, which have a likeness of Motion: as for example, A Moist Heat, and a Moist Cold, have a likeness or resemblance of Moistness;

and the same is in dry Heats and Cold: but surely, most sorts of Moistures, are some sorts of dilative Motions; and most Droughts, are some sorts of Contractive Motions: but, there are several sorts of Dilatations, Contractions, Retentions, Expulsions, and the like: for, there are Cold Contractions, Hot Contractions; Cold Dilatations, Hot Dilatations; Hot Retentions, Cold Retentions; and so of Digestions, Expulsions, and the like: But, as I said, Moist Heats, and Moist Colds, seem of a Dilative Nature; as Dry, of a Contractive Nature. But, all Cold and Heat, or Dry and Moist, may be made by one and the same Corporeal Motions: for, though the Actions may vary, the Parts may be the same: yea, the like Actions may be in different Parts. But, no Part is bound to any particular Action, having a free Liberty of Self-motion. But, concerning Hot and Cold, and the like Actions, I observe, That Extream Heat, and Extream Cold, is of a like Power, or Degree: neither can I perceive the Hot Motions to be quicker than Cold: for Water, in little quantity, shall as suddenly freeze, as any leight Fewel or Straw, burn: and Animals will as soon freeze to death, as be burned to death: and Cold is as powerful at the Poles, as Heat in the *Torrid Zone*. And 'tis to be observed, That Freezing is as quick and sudden, as Thawing: but sometimes, nay very often, Cold and Hot Motions will dispute for Power; and some sorts of Hot, with other sorts. The like Disputes are amongst several sorts of Cold Motions; Dry with Moist, Dry with Dry, Moist with Moist. And the like Disputes are also often amongst all Creatures. As for Density, it doth not make Gravity: for, there may be Dense Bodies, that are not Grave; as for example, Feathers, and Snow. Neither doth Gravity make Density: for, a quantity of Air hath some weight, and yet is not dense. But mistake me not; for, I mean by *Grave*, *Heavy*; and not for the Effects of Ascending, and Descending: for Feathers, though Dense, are more apt to ascend, than descend; and Snow, to descend. Also, all sorts of Fluidity, do not cause Moist, Liquid, or Wet; nor all Extenuations, cause Light: but, they are such and such sorts of Fluidities and Extenuations, that cause such and such Effects. And so for Heats, Colds, Droughts, Moistures, Rarities. The same for Gravities, Levities, and the like. So that, Creatures are Rare, Fluid,

Moist, Wet, Dry, Dense, Hard, Soft, Leight, Heavy, and the like, according to their Figurative Motions.

CHAP. III. Of the Change and Rechange; and of Dividing and Ioyning of the Parts of the Elements.

Of all Creatures subject to Human Perception, the Elements are most apt to Transform, *viz.* to *Change* and *Rechange*; also, to *Divide* and *Ioyne* their Parts, without altering their Innate Nature and Property. The reason is, because the Innate Figurative Motions of Elements, are not so different as those of Animals and Vegetables, whose Compositions are of many different Figurative Motions; in so much, that disjoining any Part of Animals, or Vegetables, they cannot be joined again, as they were before; at least, it is not commonly done: but, the Nature and Property of the Elements, is, That every Part and Particle are of one innate Figurative Motion; so that the least grain of Dust, or the least drop of Water, or the least spark of Fire, is of the same Innate Nature, Property, and Figurative Motions, as the whole Element; when as, of Animals, and Vegetables, almost, every Part and Particle is of a different Figurative Motion.

CHAP. IV. Of the Innate Figurative Motions of Earth.

There are many sorts of *Earth*, yet all sorts are of the same kind; that is, they are all Earth: but (in my opinion) the prime Figurative Motions of Earth, are Circles; but not dilated Circles, but contracted Circles: neither are those Circles smooth, but rugged; which is the cause that Earth is dull, or dim, and is easily divided into dusty Parts: for all, or at least, most Bodies that are smooth, are more apt to joyn, than divide; and have a Glassie Hew or Complexion; which is occasioned by the smoothness, and the smoothness occasioned by the evenness of Parts, being without Intervals:

but, according as these sorts of Circular Motions are more or less contracted, and more or less rugged, they cause several sorts of Earth.

CHAP. V. Of the Figurative Motions of Air.

There are many sorts of *Airs*, as there is of other Creatures, of one and the same kind: but, for Elemental Air, is composed of very Rare, Figurative Motions; and the Innate Motions, I conceive to be somewhat of the Nature of Water, viz. Circular Figurative Motions, only of a more Dilating Property; which causes Air, not to be Wet, but extraordinary Rare; which again causes it to be somewhat of the nature of Light: for, the Rarity occasions Air to be very searching and penetrating; also, dividable and compoundable: but, the Rarity of Air, is the cause that it is not subject to some sorts of Human Perception; but yet, not so Rare, as not to be subject to Human Respirations; which is one sort of Human Perception: for, all Parts of all Creatures, are perceptive one way, or another: but, as I said, there are many sorts of Air; as, some Cold, some Hot; some Dry, some Moist; some Sharp; some Corrupt, some Pure, some Gross; and numbers more: but, many of these sorts are rather Metamorphosed Vapours, and Waters, than pure Elemental Air: for, the pure Elemental Air, is, in my opinion, more searching and penetrating, than Light; by reason Light may be more easily eclipsed, or stopt; when as Air will search every Pore, and every Creature, to get entrance.

CHAP. VI. Of the Innate Figurative Motion of Fire.

The Innate Figurative Motions of Elemental Fire, seem the most difficult to Human Perception, and Conception: for, by the Agilness, it seems to be more pure than the other sorts of Elements; yet, by the Light, or Visibleness, it seems more gross than Air; but, by the dilating Property, it seems to be more rare than air, at least, as rare as Air. By the Glassie or Shining Property, it seems to be of Smooth and Even Parts: also, by the piercing and wounding Property, Fire seems to be composed of sharp-pointed Figurative Motions: Wherefore, the Innate Figurative Motions of Fire, are, Pure, Rare, Smooth, Sharp Points, which can move in Circles, Squares, Triangles,

Parallels, or any other sorts of Exterior Figures, without an alteration of its Interior Nature; as may be observed by many sorts of Fuels: as also, it can contract and dilate its Parts, without any alteration of its Innate Property.

CHAP. VII. Of the Productions of Elemental Fire.

It is to be observed, That Points of Fire are more numerous, and more suddenly propagating, than any other Element, or any other Creature that is subject to Human Perception. But, Sparks of Fire, resemble the Seeds of Vegetables, in this, That as Vegetables will not encrease in all sorts of Soyles, alike; neither will the Points of Fire, in all sorts of Fuel, alike. And, as Vegetables produce different Effects in several Soyls; so doth Fire on several Fuels: As for example, The Seeds of Vegetables do not work the same Effect in a Birds Crop, as in the Earth: for, there they encrease the Bird by digestion; but, in the ground, they encrease their own Issue (as I may say): So Fire, in some Fuels, doth destroy it self, and occasions the Fuel to be more consumed; when as, in other sorts of Fuel, Fire encreases extreamly. But Fire, as all other Creatures, cannot subsist single of it self, but must have Food and Respiration; which proves, Fire is not an Immaterial Motion. Also, Fire hath Enemies, as well as Friends; and some are deadly, namely, Water, or Watry Liquors. Also, Fire is forced to comply with the Figurative Motions of those Creatures it is joyned to: for, all Fuels will not burn, or alter, alike.

CHAP. VIII. Of FLAME.

Flame is the Rarest Part of Fire: and though the Fuel of Flame be of a vaporous and smoaky Substance; yet surely, there are pure Flames, which are perfect Fires: and, for proof, we may observe, That Flame will dilate and run, as it were, to catch Smoak: but, when the Smoak is above the Flame, if it be higher than the Flame can extend, it contracts back to the

Fiery Body. But, Flame doth somewhat resemble that we name *Natural Light*: but yet, in my opinion, Light is not Flame; nor hath it any Fiery Property, although it be such a sort of Extenuating or Dilating Actions, as Flame hath.

CHAP. IX. Of the two sorts of Fire most different.

There are many sorts of Fires: but two sorts are most opposite; that is, the Hot, Glowing, Burning, Bright, Shining Fire; and that sort of Fire we name a *Dead, Dull Fire*; as, Vitriol Fires, Cordial Fires, Corrosive Fires, Feverish Fires, and numerous other sorts; and every several sort, hath some several Property: as for example, There is greater difference between the Fiery Property of Oyl, and

the Fiery Property of Vitriol: for, Oyl is neither Exteriorly Hot, nor Burning; whereas Vitriol is Exteriorly Burning, though not Exteriorly Hot: but, the difference of these sorts of Fires, is, That the Actions of Elemental Fire, are to ascend, rather than to descend: and the Dull, Dead Fire, is rather apt to descend, than ascend; that is, to pierce, or dilate, either upwards, or downwards: but, they are both of Dilating and Dividing Natures. But this is to be noted, That all sorts of Heats, or Hotness, are not Fire. Also it is to be noted, That all Fires are not Shining.

CHAP. X. Of Dead or Dull Fires.

OF *Dull, Dead Fires*, some sorts seem to be of a mixt sort: as for example, Vitriol, and the like, seem to be Exteriorly, of the Figurative Motions of Fire; and Interiorly, of the Figurative Motions of Water, or of Watry Liquors: And Oyl is of Fiery Figurative Motions, Interiorly; and of Liquid Figurative Motions, Exteriorly; which is the cause that the Fiery Properties of Oyl cannot be altered, without a Total Dissolution of their Natures. But,

such sorts whose Fiery Figurative Motions are Exterior, as being not their Innate Nature, may be divided from those other Natural Parts they were joyned to, without altering their Innate Nature.

CHAP. XI. Of the Occasional Actions of Fire.

All Creatures have not only Innate figurative Motions that cause them to be such or such a sort of Creature; but, they have such and such actions, that cause such and such Effects: also, every Creature is occasioned to particular Actions, by forrein Objects; many times to improper actions, and sometimes to ruinous actions, even to the dissolution of their Nature: And, of all Creatures, Fire is the most ready to occasion the most Mischief; at least, Disorders: for, where it can get entrance, it seldom fails of causing such a Disturbance, as occasions a Ruine. The reason is, that most Creatures are porous: for, all Creatures, subsisting by each other, must of necessity have *Egress* and *Regress*, being composed of Interior and Exterior Corporeal Motions. And Fire, being the sharpest figurative Motion, is apt to enter into the smallest Pores.

But some may ask, *Whether Fire is porous it self?*

I answer: That having Respiration, it is a sufficient proof that it is Porous: for, Fire dyes if it hath not Air.

But some may say, *How can a Point be porous?*

I answer, That a Point is composed of Parts, and therefore may very well be porous: for, there is no such thing as a Single Part in Nature, and therefore, not a Single Point.

Also, some may say, *If there be Pores in Nature, there may be Vacuum.*

I answer, That, in my opinion, there is not; because there is no empty Pores in Nature: Pores signifying only an *Egress* and *Regress* of Parts.

CHAP. XII. Fire hath not the Property to Change and Rechange.

Of all the Elemental Creatures, *Fire* is the least subject to change: for, though it be apt to occasion other Creatures to alter; yet it keeps close to its own Properties, and proper Actions: for, it cannot change, and rechange, as Water can. Also, Natural Air is not apt to change and rechange, as Water: for, though it can (as all the Elements) divide and join its Parts, without altering the Property of its Nature: yet, it cannot readily alter, and alter again, its Natural Properties, as Water can. The truth is, Water and Fire, are opposite in all their Properties: but, as Fire is, of all the Elements, the furthest from altering: so Water is, of all the Elements, the most subject to alter: for, all Circular Figures are apt to variety.

CHAP. XIII. Of the Innate Figurative Motions of Water.

The Nature of *Water* is, Rare, Fluid, Moist, Liquid, Wet, Glutinous, and Glassie. Likewise, Water is apt to divide and unite its Parts, most of which Properties are caused by several sorts of Dilatations, or Extenuations: but, the Interior, or Innate Figure of Water, is a Circular Line. But yet, it is to be observed, That there are many several sorts of Waters, as there are many several sorts of Airs, Fires, and Earths, and so of all Creatures: for, some Waters are more rare than others, some more leight, and some more heavy; some more clear, and some more dull; some salt, some sharp; some bitter, some more fresh, or sweet; some have cold Effects, some hot Effects: all which is caused by the several Figurative Motions of several sorts of Waters: but, the nature of Water is such, as it can easily alter, or change, and rechange, and yet keep its Interior, or Innate Nature or Figure. But this is also to be observed, That the Dilating or Extenuating Circle of Water, is of a middle Degree, as between Two Extrems.

CHAP. XIV. The Nature or Property of Water.

Wetness, which is the Interior or Innate Property, or Nature of Water, is, in my opinion, caused by some sort of Dilatations or Extenuations. As, all Droughts, or Dryness, are caused by some sorts of Contractions; so, all Moistures, Liquors, and Wets, by Dilatations: yet, those Extenuations, or Dilatations, that cause Wet, must be of such a sort of Dilatations, as are proper to Wet; viz. Such a sort of Extenuations, as are Circular Extenuations; which do dilate, or extenuate, in a smooth, equal dilatation, from the Center, to the Circumference; which Extenuations, or Dilatations, are of a middle Degree; for otherwise, the Figure of Water might be extended beyond the Degree of Wet; or, not extended to the Degree of Wet. And it is to be observed, That there is such a Degree as only causes moistness, and another to cause liquidness, the third to cause wetness: for, though Moistness and Liquidness are in the way of Wetness; yet, they are not that which we name Wet: also, all that is Soft, or Smooth, is not Wet; nor is all that is Liquid, or Flowing, Wet: for, some sorts of Air are liquid and flowing, but not wet: nay, Flame is liquid and flowing, but yet quite opposite from wet. Dust is flowing, but neither liquid or wet, in its Nature. And Hair and Feathers are soft and smooth, but neither liquid, nor wet. But, as I said, Water is of such a Nature, as to have the Properties of Soft, Smooth, Moist, Liquid, and Wet; and is also of such flowing Properties, caused by such a sort of Extenuating Circles as are of a Middle or Mean Degree: but yet, there are many several sorts of Liquors, and Wets, as we may perceive in Fruit, Herbs, and the like: but, all sorts of Wets, and Liquors, are of a watry kind, though of a different sort. But, as I have said, all things that are Fluid, are not Wet; as, Melted Metal, Flame, Light, and the like, are fluid, but not wet: and Smoak and Oyl are of another sort of Liquidness, than Water, or Juyce; but yet they are not wet: and that which causes the difference of different sorts of Waters, and Watry Liquors, are the differences of the watry Circular Lines; as, some are edged, some are pointed, some are twisted, some are braided, some are flat, some are round, some ruff, some smooth; and so after divers several Forms or Figures: and yet are perfect Circles, and of some such a Degree of Extenuations or Dilatations.

CHAP. XV. Of the Alteration of the Exterior Figurative Motion of Water.

As I formerly said, The Figurative Motions of the Innate Nature of Water, is a sort of Extenuating; as being an equal, smooth Circle: which is the cause Water is rare, fluid, moist, liquid, and wet. But, the Exterior Figurative Motions of the watry Circle, may be edged, pointed, sharp, blunt, flat, round, smooth, ruff, or the like; which may be either divided, or altered, without any alteration of the Innate Nature, or Property: As for example, Salt-water may be made fresh, or the Salt Parts divided from the watry Circle: The like of other sorts of Waters; and yet the Nature of Water remains.

CHAP. XVI. Of OYL, and VITRIOL.

The Exterior Figurative Motions of *Oyl*, are so much like those of *Water*, as, to be fluid, smooth, soft, moist, and liquid, although not perfectly wet: but, the Interior Figurative Motions of *Oyl*, are of that sort of Fire, that we name a *Dull, Dead Fire*: and the difference between *Salt Waters*,

Vitriol or the like, and *Oyl*, is, That the Exterior Figurative Motions of *Vitriol* and *Salt Waters*, are of a sort of Fire; whereas it is the Interior Figurative Motions of *Oyl*, or the like, that are of those sorts of Fire; and that is the reason that the fiery Motions of *Oyl* cannot be altered, as the fiery Motions of *Vitriol* may. But this is to be noted, That although the Interior Figurative Motions of *Oyl*, are of such a sort of fiery Motions; yet, not just like those of *Vitriol*; and are not burning, corroding, or wounding, as *Vitriols*, *Corrosives*, and the like, are: for, those are somewhat more of the Nature of bright shining Fires, than *Oyls*.

CHAP. XVII. Of Mineral and Sulphureous Waters.

In *Sulphureous* and *Mineral Waters*, the *Sulphureous* and *Mineral* Corporeal Motions, are Exterior, and not Interior, like Salt waters: but, there are several sorts of such waters; also, some are occasionally, others naturally so affected: for, some waters running through Sulphureous, or Mineral Mines, gather, like a rowling Stone, some of the loose Parts of Gravel, or Sand; which, as they stick or cleave to the rowling Stone; so they do to the running Waters; as we may perceive by those waters that spring out of Chalk, Clay, or Lime Grounds, which will

have some Tinctures of the Lime, Chalk, or Clay; and the same happens to Minerals. But, some are naturally Sulphureous; as for example, Some sorts of hot Baths are as naturally Sulphureous, as the Sea-water is Salt: but, all those Effects of Minerals, Sulphurs, and the like, are dividable from, and also may be joyn'd to, the Body of water, without any disturbance to the nature of water; as may be proved by Salt-water, which will cause fresh Meat to be salt; and salt Meat will cause Fresh-water to be salt. As for hot Baths, those have hot figurative Motions, but not burning: and the moist, liquid, and wet Nature of water, makes it apt to joyn, and divide, to, and from other sorts of Motions; as also, to and from its own sort.

CHAP. XVIII. The Cause of the Ebbing and Flowing of the Sea.

The Nature of water is to flow; so that all sorts of waters will flow, if they be not obstructed: but it is not the Nature of Water, to ebb. Neither can water flow beyond the Power of its Quantity: for, a little water will not flow so far as a great one. But, I do not mean by flowing, the falling of water from some Descent; but, to flow upon a Level: for, as I have said, all waters do naturally flow, if they be not obstructed; but, few sorts of water, besides Sea-water, ebbs. As for the Exterior Figurative Motions of water, in the action of flowing, they are an Oval, or a half Circle, or a half Moon; where the middle parts of the half Moon, or Circle, are fuller than the two Ends. Also, the figure of a half Moon, or half Circle, is concave on the inside, and convex on the outside of the Circle: but, these Figurative Motions, in a great quantity of water, are bigg and full, which we name *Waves of Water*; which waves flowing fast upon each other, presses each other forward, until such

time as the half Circle divides: for, when the Bow of the half Circle is overbent, or stretched, it divides into the middle, which is most extended: and when a half Circle (which is a whole wave of water) is divided, the divided Parts fall equally back on each side of the flowing waves: so, every wave dividing, after that manner, in the full extension, it causes the motion of ebbing, that is, to flow back, as it flow'd forward: for, the divided Parts falling back, and joining as they meet, makes the head of the half Circle, where the Ends of the half Circle were; and the Convex, where the Concave was; by which action, the ebbing Parts are become the flowing Parts. And the reason that it ebbs and flows by degrees, is, That the flowing half Circles require so much time to be at the utmost extension. Also, every wave, or half Circle, divides not all at one time, but one after another: for, two Bodies cannot be in one place at one point of time; and until the second, third, and so the rest, flow as far as the first, they are not at their full extension. And thus the Sea, or such a great Body of Water, must flow, and ebb, as being its nature to flow; and the flowing Figure, being overextended, by endeavouring to flow beyond its power, causes a dividing of the Extended Parts, which is the Cause of the Ebbing.

But, whether this Opinion of mine, be as probable as any of the former Opinions concerning the Ebbing and Flowing of the Sea, I cannot judg: but I would not be mistaken; for the flowing of the water, is according to its Quantity; for, the further it flows, the fainter, or weaker it is.

CHAP. XIX. Of OVERFLOWS.

As for *Overflows*, there be many; and many more would be, if the waters were not hindred and obstructed by Man's Inventions. But, some Overflows are very Uncertain and Irregular; others, Certain and Regular, as, the flowing of *Nilus* in *Egypt*: but as for the distance of time of its flowing, it may proceed from the far Journey of those flowing-waters: and, the time of its ebbing, may be attributed to the great Quantity of Water; so that the great quantity of water, will cause a longer or a shorter time in the flowing or ebbing; and certainly the waters are as long a flowing back, as flowing forward.

As for Spring Tides, they are only in such a time when there is a Naturall Issue of a greater quantity of water: so that Spring-Tides are but once a Month, and Single-Tides in so many hours: but, many several occasions, may make the Tides to be more or less full.

As for Double-Tides, they are occasioned through the Irregular dividing of the Half-Circle; as, when they divide not orderly, but faster than they orderly should do; which, falling back in a Crowd, and being, by that means, obstructed, so that they cannot get forward, they are necessitated to flow, where they ebb'd.

The reason the Tides flow through Streams of Running-waters, is, That the Tide is stronger than the Stream: but, if the Stream and Tides pass through each other, then the Tide and Stream are somewhat like Duellers together, which make Passes and Passages for their conveniency.

CHAP. XX. Of the Figure of Ice and Snow.

A Circle may not only extend and contract it self without dividing; but may draw it self into many several Figures, as Squares, or Triangles: as also, into many other Figures mix'd of Squares, Triangles, Cubes, or the like; being partly one, and partly, another; and into other several ways, and after several manners; which is the reason, Water may appear in many several Postures of Snow, Ice, Hail, Frost, and the like: and, in my Opinion, when the Water-Circle is Triangular, it is Snow; when the Circle is Square, it is Ice: as for Hail, they are but small pieces of Ice; that is, small Parts, or few Drops of Water, changed into Ice; and those several Parts moving after several manners, make the Exterior Figures, after several shapes; as, great Bodies of Ice will be of many several shapes, occasioned by many or fewer Parts, and by the several Postures of those Parts: but, such Figures, though they are of Ice, yet, are not the Innate Figures of Ice. The same is to be said of Snow. But, the reason of these my Opinions concerning the Figures of Ice and Snow, is, That Snow is leighter than the Water it self; and Ice is heavier, at least, as heavy. And the reason Snow is so leight, is, That a Triangular Figure hath no poyse, being an odd Figure; whereas a Square is

poysed by Even and Equal Lines, and just Number of Points, as, Two to Two: but, a Triangle is Two to One. Also, a Circle is a poysed Figure, as being equal every way, from the Center to the Circumference; and from the Circumference to the Center, all the Lines drawing to one Point. But, mistake me not; for I treat (concerning the Figures of Snow and Ice) only of those Figures that cause Water to be Snow or Ice; and not of the Exterior Figures of Snow and Ice, which are occasioned by the Order or Disorder of Adjoining Parts: for, several Parts of Water, may order themselves into numerous several Figures, which concern not the nature of Water, as it is Water, Snow, or Ice: As for example, Many Men in a Battel, or upon Ceremony, joyn into many several Figures or Forms; which Figures or Forms, are of no concern to their Innate Nature. Also, the several Figures or Forms of several Houses, or several sorts of Building in one House, are of no concern to the Innate Nature of the Materials. The like for the Exterior Figures of Ice and Snow; and therefore *Microscopes* may deceive the Artist, who may take the Exterior for the Interior Figure; but there may be great difference between them.

CHAP. XXI. Of the Change and Rechange of Water.

Water being of a Circular Figurative Motion, is, as it were, but one Part, having no divisions; and therefore can more easily change and rechange it self into several Postures, viz. into the Posture of a Triangle, or Square; or can be dilated or extended into a larger compass, or contracted into a lesser compass; which is the cause it can turn into Vapour and Vaporous Air; or into Slime, or into some grosser Figure: For example, Water can extend it self beyond the proper degrees of Water, into the degree of Vapour; and the Circle, extending further than the degree of a Vaporous Circle, is extended into a Vaporous Air; and if the Vaporous Airy Circle be extreamly extended, it becomes so small, as it becomes to be a sharp Edg, and so, in a degree, next to Fire; at least, to have a hot Effect: but, if it extends further than an Edg, the Circle breaks into Flashes of Fire, like Lightning, which is a flowing Flame: for, being produced from Water, it hath the property of Flowing, or Streaming, as Water hath, as we may perceive by the Effects of

some few Parts of Water flung on a bright Fire; for those few drops of Water being not enough to quench the Fire, straight dilate so extreamly, that they break into a Flame; or else cause the Fire to be more brisk and bright: and as the Water-Circle can be turned into Vapour, Air, and Flame, by Extension; so, it can be turned into Snow, Hail, or Ice, by Contraction.

CHAP. XXII. Of Water Quenching Fire; and Fire Evaporating Water.

There is such an Antipathy betwixt *Water* and *Fire*, (I mean bright shining Fire) that they never meet Body to Body, but Fire is in danger to be quenched out, if there be a sufficient Quantity of Water. But it is to be observed, That it is not the actual Coldness of Water, that quenches out Fire; for, Scalding-water will quench out Fire: wherefore, it is the Wetness that quenches out Fire; which Wetness choaks the Fire, as a Man that is drown'd: for, Water being not fit for Man's Respiration, because it is too thick, choaks and smuthers him; and the same doth Water to Fire: for, though Air is of a proper temper for Respiration, both to some sorts of Animals, such as Man; as also, to Fire: yet, Water is not: which is most proper for other sorts of Animals, namely, Fish; as also, for some sorts of Animals that are of a mixt kind or sort, partly Fish, and partly Flesh: to which sort of Creatures, both Air and Water are both equally proper for their Respiration; or, their Respiration equal to either: for certainly, all sorts of Creatures have Respiration, by reason all Creatures subsist by each other; I say, *By each other*, not *Of each other*. But, there are many several sorts and kinds of Respirations; as concerning Water and Fire, though a sufficient quantity of Water, to Fire, doth always choak, smuther, or quench out the Fire's Life, if joyn'd Body to Body; yet, when there is another Body between those two Bodies, water is in danger to be infected with the Fire's heat; the Fire first infecting the Body next to it; and that Body infecting the Water: by which Infection, Water is consumed, either by a languishing Hectick Fever; or, by a raging Boyling Fever; and the Life of Water evaporates away.

CHAP. XXIII. Of Inflamable Liquors.

There are many Bodies of mixt Natures; as for example, Wine, and all Strong Liquors, are partly of a watry Nature, and partly of a fiery Nature; but, 'tis of that sort we name a *Dead*, or *Dull Fire*: but, being of such a mixt Nature, they are both apt to quench Bright Fire, as also, apt to burn or flame; so that such sorts are both Inflamable, and Quenchable. But, some have more of the fiery Nature; and others more of the watry Nature; and, by those Effects, we may perceive, that not only different, but opposite Figurative Motions, do well agree in one Society.

CHAP. XXIV. Of THUNDER.

I observe, that all Tempestuous Sounds have some resemblances to the flowing of waters, either in great and ruffling waves; or, when the waters flow in such sort, as to break in pieces against hard and rugged Rocks; or run down great Precipices, or against some Obstruction. And the like Sound hath the Blowings of Wind, or the Clappings of Thunder; which causes me to be of opinion, That Thunder is occasioned by a Discord amongst some Water-Circles in the Higher Region; which, pressing and beating upon each other in a confused manner, cause a confused Sound, by reason all Circles are Concave within the Bow, and Convex without; which is a Hollow Figure, although no Vacuum: which Hollow Figure, causes quick Repetitions and Replies; which Replies and Repetitions, we name Rebounds but, Replies are not Rebounds; for, Rebounds are Pressures and Re-actions; whereas Repetitions are without Pressure, but Re-action is not: and, Replies are of several Parts; as, one Part to reply to another.

But for *Thunder*, it is occasioned both by Pressures and Re-actions; as also, Replies of Extended Water-Circles, which make a kind or sort of Confusion, and so a confused Sound, which we name *Horrid*; and, according to their Discord, the Sound is more or less terrifying, or violent. But this is to be noted, That as *Thunder* is caused by undivided or broken Circles; so *Lightning* is caused by broken or divided Circles, that are extended beyond the Power of the Nature of the Water-Circle; and when the Circle is

extreamly extended, it divides it self into a straight Line, and becomes a flowing Flame.

CHAP. XXV. Of Vapour, Smoak, Wind, and Clouds.

Vapour and *Smoak* are both fluid Bodies: but, *Smoak* is more of the Nature of Oyl, than Water; and Vapour more of the Nature of Water, than Oyl; they are dividable: and may be join'd, as other Elements: also, they are of a Metamorphosing Nature, as to change and rechange; but, when they are Metamorphosed into the form of Air, that Air is a gross Air, and is, as we say, a corruptible Air. As for Vapour, it is apt to turn into Wind: for, when it is rarified beyond the Nature of Vapour, and not so much as into the Nature of Air, it turns into some sorts of Wind. I say, some sorts: and certainly, the strongest Winds are made of the grossest Vapours. As for Smoak, it is apt to turn into some sorts of Lightning; I say, apt: for, both Vapour and Smoak can turn into many sorts of Metamorphosed Elements. As for Wind, it proceeds either from Rarified Vapour, or Contracted Air. And there are many sorts of Vapours, Smoaks, and Winds; all which sorts of Vapours and Smoaks, are apt to ascend: but, Wind is of a more level action. As for Clouds, they cannot be composed of a Natural Air; because Natural Air is too rare a Body to make Clouds. Wherefore, Clouds are composed of Vapour and Smoak: for, when Vapour and Smoak ascends up high without transformation, they gather into Clouds, some higher, some lower, according to their purity: for, the purer sort (as I may say for expression-sake) ascends the highest, as being the most agil. But, concerning the Figurative Motions of Vapour and Smoak, they are Circles; but of Winds, they are broken Parts of Circular Vapours: for, when the Vaporous Circle is extended beyond its Nature of Vapour, the Circumference of the Circle breaks into perturbed Parts; and if the Parts be small, the wind is, in our perception, sharp, pricking, and piercing: but, if the Parts are not so small, then the wind is strong and pressing: but wind, being rarified Vapour, is so like Air, as it is not perceived by human sight, though it be perceived by human touch. But, as there are hot vapours, cold vapours, sharp vapours, moist vapours, dry vapours, subtil vapours, and the like; so there is such

sorts of winds. But, pray do not mistake me, when I say, that some sorts of winds are broken and perturbed Circles, as if I meant, such as those of Lightning: for, those of Lightning, are extended beyond the degree of Air; and those of Vapours, are not extended to the degree of Air: also, those of Lightning, are not perturbed; and those of Wind, are perturbed. Again, those of Lightning, flow in Streams of smooth, small, even Lines; those of Wind, in disordered Parts and Fragments.

CHAP. XXVI. Of WIND.

Wind and *Fire* have some resemblance in some of their particular actions: as for example, Wind and Fire endeavour the disturbance of other Creatures, occasioning a separating and disjoining of Parts. Also, Wind is both an Enemy and Friend to Fire: for Wind, in some sorts of its actions, will assist Fire; and in other actions, dissipates Fire, nay, blows it out: but certainly, the powerful Forces of Wind, proceed not so much from Solidity, as Agility: for, soft, weak, quick Motions, are far more powerful, than strong, slow Motions; because, quick Replies are of great Force, as allowing no time of respite. But this is to be observed, That Wind hath some watry Effects: for, the further water flows, the weaker and fainter it is: so the Wind, the further it blows, the weaker and fainter it is. But this is to be observed, That according to the agilness or slowness of the Corporeal Motions; or, according to the number; or, according to the manner of the compositions, or joynings, or divisions; or, according to the regularity or irregularity of the Corporeal Figurative Motions, so are the Effects.

CHAP. XXVII. Of LIGHT.

Water, Air, Fire, and Light; are all Rare and Fluid Creatures; but they are of different sorts of Rarities and Fluities: and, though Light seems to be extreamly Rare and Fluid; yet, Light is not so Rare and Fluid, as pure Air is,

because it is subject to that sort of Human Perception we name *Sight*; but yet, it is not subject to any of the other Perceptions: and, pure Air is only subject to the Perception of Respiration, which seems to be a more subtil Perception than Sight; and that occasions me to believe, That Air is more Rare and Pure, than Light: but howsoever, I conceive the Figurative Motions of Light, to be extraordinary even, smooth, agil Lines of Corporeal Motions: but, as I said before, there are many sorts of Lights that are not Elemental Lights; as, Glow-worms Tails, Cats Eyes, Rotten Wood, Fish Bones, and that Human Light which is made in Dreams, and Infinite other Lights, not subject to our Perception: which proves, That Light may be without Heat. But, whether the Light of the Sun, which we name *Natural Light*, is naturally hot, may be a dispute: for, many times, the Night is hotter than the Day.

CHAP. XXVIII. Of DARKNESS.

The Figurative Motions of *Light* and *Darkness*, are quite opposite; and the Figurative Motions of Colours, are as a Mean between both, being partly of the Nature of both: but, as the Figurative Motions of Light, in my opinion, are rare, straight, equal, even, smooth Figurative Motions: those of Darkness are uneven, ruff, or rugged, and more dense. Indeed, there is as much difference between Light and Darkness, as between Earth and Water; or rather, between Water and Fire; because each is an Enemy to other; and, being opposite, they endeavour to out-power each other. But this is to be noted, That Darkness is as visible to Human Perception, as Light; although the Nature of Darkness is, To obscure all other Objects besides it self: but, if Darkness could

not be perceived, the Optick Perception could not know when it is dark; nay, particular dark Figurative Motions, are as visible in a general Light, as any other Object; which could not be, if Darkness was only a privation of Light, as the Opinions of many Learned Men are: but, as I said before, Darkness is of a quite different Figurative Motion, from Light; so different, that it is just opposite: for, as the property of Light is to divulge Objects; so, the property of Darkness is to obscure them: but, mistake me not; I mean,

that Light and Darkness have such properties to our Perception: but, whether it is so to all Perceptions, is more than I know, or is, as I believe, known to any other Human Creature.

CHAP. XXIX. Of COLOURS.

AS for *Colour*, it is the same with *Body*: for surely, there is no such thing in Nature, as a Colourless Body, were it as small as an Atom; nor no such thing as a Figureless Body; or such a thing as a Placeless Body: so that Matter, Colour, Figure, and Place, is but one thing, as one and the same Body: but Matter, being self-moving, causes varieties of Figurative Actions, by various changes. As for Colours, they are only several Corporeal Figurative Motions; and as there are several sorts of Creatures, so there are several sorts of Colours: but, as there are those, Man names Artificial Creatures; so there are Artificial Colours. But, though to describe the several Species of all the several sorts of Colours, be impossible; yet we may observe, that there is more variety of Colours amongst Vegetables and Animals, than amongst Minerals and Elements: for, though the Rain-bow is of many fine Colours; yet, the Rain-bow hath not so much Variety, as many particular Vegetables, or Animals have; but every several Colour, is a several Figurative Motion; and the Brighter the Colours are, the Smoother and Evener are the Figurative Motions. And as for Shadows of Colours, they are caused when one sort of Figurative Motions is as the Foundation: for example, If the Fundamental Figurative Motion, be a deep Blew, or Red, or the like, then all the variations of other Colours have a tincture. But, in short, all Shadows have a ground of some sort of dark Figurative Motions. But, the Opinions of many Learned Men, are, That all Colours are made by the several Positions of Light, and are not inherent in any Creature; of which Opinion I am not: For, if that were so, every Creature would be of many several Colours; neither would any Creature produce after their own *Species*: for, a Parrot would not produce so fine a Bird as her self; neither would any Creature appear of one and the same Colour, but their Colour would change according to the Positions of Light; and in a dark day, in my opinion, all fine coloured Birds, would appear like Crows; and fine

coloured Flowers, appear like the Herb named Night-shade; which is not so. I do not say, That several Positions of Light may not cause Colours; but I say, The Position of Light is not the Maker of all Colours; for, *Dyers* cannot cause several Colours by the Positions of Light.

CHAP. XXX. Of the Exterior Motions of the Planets.

By the *Exterior Motions of the Planets*, we may believe their Exterior Shape is Spherical: for, it is to be observed, That all Exterior Actions are according to their Exterior Shape: but, by reason Vegetables and Minerals have not such sorts of Exterior Motions or Actions, as Animals; some Men are of opinion, they have not Sensitive Life; which opinion proceeds from a shallow consideration: neither do they believe the Elements are sensible, although they visibly perceive their Progressive Motions; and yet believe all sorts of Animals to have sense, only because they have Progressive Motions.

CHAP. XXXI. Of the Sun, and Planets, and Seasons.

The Sun, Moon, Planets, and all those glittering Starrs we see, are several sorts of that Man names *Elemental Creatures*: but Man, having not an infinite Perception, cannot have an infinite perceptive knowledg: for, though the Rational Perception is more subtil than the Sensitive; yet, the particular Parts cannot perceive much further than the Exterior Parts of Objects: but, Human Sense and Reason cannot perceive what the Sun, Moon, and Starrs are; as, whether solid, or rare; or, whether the Sun be a Body of Fire; or the Moon, a Body of Water, or Earth; or, whether the Fixed Starrs be all several Suns; or, whether they be other kinds or sorts of Worlds. But certainly, all Creatures do subsist by each other, because Nature seems to be an Infinite united Body, without *Vacuum*. As for the several Seasons of the Year, they are divided into Four Parts: but the several

Changes and Tempers of the Four Seasons, are so various, altering every moment, as it would be an endless work, nay, impossible, for one Creature to perform: for, though the *Almanack-makers* pretend to foreknow all the variations of the Elements; yet, they can tell no more than just what is the constant and set-motions; but not the variations of every Hour, or Minute; neither can they tell any thing, more than their Exterior Motions.

CHAP. XXXII. Of Air corrupting Dead Bodies.

Some are of opinion, *That Air is a Corrupter, and so a Dissolver of all dead Creatures, and yet is the Preserver of all living Creatures.* If so, Air hath an Infinite Power: but, all the reason I can perceive for this Opinion, is, That Man perceives, that when any Raw (or that we name *Dead*) *Flesh*, is kept from the air, it will not stink, or corrupt, so soon as when it is in the air: but yet it is well known, that extream cold air will keep *Flesh* from corrupting.

Another Reason is, That a Flye entomb'd in *Amber*, being kept from air, the Flye remains in her Exterior Shape as perfectly as if she were alive.

I answer, The cause of that may be, that the Figurative Motions of *Amber*, may sympathize with the Exterior consistent Motions of the Fly, which may cause the Exterior Shape of the Flye to continue, although the Innate Nature be altered. But Air is, as all other Creatures are, both Beneficial, and Hurtful to each other; for Nature is poysed with Opposites: for we may perceive, that several Creatures are both Beneficial and Hurtful to each other: as for example, A Bear kills a Man; and, on the other side, a Bear's

Skin will cure a Man of some Disease. Also, a *Wild-Boar* will kill a Man; and the *Boar's* *Flesh* will nourish a Man. Fire will burn a Man, and preserve a Man; and Millions of such Examples may be proposed. The same may be said of Air, which may occasion Good or Evil to other Creatures; as, the *Amber* may occasion the death of a Fly; and, on the other side, may occasion the Preservation, or Continuation of the Fly's Exterior Figure, or Form: but, Nature being without *Vacuum*, all her Parts must be be joined; and her Actions being poysed, there must be both Sympathetical, and Antipathetical actions, amongst all Creatures.

The Thirteenth Part.

CHAP. I. Of the Innate Figurative Motions of Metals.

All sorts of *Metals*, in my opinion, are of some sorts of Circular Motions; but not like that sort, that is Water: for, the Water-Circle doth extend outward, from the Center; whereas, in my opinion, the Circular Motion of Metal, draws inward, from the Circumference. Also, in my opinion, the Circular Motions are dense, flat, edged, even, and smooth; for, all Bright and Glassie Bodies are smooth: and, though Edges are wounding Figures; yet, Edges are rather of the Nature of a Line, than of a Point. Again, all Motions that tend to a Center, are more fixt than those that extend to a Circumference: but, it is according to the degree of their Extensions, that those Creatures are more or less fixt; which is the cause that some sorts of Metals are more fixt than others; and that causes Gold to be the most fixt of all other sorts of Metals; and seems to be too strong for the Effects of Fire. But this is to be noted, That some Metals are more near related to some sort, than other: as for example, There is no Lead, without some Silver; so that Silver seems to be but a well-digested Lead. And certainly, Copper hath some near relation to Gold, although not so near related, as Lead is to Silver.

CHAP. II. Of the Melting of Metals.

Metals may be occasioned, by Fire, to slack their Retentive Motions, by which they become fluid; and as soon as they are quit of their Enemy, *Fire*, the Figurative Motions of Metal return to their proper Order: and this is the reason that occasions Metal to melt, which is, to flow: but yet, the Flowing Motion is but like the Exterior, and not the Innate actions of: for, the

Melting actions do not alter the Innate actions; that is, they do not alter from the Nature of being Metal: but, if the Exterior Nature be occasioned, by the Excess of those Exterior actions, to alter their Retentive actions, then Metal turns to that we name *Dross*; and as much as Metal loses of its weight, so much of the Metal dissolves; that is, so much of those Innate motions are quite altered: but, Gold hath such an Innate Retentiveness, that though Fire may cause an extream alteration of the Exterior actions; yet, it cannot alter the Interior motions. The like is of Quick-silver. And yet Gold is not a God, to be Unalterable, though man knows not the way, and Fire has not the power to alter the Innate Nature of Gold.

CHAP. III. Of Burning, Melting, Boyling, and Evaporating.

Burning, Melting, Boyling, and Evaporating, are, for the most part, occasioned by Fire, or somewhat that is, in effect, Hot: I say, *occasioned*, by reason they are not the actions of Fire, but the actions of those Bodies that melts, boyls, evaporates, or burns; which being near, or joyned to Fire, are occasioned so to do: as for example, Put several sorts of Creatures, or Things, into a Fire, and they shall not burn alike: for, Leather and Metal do not burn alike; for Metal flows, and Leather shrinks up, and Water evaporates, and Wood converts it self, as it were, into Fire; which other things do not; which proves, That all Parts act their own actions. For, though some Corporeal motions may occasion other Corporeal motions to act after such or such a manner; yet, one Part cannot have another Part's motion, because Matter can neither give nor take motion.

CHAP. IV. Of STONE.

All Minerals seem to be some kinds of Dense and Retentive motions: but yet, those kinds of Dense and Retentive motions, seem to be of several sorts; which is the cause of several sorts of Minerals, and of several sorts of

Stones and Metals. Also, every several sort, hath several sorts of Properties: but, in my opinion, some sorts are caused by Hot Contractions and Retentions; others, by Cold Contractions and Retentions; as also, by Hot or Cold Densations: and the reason why I believe so, is, That I observe that many Artificial Stones are produced by Heat: but Ice, which is but in the first Degree of a Cold Density, seems somewhat like transparent Stones; so that several sorts of Stones, are produced by several sorts of Cold and Hot Contractions and Densations.

CHAP. V. Of the LOADSTONE.

As for the *Loadstone*, it is not more wonderful in attracting Iron, than Beauty, which admirably attracts the Optick Perception of Human Creatures: and who knows, but the North and South Air may be the most proper Air for the Respiration of the *Loadstone*; and, that Iron may be the most proper Food for it. But, by reason there hath been so many Learned Men puzzled in their Opinions concerning the several Effects of the *Loadstone*, I dare not venture to treat of the Nature, and Natural Effects of that Mineral; neither have I had much experience of it: but I observe, That Iron, and some sorts of Stone, are nearly allied; for, there is not any Iron, but what is growing, or is intermixt and united in some sorts of Stone, as that which we call *Iron-stone*. Wherefore, it is no wonder if the *Loadstone*, and Iron, should be apt to embrace one another.

CHAP. VI. Of Bodies, apt to ascend or descend.

There are so many several Causes that occasion some sorts of Creatures to be apt to *ascend*, and others to *descend*, as they are neither known, or can be conceived by one finite Creature: for, it is not Rarity or Density, that causes Levity and Gravity; but, the Frame or Form of a Creature's Exterior Shape, or Parts. As for example: A Flake of Snow is as Rare as a Downy

Feather; yet, the Feather is apt to ascend, and the Flake of Snow to descend. Also Dust, that is hard and dense, is apt to ascend; and Water, that is soft and rare, is more apt to descend. Again, a Bird, that is both a bigger, and a more dense Creature by much, than a small Worm; yet, a Bird can flye up into the air, when as a leight Worm cannot ascend, or flye, having not such a sort of Shape. Also, a great heavy Ship, as big as an ordinary House, fraughted with Iron, will swim upon the face of the Water; when as a small Bullet, no bigger than a *Hasle-Nut*, will sink to the bottom of the Sea. A great Bodied Bird will flye up into the air; when as a small Worm lies on the earth, with a slow kind of crawling, and cannot ascend. All which is caused by the manner of their Shapes, and not the matter of Gravity and Levity.

CHAP. VII. Why Heavy Bodies descend more forcibly than Leight Bodies ascend.

Although the manner of the Shape of several Creatures, is the chief cause of their *Ascent*, and *Descent*; yet, Gravity and Levity, doth occasion more or less Agility: for, a Heavy Body shall descend with more force, than a Leight Body ascend: and the reason is, not only that there may be more Parts in a Heavy Body, than a Leight; but, that in a Descent, every Corporeal Motion seems to press upon each other; which doubles and trebles the Strength, Weight, and Force, as we may perceive in the Ascending and Descending of the Flight of Birds, especially of Hawks; of which, the weight of the Body is some hindrance to the Ascent, but an advantage to the Descent: but yet, the Shape of the Bird hath some advantage by the Weight, in such sort, that the Weight doth not so much hinder the Ascent, as it doth assist the Descent.

CHAP. VIII. Of several sorts of Densities and Rarities, Gravities and Levities.

There are different sorts of Densities and Rarities, Softness and Hardness, Levities and Gravities: as for example, The density of Earth is not like the density of Stone; nor the density of Stone, like the density of Metal: nor are all the Parts of the Earth dense alike; nor all Stones, nor all Metals; as we may perceive in Clay, Sand, Chalk, and Lime-Grounds. Also, we may perceive difference between Lead, Tynne, Copper, Iron, Silver, and Gold; and between Marble, Alabaster, Walling-Stone, Diamonds, Crystals, and the like: and so much difference there is between one and the same kind, that some particulars of one sort, shall more resemble another kind, than their own: as for example, Gold and Diamonds resemble each other's Nature, more than Lead doth Gold; or Diamonds, Crystal; I say, in their Densities. Also, there is a great difference of the Rarity, Gravity, and Levity of several sorts of Waters, and of several sorts of Air.

CHAP. IX. Of VEGETABLES.

Vegetables are of numerous sorts, and every sort of very different Natures: as for example, Some are Reviving Cordials; others, Deadly Poyson; some are Purgers, others are Nourishers: some have Hot Effects, some Cold; some Dry, some Moist; some bear Fruit, some bears no Fruit; some appear all the year Young; others appear but part of the year Young, and part Old; some are many years a producing; others are produced in few hours; some will last many hundred years; others will decay in the compass of few hours: some seem to dye one part of the year, and revive again in another part of the year: some rot and consume in the Earth, after such a time; and will continue in perfection, if parted from the Earth. Others will wither and decay, as soon as parted from the Earth. Some are of a dense Nature, some of a rare Nature; some grow deep into the Earth; others grow high out of the Earth; some will only produce in dry Soyls, some in moist: some will produce only in Water, as we may perceive by some Ponds; others on Houses of Brick or Stone. Also, some grow out of Stone; as, many Stones will have a green Moss: some are produced by sowing their Seed into the Earth; others, by setting their Roots, or Slips, into the Earth: others again, by joyning or engrafting one Plant into another: so that there is much

variety of Vegetables, and those of such different Natures, that they are not only different Sorts, but are variety of Effects of one and the same sort; and it requires not only the study of one Human Creature, or many Human Creatures; but, of all the Human Creatures in all Nations and Ages, to know them; which is the reason, that those that have writ of the Natures of Herbs, Flowers, Roots, and Fruits, may be much mistaken. But I, living more constantly in my Study, than in my Garden, shall not venture to treat much of the particular Natures, and Natural Effects of Vegetables.

CHAP. X. Of the Production of Vegetables.

Tis no wonder, that some sorts of *Vegetables* are produced out of Stone or Brick, (as some that will grow on the top of Houses) by reason that Brick is made of Earth, and Stone is generated in the Bowels of the Earth; which shows they are of an Earthly Nature or Substance. Neither is it a wonder that Vegetables will grow upon some sorts of Water, by reason some sorts of Waters may be mixt with some Parts of Earth. But, I have been credibly informed, That a Man whose Legg had been cut, and

a Seed of an Oat being gotten into the Wound by chance, the Oat did sprout out into a green Blade of Grass: which proves, that Vegetables may be produced in several Soyls. But 'tis probable, that though many sorts of Vegetables may sprout, as Barly in Water; yet, they cannot produce any of the off-spring of the same Sort or Kind.

But, my Thoughts are, at this present, in some dispute; as, Whether the Earth is a Part of the Production of Vegetables, as being the Breeder? or, whether the Earth is only Parts of Respiration, and not Parts of Production; and so, rather Breathing-Parts, than Breeding-Parts, as Water to Fishes?

But, if so, then every particular Seed must encrease, not only by a bare Transformation of their Parts into the first Form of Production; but, by division of their united Parts, must produce many other Societies of the same sort; as Religious Orders, where one Convent divides into many Convents of the same Order; which occasions a numerous Encrease. So the several Parts of one Seed, may divide into many Seeds of the same sort, as

being of the same *Species*; but then, every Part of that Seed, must be encreased by additional Parts; which must be, by Nourishing Parts: which Nourishing Parts are, in all probability, Earthy Parts; or, at least, partly of Earthy Parts; and partly, of some of the other Elemental Parts: but, as I have often said, all Creatures in Nature are Assisted, and do Subsist, by each other.

CHAP. XI. Of Replanting Vegetables.

Replanting of Vegetables, many times, occasions great Alterations; in so much as a Vegetable, by often Replanting, will be so altered, as to appear of another sort of Vegetable: the reason is, that several sorts, or parts of Soyls, may occasion other sorts of Actions, and Orders, in one and the same Society. But this is to be noted in the Lives of many Animals, That several sorts of Food, make great alterations in their Temper and Shape; though not to alter their Species, yet so as to cause them to appear worse or better: but, this is most visible amongst Human Creatures, whom some sorts of Food will make weak, sick, faint, lean, pale, old, and withered: other sorts of Food will make them strong, and healthy, fat, fair, smooth, and ruddy. So some sorts of Soyls will cause some Vegetables to be larger, brighter, smoother, sweeter, and of more various and glorious Colours.

CHAP. XII. Of Artificial Things.

Artificial Things, are Natural Corporeal Figurative Motions: for, all Artificial Things are produced by several produced Creatures. But, the differences of those Productions we name *Natural* and *Artificial*, are, That the Natural are produced from the Producer's own Parts; whereas the Artificial are produced by composing, or joyning, or mixing several Forrein Parts; and not any of the particular Parts of their composed Society: for, Artificial things are not produced as Animals, Vegetables, Minerals, or the

like: but only, they are certain seral Mixtures of some of the divided, or dead Parts, as I may say, of Minerals, Vegetables, Elements, and the like. But this is to be noted, That all, or at least, most, are but Copied, and not Originals.

But some may ask, *Whether Artificial Productions have Sense, Reason, and Perception?*

I answer: That if all the Rational and Sensitive Parts of Nature, are Perceptive, and that no part is without Perception; then all Artificial Productions are Perceptive.

CHAP. XIII. Of several Kinds and Sorts of Species.

According to my Opinion, though the *Species* of this World, and all the several Kinds and Sorts of *Species* in this World, do always continue; yet, the particular Parts of one and the same Kind or sort of *Species*, do not continue: for, the particular Parts are perpetually altering their Figurative Actions. But, by reason some Parts compose or unite, as well as some Parts dissolve or disunite; all kinds and sorts of *Species*, will, and must last so long as Nature lasts. But mistake me not, I mean such kinds and sorts of *Species* as we name Natural, that is, the Fundamental Species; but not such *Species*, as we name Artificial.

CHAP. XIV. Of Different WORLDS.

Tis probable if Nature be Infinite, there are several kinds and sorts of those Species, Societies, or Creatures, we name *Worlds*; which may be so different from the Frame, Form, Species, and Properties of this World, and the Creatures of this World, as not to be any ways like this World, or the Creatures in this World. But mistake me not, I do not mean, not like this World, as it is Material and Self-moving; but, not of the same Species, or Properties: as for example, That they have not such kind of Creatures, or

their Properties, as Light, Darkness, Heat, Cold, Dry, Wet, Soft, Hard, Leight, Heavy, and the like.

But some may say, *That is impossible: for, there can be no World, but must be either Light or Dark, Hot or Cold, Dry or Wet, Soft or Hard, Heavy or Leight; and the like.*

I answer, That though those Effects may be generally beneficial to most of the Creatures in this World; yet, not to all the Parts of the World: as for example, Though Light is beneficial to the Eyes of Animals; yet, to no other Part of an Animal Creature. And, though Darkness is obstructive to the Eyes of Animals; yet, to no other Parts of an Animal Creature. Also, Air is no proper Object for any of the Human Parts, but Respiration. So Cold and Heat, are no proper Objects for any Part of a Human Creature, but only the Pores, which are the Organs of Touch. The like may be said for Hard and Soft, Dry and Wet: and since they are not Fundamental actions of Nature, but Particular, I cannot believe, but that there may be such Worlds, or Creatures, as may have no use of Light, Darkness, and the like: for, if some Parts of this World need them not, nor are any ways beneficial to them, (as I formerly proved) surely a whole World may be, and subsist without them: for these Properties, though they may be proper for the Form or Species of this World; yet, they may be no ways proper for the Species of another kind or sort of World: as for example, The Properties of a Human Creature are quite different from other kinds of Creatures; the like may be of different Worlds: but, in all Material Worlds, there are Self-moving Parts, which is the cause there is self-joyning, uniting, and composing; self dividing, or dissolving; self-regularities, and self-irregularities: also, there is Perception amongst the Parts or Creatures of Nature; and what Worlds or Creatures soever are in Nature, they have Sense and Reason, Life and Knowledg: but, for Light and Darkness, Hot and Cold, Soft and Hard, Leight and Heavy, Dry and Wet, and the like; they are all but particular actions of particular Corporeal Species, or Creatures, which are finite, and not infinite: and certainly, there may be, in Nature, other Worlds as full of varieties, and as glorious and beautiful as this World; and are, and may be more glorious or beautiful, as also, more full of variety than this World, and yet be quite different in all kinds and sorts, from this World: for, this is to be noted, That the different kinds and sorts of Species, or Creatures, do not make Particulars more or less perfect, but according to their kind. And one thing I

desire, That my *Readers* would not mistake my meaning, when I say, *The Parts dissolve*: for, I do not mean, that Matter dissolves; but, that their particular Societies dissolve.

APPENDIX TO THE GROUNDS OF Natural Philosophy.

The First Part.

CHAP. I. Whether there can be a Substance, that is not a Body.

What a *Substance*, that is not *Body*, can be, (as I writ in the First Chapter of this Book) I cannot imagine; nor, that there is any thing between *Something* and *Nothing*.

But, some may say, *That Spiritual Substances are so*. I answer: That Spirits must be either Material, or Immaterial: for, it is impossible for a thing to be between Body and no Body.

Others may say, *There may be a Substance, that is not a Natural Substance; but, some sort of Substance that is far more pure than the purest Natural Substance*.

I answer: Were it never so pure, it would be in the List or Circle of Body: and certainly, the purest Substance, must have the Properties of Body, as, to be divisible, and capable to be united and compounded; and being divisible and compoundable, it would have the same Properties that grosser Parts have: but, if there be any difference, certainly the purest Substance would be more apt to divide and unite, or compound, than the grosser sort. But, as to those sorts of Substance, which some Learned Men have imagined; in my opinion, they are but the same sort of Substance that the Vulgar call,

Thoughts, and I name, the *Rational Parts*; which, questionless, are as truly Body, as the grossest Parts in Nature: but, most Human Creatures are so troubled with the Thoughts of Dissolving, and Dis-uniting, that they turn Fancies and Imaginations, into Spirits, or Spiritual Substances; as if all the other Parts of their Bodies, should become Rational Parts; that is, that all their Parts should turn into such Parts as Thoughts, which I name, the *Rational Parts*. But that Opinion is impossible: for, Nature cannot alter the nature of any Part; nor can any Part alter its own Nature; neither can the Rational Parts be divided from the Sensitive and Inanimate Parts, by reason those Three sorts constitute but one Body, as being Parts of one Body. But, put the case that the Rational Parts might divide and subsist without the Sensitive and Inanimate Parts; yet, as I said, they must of necessity have the Properties and Nature of a Body, which is, to be divisible, and capable to be united, and so to be Parts: for, it is impossible for a Body, were it the most pure, to be indivisible.

CHAP. II. Of an IMMATERIAL.

I Cannot conceive how an Immaterial can be in Nature: for, first, An Immaterial cannot, in my opinion, be naturally created; nor can I conceive how an Immaterial can produce particular Immaterial Souls, Spirits, or the like. Wherefore, an Immaterial, in my opinion, must be some uncreated Being; which can be no other than GOD alone. Wherefore, Created Spirits, and Spiritual Souls, are some other thing than an Immaterial: for surely, if there were any other Immaterial Beings, besides the Omnipotent God, those would be so near the Divine Essence of God, as to be petty gods; and numerous petty gods, would, almost, make the Power of an Infinite God. But, God is Omnipotent, and only God.

CHAP. III. Whether an Immaterial be Perceivable.

Whatsoever is Corporeal, is Perceivable; that is, may be perceived in some manner or other, by reason it hath a Corporeal Being: but, what Being an Immaterial hath, no Corporeal can perceive. Wherefore, no Part in Nature can perceive an Immaterial, because it is impossible to have a perception of that, which is not to be perceived, as not being an Object fit and proper for Corporeal Perception. In truth, an Immaterial is no Object, because no Body.

But some may say, that, *A Corporeal may have a Conception, although not a Perception, of an Immaterial.*

I answer, That, surely, there is an innate Notion of God, in all the Parts of Nature; but not a perfect knowledg: for if there was, there would not be so many several Opinions, and Religions, amongst one Kind, or rather, sort of Creatures, as Mankind, as there are; insomuch, that there are but few of one and the same Opinion, or Religion: but yet, that Innate Notion of God, being in all the Parts of Nature, God is infinitely and eternally worshipped and adored, although after several manners and ways; yet, all manners and ways, are joyned in one Worship, because the Parts of Nature are joyned into one Body.

CHAP. IV. Of the Differences between God, and Nature.

God is an Eternal Creator; Nature, his Eternal Creature. GOD, an Eternal Master: Nature, GOD's Eternal Servant. GOD is an Infinite and Eternal Immaterial Being: Nature, an Infinite Corporeal Being. GOD is Immovable, and Immutable: Nature, Moving, and Mutable. GOD is Eternal, Indivisible, and of an Incompoundable Being: Nature, Eternally Divisible and Compoundable. GOD, Eternally Perfect: Nature, Eternally Imperfect. GOD, Eternally Inalterable: Nature Eternally Alterable. GOD, without Error: Nature, full of Irregularities. GOD knows exactly, or perfectly, Nature: Nature doth not perfectly know GOD. GOD is Infinitely and Eternally worshipped: Nature is the Eternal and Infinite Worshipper.

CHAP. V. All the Parts of Nature worship God.

All Creatures (as I have said) have an Innate Notion of GOD; and as they have a Notion of God, so they have a Notion to worship GOD: but, by reason Nature is composed of Parts; so is the Infinite Worship to God: and, as several Parts are dividing and uniting after several kinds, sorts, manners and ways; so is their Worship to GOD: but, the several manners and ways of Worship, make not the Worship to GOD less: for certainly, all Creatures Worship and Adore GOD; as we may perceive by the Holy Scripture, where it says, *Let the Heavens, Earth, and all that therein is, praise God.* But 'tis probable, that some of the Parts being Creatures of Nature, may have a fuller Notion of GOD than others; which may cause some Creatures to be more Pious and Devout, than others: but, the Irregularity of Nature, is the cause of Sin.

CHAP. VI. Whether GOD's Decrees are limited.

In my opinion, though God is Inalterable, yet no ways bounded or limited: for, though GOD's Decrees are fixt, yet, they are not bound: but, as GOD hath an Infinite Knowledg, He hath also an Infinite Fore-knowledg; and so, fore-knows Nature's Actions, and what He will please to decree Nature to do: so that, GOD knows what Nature can act, and what she will act; as also, what He will decree: and this is the cause, that some of the Creature's or Parts of Nature, especially Man, do believe *Predestination*. But surely, GOD hath an Omnipotent Divine Power, which is no ways limited: for GOD, being above the nature of Nature, cannot have the Actions of Nature, because GOD cannot make Himself no GOD; neither can He make Himself more than what he is, He being the All-powerful, Omnipotent, Infinite, and Everlasting Being.

CHAP. VII. Of GOD's Decrees concerning the particular Parts of Nature.

Though Nature's Parts have Free-will, of Self-motion; yet, they have not Free-will to oppose *GOD's Decrees*: for, if some Parts cannot oppose other Parts, being over-power'd, it is probable, that the Parts of Nature cannot oppose the All-powerful Decrees of *GOD*. But, if it please the All-powerful *GOD* to permit the Parts of Nature to act as they please, according to their own natural Will; and, upon condition, if they act so, they shall have such Rewards as Nature may be capable to receive; or such Punishments as Nature is capable of; then the Omnipotent *GOD* doth not predestinate those Rewards, or Punishments, any otherwise than the Parts of Nature do cause by their own Actions. Thus all Corporeal Actions, belong to Corporeal Parts; but, the Rewards and Punishments, to *GOD* alone: but, what those Punishments and Blessings are, no particular Creature is capable to know: for, though a particular Creature knows there is a *GOD*; yet, not what *GOD* is: so, although particular Creatures know there are Rewards and Punishments; yet, not what those Rewards and Punishments are. But mistake me not; for I mean the general Rewards and Punishments to all Creatures: but 'tis probable, that *GOD* might decree Nature, and her Parts, to make other sorts of Worlds, besides this World; of which Worlds, this may be as ignorant, as a particular Human Creature is of *GOD*. And therefore, it is not probable (since we cannot possibly know all the Parts of Nature, of which we are parts) that we should know the Decrees of *GOD*, or the manners and ways of Worship, amongst all kinds and sorts of Creatures.

CHAP. VIII. Of the Ten Commandments.

In my opinion, the Notions Man hath of *GOD's Commands* concerning their Behaviour and Actions to Himself, and their Fellow-Creatures, is the very same that Moses writ, and presented to all those of whom he was Head and Governour. But, mistake me not, I mean only the *Ten Commandments*; which Commandments are a sufficient Rule for all Human Creatures: and certainly, *GOD* had decreed, that Moses should be a wise Man, and should

publish these wise Commands. But, the Interpretation of the Law must be such, as not to make it no such Law: but, by reason Nature is as much Irregular, as Regular, Human Notions are also Irregular, as much as Regular; which causes great variety of Religions: and their Actions being also Irregular, is the cause that the practise of Human Creatures is Irregular; and that occasions Irregular Devotions, and is the cause of SIN.

CHAP. IX. Of Several RELIGIONS.

Concerning *the several Religions*, and several Opinions in Religions, which are like several Kinds and Sorts; the Question is,

Whether all Mankind could be perswaded to be of one Religion, or Opinion?

The opinion of the Minor part of my Thoughts, was, That all men might be perswaded.

And, the opinion of the Major part of my Thoughts, was, That Nature, being divisible and compoundable, and having Free-will, as well as Self-motion; and being Irregular, as well as Regular; as also, Variable, taking delight in variety; it was impossible for all Mankind to be of one *Religion*, or *Opinion*.

The opinion of the Minor part of my Thoughts, 246,\$ was, That the Grace of GOD could perswade all Men to one Opinion.

The Major part of my Thoughts was of opinion, That GOD might decree or command Nature: but, to alter Nature's nature, could not be done, unless GOD, by his Decree, would annihilate this Nature, and create another Nature, and such a Nature as was not like this Nature: for, it is the nature of this Material Nature, to be Alterable; as also, to be Irregular, as well as Regular; and, being Regular, and Irregular, was a fit and proper Subject for GOD's Justice, and Mercies; Punishments, and Rewards.

CHAP. X. Of Rules and Prescriptions.

As Saint Paul said, *We could not know Sin, but by the Law*; so, we could not know what Punishment we could or should suffer, but by the Law; not only Moral, but Divine Law.

But, some may ask, *What is Law*?

I answer: Law is, Limited Prescriptions and Rules.

But, some may ask, *Whether all Creatures in Nature, have Prescriptions and Rules*?

I answer: That, for any thing Man can know to the contrary, all Creatures may have some Natural Rules: but, every Creature may chuse whether they will follow those Rules; I mean, such Rules as they are capable to follow or practise: for, several kinds and sorts of Creatures, cannot possibly follow one and the same Prescription and Rule. Wherefore, Divine Prescriptions and Rules, must be, according to the sorts and kinds of Creatures; and yet, all Creatures may have a Notion, and so an Adoration of God, by reason all the Parts in Nature, have Notions of God. But, concerning particular Worships, those must be Prescriptions and Rules; or else, they are according to every particular Creature's conception or choice.

CHAP. XI. Sins and Punishments, are Material.

As all Sins are Material, so are Punishments: for, Material Creatures, cannot have Immaterial Sins; nor can Material Creatures be capable of Immaterial Punishments; which may be proved out of the Sacred Scripture: for, all the Punishments that are declared to be in Hell, are Material Tortures: nay, Hell it self is described to be Material; and not only Hell, but Heaven, is described to be Material. But, whether Angels, and Devils, are Material, that is not declared: for, though they are named Spirits, yet we know not whether those Spirits be Immaterial. But, considering that Hell and Heaven is described to be

Material, it is probable, Spirits are also Material: nay, our blessed Saviour Christ, who is in Heaven, with God the Father, hath a Material Body; and in that Body will come attended by all the Hosts of Heaven, to judg the quick and the dead; which quick and dead, are the Material Parts of Nature: which could not be actually judged and punished, but by a Material Body, as Christ hath. But, pray mistake me not; I say, They could not be actually judged and punished; that is, not according to Nature, as Material Actions: for, I do not mean here, Divine and Immaterial Decrees. But Christ, being partly Divine, and partly Natural; may be both a Divine and Natural Judg.

CHAP. XII. Of Human Conscience.

The Human Notions of GOD, Man calls *Conscience*: but, by reason that Nature is full of Varieties, as having Self-moving Parts; Human Creatures have different Notions, and so different Consciences, which cause different Opinions and Devotions: but, Nature being as much compoundable as dividable, it causes unity of some, as also, divisions of other Opinions, which is the cause of several Religions: which Religions, are several Communities and Divisions. But, as for Conscience, and holy Notions, they being Natural, cannot be altered by force, without a Free-will: so that the several Societies, or Communicants, commit an Error, if not a Sin, to endeavour to compel their Brethren to any particular Opinion: and, to prove it is an Error, or Sin, the more earnest the *Compellers* are, the more do the *Compelled* resist; which hath been the cause of many Martyrs. But surely, all Christians should follow the Example of Christ, who was like a meek Lamb, not a raging Lyon: neither did Christ command his Apostles to Persecute; but, to suffer Persecution patiently. Wherefore, *Liberty of Conscience* may be allowed, conditionally, it be no ways a prejudice to the Peaceable Government of the State or Kingdom.

The Second Part.

CHAP. I. Whether it is possible there could be Worlds consisting only of the Rational Parts, and others only of the Sensitive Parts.

The Parts of my Mind did argue amongst themselves, *Whether there might not be several kinds and sorts of Worlds in Infinite Nature?*

And they all agreed, That probably there might be several kinds and sorts of Worlds.

But afterwards, the Opinion of the Major parts of my Mind, was, That it is not possible: for, though the Rational parts of Nature move free, without Burdens of the Inanimate Parts; yet, being Parts of the same Body, (viz. of the Body of Nature) they could not be divided from the Sensitive and Inanimate Parts; nor the Sensitive and Inanimate Parts, from the Rational.

The Opinion of the Minor Parts of my Mind, was, That a Composed World, of either degree, was not a division from the Infinite Body of Nature, though they might divide so much, as to compose a World meerly of their own Degree.

The Major's Opinion was, That it was impossible; because the three Degrees, Rational, Sensitive, and Inanimate, were naturally joyned as one Body, or Part.

The Minor's Opinion was, That a World might be naturally composed only of Rational Parts, as a Human Mind is only composed of Rational Parts; or, as the Rational Parts of a Human Creature, could compose themselves into several Forms, viz. into several sorts and kinds of Worlds, without the assistance of the Sensitive or the Inanimate Parts: for, they fancy Worlds which are composed in Human Minds, without the assistance of the Sensitive.

The Major Part agreed, That the Rational Corporeal Actions, were free; and all their Architects were of their own Degree: but yet, they were so joyned in every Part and Particle, to the Sensitive and Inanimate, as they could not separate from these two Degrees: for, though they could divide and unite from, and to Particulars, as either of their own Degrees, or the other Degrees; yet, the Three Degrees being but as one united Body, they could

not so divide, as not to be joyned to the other Degrees: for, it was impossible for a Body to divide it self from it self.

After this Argument, there followed another; *That, if it were possible there could be a World composed only of the Rational Parts, without the other two Degrees; Whether that World would be a Happy World?*

The Major Part's Opinion was, That, were it possible there could be such unnatural Divisions, those divide Parts would be very unhappy: for, the Rational Parts would be much unsatisfied without the Sensitive; and the Sensitive very dull without the Rational: also, the Sensitive Architects would be very Irregular, wanting their Designing Parts, which are the Rational Parts.

Upon which Argument, all the Parts of my Mind agreed in this Opinion, That the Sensitive was so Sociable to the Rational, and the Rational so Assisting to the Sensitive, and the Inanimate Parts so necessary to the Sensitive Architects, that they would not divide from each other, if they could.

CHAP. II. Of Irregular and Regular Worlds.

Some Parts of my Mind were of opinion, *That there might be a World composed only of Irregularities; and another, only of Regularities: and some, that were partly composed of the one, and the other.*

The Minor Part's Opinion was, That all Worlds were composed partly of the one, and partly of the other; because all Nature's Actions were poysed with Opposites, or Contraries: wherefore, there could not be a World only of Irregularities, and another of Regularities.

The Major Part's Opinion, was, That Nature's Actions were as much poysed by the contrary Actions of Two Worlds, as by the contrary Actions of the Parts of One World, or one Creature: As for example, The Peace and Trouble, Health and Sickness, Pain and Ease, and the like, of one Human Creature; and so of the contrary Natures of several kinds and sorts of Creatures of one and the same World.

After which Discourse, they generally agreed, There might be Regular and Irregular Worlds; the one sort to be such happy Worlds, as that they might be named Blessed Worlds; the other so miserable Worlds, as might be named Cursed Worlds.

CHAP. III. Whether there be Egress and Regress between the Parts of several Worlds.

There arose a Third Argument, *viz. Whether it was possible for some of the Creatures of several Worlds, to remove, so as to remove out of one World, into another?*

The Major Part's Opinion was, That it was possible for some Creatures: for, if some particular Creatures could move all over the World, of which they were a part, they might divide from the Parts of the World they were of, and joyn with the Parts of another World.

The Minor Part's Opinion was, That they might travel all over the World they were part of, but not to joyn with the Parts of another World, to which they belong not.

The Major's Opinion was, That every Part and Particle, belonged to the Infinite Body of Nature, and therefore not any Part could account it self not of the Infinite Body; and being so, then every Part of Nature may joyn, and divide from and to particular Parts, as they please, if there were not Obstructions and Hindrances, and some Parts did not obstruct other Parts: Wherefore, if there were not Obstructions, there might be Egress and Regress amongst the particular Parts of several Worlds.

The Minor's Opinion was, That if it could be according to the Major's Opinion, it would cause an Infinite Confusion in Infinite Nature: for, every Creature of every World, was composed according to the Nature and Compositions of the World they were of: wherefore, the Products of one kind or sort of Worlds, would not be sutable, agreeable, and Regular, to the productions of another kind.

The Major Part's Opinion was, That it was impossible, since Nature is one united Body, without *Vacuum*, but that the Parts of all Worlds must have Egress and Regress.

CHAP. IV. Whether the Parts of one and the same Society, could, after their Dissolution, meet and unite.

The Fifth Argument, was partly of the same Subject, viz. *Whether the particular Parts of a Creature, (such as a Human Creature is) could travel out of one World into another, after the dissolution of his Human Life?*

The Major Part's Opinion was, That they could.

The Minor's Opinion was, They could not; because the particular Parts so divided and joyned to and from other particular Parts and Societies, as it was impossible, if they would, so to agree, as to divide from those Parts and Societies they are joyned to, and from those they must joyn with, to meet in another World, and joyn as they would, in the same Society they were of, when the whole Society is dissolved. Neither can Parts divide and joyn, as they would: for, though Self-moving Parts have a Free-will to move; yet, being subject to Obstructions, they must move as they can: for, no particular Part hath an absolute Power. Wherefore, the Dispersed Parts of a Dissolved Society, cannot meet and joyn as they would. Besides, every Part is as much affected to one Sort, Kind, or Particular, they are Parts of, as to another. Besides, the Knowledg of every Part alters, according as their Actions alter: so that the Parts of one and the same Society, after division, have no more knowledg of that Society.

CHAP. V. Whether, if a Creature being Dissolved, and could Unite again, would be the same.

The Sixth Argument was, *That, put the case it were possible all the several Parts belonging to one and the same Society; as for example, To one Human Creature, after his Human Life was dissolved, and his Parts dispersed, and afterwards, all those Parts meeting and uniting; Whether that Human Creature would be the same?*

The Minor Part's Opinion was, That it could not be the same Society: for, every Creature was according to the nature of their Kind or Sort; and so according to the Form and Magnitude of one of their Kind or Sort.

The Major Part's Opinion was, That though the Nature of every particular Creature had such Forms, Shapes, and Properties, as was natural to that sort of Creatures they were of; yet, the Magnitude of particular Creatures of one and the same sort, might be very different.

The Minor Part's Opinion was, That if all the Parts of one Society, as for example, a Man, from the first time of his Production, to the time of his Dissolution, should, after division, come to meet and unite; that Man, or any other Creature, would be a Monstrous Creature, as having more Parts than was agreeable to the nature of his Kind.

The Major Part's Opinion was, That though the Society, viz. The MAN, would be a Society of greater Magnitude; yet, not any ways different from the Nature of his Kind.

CHAP. VI. Of the Resurrection of Human Kind.

The Seventh Argument, was, *Whether all the particular Parts of every Human Creature, at the time of the Resurrection, be, to meet and joyn, as being of one and the same Society?*

The Minor Part's Opinion was, They shall not: for, if all those Parts that had been of the same Body and Mind of one Man, from his first Production, to the last of his Dissolution; or, from his Birth, to the time of his Death, (supposing him to have liv'd long) should meet and joyn, as one Society, that is, as one Man; that Man, at the time of his Resurrection, would be a Gyant; and if so, then old Men would be Gyants; and young Children, Dwarfs.

The Major Part's Opinion was, That, if it was not so, then every particular Human Society would be imperfect at the time of their Resurrection: for, if they should only rise with some of their Parts, as (for example) when they were in the strength of their Age, then all those Parts that had been either before, or after that time, would be unjustly dealt with, especially if Man be the best Product in Nature. Besides, if a dead Child did rise a Man, as at his most perfect Age, it could not be said, He rises according to a Natural Man,

having more Parts than by Nature he ever had; and an old Man, fewer Parts than naturally he hath had: so, what by Adding and Diminishing the Parts of particular Men, it would not cause only Injustice; but, not any particular Human Creature, would be the same he was.

CHAP. VII. Of the Dissolution of a World.

The Eighth Argument was, *That when all Human Creatures that were dissolved, should rise, Whether the World they were of, should not be dissolved?*

All the Parts of my Mind agreed, That when all the Human Creatures that had been dissolved, should rise, the whole World, besides themselves, must also dissolve, by reason they were Parts of the World: for, when all those numerous dissolved and dispersed Parts, did meet and joyn, the World wanting those Parts, could not subsist: for, the Frame, Form, and Uniformity of the World, consisted of Parts; and those Parts that have been of the Human Kind, are, at several times, of other kinds and sorts of Creatures, as other sorts and kinds are of Human Kind; and all the Sorts and Kinds, are Parts of the World: so that the World cannot subsist, if any kind or sort of Creatures, that had been from the first time of the Creation, should be united; I mean, into one and the same sort or kind of Creatures; as it would be, if all those that are Quick, and those that have been *Dissolved*, (that is, have been dead) should be alive at one time.

CHAP. VIII. Of a New Heaven, and a New Earth.

The Ninth Argument was, *That if a World could be dissolved, and that the Human Creatures should rise, and reunite; what World should they reside in?*

All the Parts of my Thoughts generally agreed, That the Omnipotent GOD would command the Parts of His Servant NATURE, to compose other

Worlds for them, into which Worlds they should be separated; the Good should go into a Blessed World; the Bad, into a Cursed World: and the Sacred Scripture declares, That there shall be a *New Heaven*, and a *New Earth*; which, in their opinion, was a Heaven and a Hell, for the Blessed and Cursed Human Kind of this World.

CHAP. IX. Whether there shall be a Material Heaven and Hell.

The Tenth Argument was, *Whether the Heaven and Hell that are to be produced for the Blessed and Cursed, shall be Material?*

The Minor Part's Opinion was, That they shall not be Material.

The Major Parts were of opinion, They shall be Material, by reason all those Creatures that did rise, were Material; and being Material, could not be sensible either of Immaterial Blessings, or Punishments: neither could an Immaterial World, be a fit or proper Residence for Material Bodies, were those Bodies of the purest Substance. But, whether this Material Heaven and Hell, shall be like other Material Worlds, the Parts of my Mind could not agree, and so not give their Judgment. But, in this they all agreed, That the Material Heaven and Hell, shall not have any other Animal Creatures, than those that were of Human kind, and those not produced, but raised from Death.

But when they came to argue, Whether there might be Elements, Minerals, and Vegetables, they could not agree; but some did argue, and offer to make proof, That there might be Mines of Gold, and Rocks of Diamonds, Rubies, and the like; all which, were Minerals. Also, some were of opinion, there were Elements: for, Darkness and Light, are Elemental Effects: and, if Hell was a World of Darkness; and Heaven, a World of Light; it was probable there were Elements.

CHAP. X. Concerning the Joys or Torments of the Blessed and Cursed, after they are in Heaven, or Hell.

As for the *Joys of Heaven*, and the *Torments of Hell*, all the Parts of my Mind agreed, they could not conceive any more probably, than those they had formerly conceived: which former Conceptions they had occasioned the Sensitive Parts to declare; and having been formerly divulged in the Book of my *Orations*, their Opinion was, *That it would be a superfluous Work to cause them to be repeated in this Book*. But, the Ground or Foundation of those Conceptions, is, That God may decree, *That both the Sensitive and Rational Parts of those that are restored to Life, should move in variety of Perceptions, or Conceptions, without variety of Objects: and, that those Creatures (viz. Human Creatures) that are raised from Death to Life, should subsist without any Forrein Matter, but should be always the same in Body and Mind, without any Traffick, Egress, or Regress of Forrein Parts*. And the proof, that the Sensitive and Rational Parts of Human Creatures, may make Perceptions, or rather Conceptions, without Forrein Objects, is, *That many men in this world have had Conceptions, both amongst the Rational and Sensitive, which Man names Visions, or Imaginations; whereof some have been Pleasing and Delightful; others, Displeasing, and Dreadful*.

The Third Part.

The PREAMBLE.

The Parts of my Mind, after some time of respite from *Philosophical Arguments*, delighting in such harmless Pastimes; did begin to argue about a *Regular and Irregular World*; having formerly agreed, there might be such Worlds in Nature; and that the Regular Worlds, were Happy Worlds; the Irregular, Miserable Worlds. But, there was some division amongst the Parts

of my Mind, concerning the choice of their Arguments; as, Whether to argue, first, of the particular Parts of the Regular, or of the Irregular World. But, at last, they agreed to argue, first, of the Regular World. But, pray mistake not these Arguments; for they are not Arguments of such Worlds as are for the reception of the Blessed and Cursed Humans, after their Resurrections: but, such as these Worlds we are of, only freely Regular, or Irregular. Also, though I treat but only of one Regular World, and one Irregular World; yet, my opinion is, there may be a great many Irregular Worlds, and a great many Regular Worlds, of several kinds and sorts: but, these I shall treat of, are such as are somewhat like this World we are of.

CHAP. I. Of the Happy and Miserable Worlds.

The First Argument was, *Whether there might not be such Worlds in Nature, as were in no kind or sort like this World we are of?*

They all agreed, That it was probable there was.

The Second Argument was, *Whether it was probable that the Happy and Miserable Worlds were, in any kind, like this we are of.*

They all agreed, It was probable that this World was somewhat like both one, and the other; and so, both those were somewhat like this: for, as the *Happy World* was no ways Irregular; and the *Miserable World* no ways Regular: so this World we are of, was partly Irregular, and partly Regular; and so it was a *Purgatory World*.

CHAP. II. Whether there be such kinds and sorts of Creatures in the Happy and Blessed World, as in this World.

The Third Argument was, *Whether it was probable, the Happy and Miserable Worlds, had Animal, Vegetable, Mineral, and Elemental Kinds?*

They agreed, It was probable there were such Kinds: but yet, those Kinds, and particular sorts of those Kinds, might be different from those of this World.

The Fourth Argument was, *Whether there was Human sorts of Creatures in those Worlds.*

They all agreed, There was.

CHAP. III. Of the Births and Deaths of the Heavenly World.

The Fifth Argument was, *Whether there could be Births and Deaths in the Happy World?*

Some Parts of my Mind were of opinion, That if there was so Regular a World, as that there were no Irregularities in it, there could not be *Deaths*: for, Death was a Dissolution; and if there was no Death, there could be no Birth, or Production: for, if any particular sort of Creatures should Encrease, and never dissolve, they would become Infinite; which every particular kind or sort of Creatures, may be, for time, and be Eternal; as also, be Infinite for Number; because, as some dissolve, others are produced. And so, if particular sorts or kinds of Creatures, be Eternal; the particular Production and Dissolution, is Infinite: but, if any Sort, or Kind, should encrease, without decrease, not any particular World could contain them: As for example, If all the Human Creatures that have been produced from our Father *Adam*, (which hath not been above Six thousand years) should be alive, this World could not contain them; much less, if this World, and the Human sorts of Creatures, had been of a longer date. And besides, if there should be a greater Encrease, by the Number of Human Creatures: in truth, the numerous Encrease, would have caused Mankind, in the space of Six thousand years, to be almost Infinite.

But, the Minor Parts of my Mind was of opinion, That then the *Happy World* could not be so perfectly Regular, if there was Death.

The Major Part's opinion was, That some sorts of Deaths were as regular, as the most Regular Births: for, though Diseases were caused by Irregular

Actions, yet, Death was not: for, as it is not Irregular, to be old; so it is not Irregular, to dye. But, this Argument broke off for that time.

CHAP. IV. Whether those Creatures could be named Blessed, that are subject to dye.

The Sixth Argument was, *Whether those Creatures could be called Blessed, or Happy, that are subject to dye?*

The Major Parts of my Mind was of opinion, That, if Death was as free from Irregularities, as Birth; then it was as happy to Dye, as to be Born.

The Minor Parts were of opinion, That though Dissolution might be as Regular as Composition; yet, it was an Unhappiness for every particular Society, to be dissolved.

The Major Part's Opinion, was, That though the particular Societies were dissolved; yet, by reason the general Society of the Kind, did continue, it was not so much Unhappiness; considering, particular Parts, or Creatures, did make the General Society; and not, the General, the Particular Societies: so that, the Parts of the Particulars, remained in the General, as in the Kind of Sort.

The Minor Parts were of opinion, That the Particulars of the same Kind or Sort, (as *Mankind*) did contribute but little to the General: for, other sorts of Creatures did contribute more than they; only Mankind was the Occasion, or Contributor of the First Foundation, but no more: but, the other Parts or Creatures of the World, did contribute more to their Kind, than the Creatures of the same Kind did: and, as other Kinds, and Sorts, did contribute to Mankind; so Mankind, to other Kinds or Sorts: for, all Kinds and Sorts, did contribute to the Subsistence and Assistance of each other.

The Major Part's Opinion was, That if all the Parts of a World did assist each other, then Death could be no Unhappiness, especially in the Regular World; by reason all Creatures in that World, of what Kind or Sort soever, was Perfect and Regular: so that, though the particular Human Creatures did dissolve from being Humans; yet, their Parts could not be Unhappy,

when they did unite into other Kinds, and Sorts, or particular Societies: for, those other sorts and kinds of Creatures, might be as happy as Human Creatures.

CHAP. V. Of the Productions of the Creatures of the Regular World.

The Seventh Argument was, of Productions of the Creatures of the Regular World, *viz. Whether their Productions were frequent, or not?*

The Minor Part's Opinion, was, That they were frequent.

The Major Part's Opinion, was, That they were not *frequent*, or *numerous*, by reason the World was Regular, and so all the Productions or Generations, were Regular; but could not exceed such a Number as was, regularly, sufficient for a World, of such a Dimension as the Regular World; and according to the Dimensions, must the Society or Creatures be, let them be large or little.

CHAP. VI. Whether the Creatures in the Blessed World, do Feed, and Evacuate.

The Eighth Argument, was, *Whether the Blessed Humans, in the Happy World, did Eat, and Evacuate?*

They agreed, That, if they did feed, they must evacuate.

Then there was a Dispute, *Whether those Happy Creatures did eat?*

They all agreed, That, if they were Natural Human Creatures, they had Natural Appetites: but, by reason there were no Irregularities in this World, the Human Creatures had not any Irregular Appetites, nor Irregular Digestions; Irregular Passions, or Irregular Pastimes.

Then there arose a Dispute, *Whether those Blessed Creatures did sleep?*

Some were of opinion, They did not sleep: for, Sleep was occasioned through a weariness of the Sensitive Organs, making perceptions of Forrein Objects; and all Weariness, or Tiredness, was Irregular.

The Major part of my Mind, was of a contrary Opinion; because the delight of Nature, is in Variety: and therefore, Regular Sleeps were delightful.

The Minor was of opinion, That Sleep was like Death, and therefore it could not be Happy.

But, at last, they did conclude, That Sleep, being a soft and quiet Repose, (as being retired from all Actions concerning Forrein Parts, and had only Actions at Home, and of private Affairs; and that all the Parts of Body and Mind, were then most sociable amongst themselves) that the Blessed Humans did sleep.

CHAP. VII. Of the Animals, and of the Food of the Humans of the Happy World.

The Ninth Argument, was, *Whether there were all sorts of Animals in the Regular World?* All the Parts of my Mind agreed, That if there were such Creatures as Human Creatures, it was probable there was other Animal Creatures: but, by reason there was no Irregularities, there could not be Cruel or Ravenous Animal Creatures: for, a Lyon, Leopard, or Wolf, in that World, would be as harmless as a Sheep in this; and all Kites, Hawks, and the like ravenous Birds, would be as harmless as those Birds that only feed on the Berries, and Fruit of the Earth.

CHAP. VIII. Whether it is not Irregular, for one Creature to feed on another.

The Tenth Argument was, *Whether it was not Irregular, for one Creature to feed on another?*

Some were of opinion, That it was natural for one Creature to subsist by another, and to assist each other; but not cruelly to destroy each other.

Upon this Argument, the Parts of my Mind divided into a Minor and a Major part.

The Minor Part's opinion, was, That, since all the Creatures in Nature, had Life; then, all Creatures that did feed, did destroy each other's Life.

The Major Part's Opinion, was, That they might be assisted by the Lives of other Creatures, and not destroy their Lives: for, Life could not be destroyed, though Lives might be occasionally alter'd: but, some Creatures may assist other Creatures, without destruction or dissolution of their Society: as for example, The Fruits and Leaves of Vegetables, are but the Humorous Parts of Vegetables, because they are divisible, and can encrease and decrease, without any dissolution of their Society; that is, without the dissolution of the Plant. Also, Milk of Animals, is a superfluous Humor of Animals: and, to prove it to be a superfluous Humor, I alledg, That much of it oppresses an Animal. The same I say of the Fruits and Leaves of many sorts of Vegetable Creatures. Besides, it is natural for such sorts of Creatures to have their Fruits and Leaves to divide from the Stock.

The Minor Part's Opinion, was, That the Milk of Animals, and the Fruits of Vegetables, and the Herbs of the Earth, had as much Life as their Producers.

The Major Part's Opinion, was, That though they had as much Life as their Producers; yet, it was natural for such off-springs to change and alter their Lives, by being united to other sorts of Creatures: as for example, An Animal eats Fruit and Herbs; and those Fruits and Herbs convert themselves into the nature of those Animals that feed of them. The same is of Milk, Eggs, and the like; out of which, a condition of Life is endeavoured for: and, for proof, such sorts of Creatures account an Animal Life the best; and therefore, all such superfluous Parts of Creatures, endeavour to unite into an Animal Society; as we may perceive, that Fruits and Herbs, are apt to turn into Worms, and Flies; and some Parts of Milk, as Cheese, will turn into Maggots; so that when Animals feed of such Meats, they occasion those Parts they feed on, to a more easie Transformation; and not only such Creatures, but Humans also, desire a better Change: for, what Human would not be a glorious Sun, or Starr?

After which Discourse, all the Parts of my Mind agreed unanimously, That Animals, and so Human Creatures, might feed on such sorts of Food, as aforesaid; but not on such Food as is an united Society: for, the Root and Foundation of any kind and sort of Creature, ought not to be destroyed.

CHAP. IX. Of the Continuance of Life in the Regular World.

The Opinion of the Parts of my Mind, was, That, it was probable, that all Societies in the Regular World, (that is, all such Parts of Nature as are united into particular Creatures) are of long life, by reason there are no Irregularities to destroy them, before their natural time.

But then a Dispute was raised amongst the Parts of my Mind, concerning the natural time, that is, the proper time of the Lives of those Creatures: for, all Creatures were not of the same time of Production; nor, after their Production, of the same time of Continuance. But the Parts of my Mind concluded, That though they could not judg by observation of any Creature, no, not of their own Sort; yet, they did

believe they could judg better of Human Creatures, as being, at that time, of a Human Society, than of any other: but, by reason they were of this World (that is, Irregular in part) they did believe they might very much err in their Judgment, concerning the continuance of Human Lives, in the *Happy World*. But, after much debate, they concluded, That a Human Creature, in the Regular World, might last as long as the Productions did not oppress or burden that World, (for that would be irregular) but how long a time that might be, they could not possibly conceive or imagine.

CHAP. IX. Of the Excellency and Happiness of the Creatures of the Regular World.

The Parts of my Mind could not possibly, being Parts of a Purgatory World, conceive the happy condition of all Creatures in the Regular World; but

only, conceiving there was no Irregularities, they did also conceive, that all Creatures there, must be in perfection; and that the Elemental Creatures were purer, without drossie mixtures; so that their Earth must needs be so fruitful, that it produces all sorts of excellent Vegetables, without the help of Art; and their Minerals as pure, as all sorts of Stone that are transparent, and as hard as Diamonds; the Gold and Silver, more pure than that which is refined in our World. The truth is, that, in their Opinions, the meanest sorts of Metal in the Regular World, were more pure than the richest sort in this World: so that then, their richest Metal must be as far beyond ours, as our Gold is beyond our Iron, or Lead. As for the Elemental Waters in the Regular World, they must be extraordinary smooth, clear, flowing, fresh, and sweet; and the Elemental Air only, a most pure, clear, and glorious Light; so that there could be no need of a Sun: and, by reason all the Air was a Light, there could be no Darkness; and so, no need of a Moon, or Starrs. The Elemental Fire, although it was Hot, yet it was not Burning. Also, there could neither be scorching Heats, nor freezing Colds, Storms, nor Tempest: for, all Excess is irregular. Neither could there be Clouds, because no Vapours. But, not to be tedious; it was my Mind's Opinion, That all the Parts of the Happy World, being Regular, they could not obstruct each other's Designs or Actions; which might be a cause, that both the Sensitive and Rational Parts may not only make their Societies more curious, and their Perceptions more perfect; but their Perceptions more subtile: for, all the actions of that World being Regular, must needs be exact and perfect; in so much, that every Creature is a perfect Object to each other; and so every Creature must have, in some sort, a perfect Knowledg of each other.

CHAP. XI. Of Human Creatures in the Regular World.

The Opinion of my Mind, was, That the *Happy World*, having no Irregularities, all Creatures must needs be Excellent, and most Perfect, according to their Kind and Sort; amongst which, are Human Creatures, whose Kinds, or Sorts, being of the Best, must be more excellent than the rest, being Exactly formed, and Beautifully produced: there being, also, no

Irregularities, Human Creatures cannot be subject to Pains, Sickness, Aversions, or the like; or, to Trepidations, or Troubles; neither can their Appetites, or Passions, be irregular: wherefore, their Understanding is more clear, their Judgments more poysed: and by reason their Food is Pure, it must be Delicious, as being most tastable: also, it must be wholesome, and nourishing; which occasions the Parts of Body and Mind, to be more Lively and Pleasant.

CHAP. XII. Of the Happiness of Human Creatures in the Material World.

The Happiness that Human Creatures have in the *Regular World*, is, That they are free from any kind or sort of Disturbance, by reason there are

no Irregular Actions; and so, no Pride, Ambition, Faction, Malice, Envy, Suspition, Jealousie, Spight, Anger, Covetousness, Hatred, or the like; all which, are Irregular Actions among the Rational Parts: which occasions Treachery, Slander, false Accusations, Quarrels, Divisions, Warr, and Destruction; which proceeds from the Irregularities of the Sensitive Parts, occasioned by the Rational, by reason the Sense executes the Mind's Designs: but, there are no Plots or Intrigues, neither in their State, nor upon their Stage; because, though they may act the parts of Harmless Pleasures; yet, not of Deceitful Designs: for, all Human Creatures, live in the Regular World, so united, that all the particular Human Societies, (which are particular Human Creatures) live as if they were but one Soul, and Body; that is, as if they were but one Part, or particular Creature. As for their pleasures, and pleasant Pastimes; in my opinion, they are such, as not any Creature can express, unless they were of that World, or Heaven: for, all kinds and sorts of Creatures, and all their Properties or Associations, in this World we are of, are mixt; as, partly Irregular; and partly, Regular; and so it is but a *Purgatory-World*. But surely, all Human Creatures of that World, are so pleasant and delightful to each other, as to cause a general Happiness.

The Fourth Part.

CHAP. I. Of the Irregular World.

After the Arguments and Opinions amongst the Parts of my Mind, concerning a Regular World; their Discourse was, of an *Irregular World*: Upon which they all agreed, That if there was a World that was not in any kind or sort, Irregular; there must be a World that was not in any kind or sort, Regular. But, to conceive those Irregularities that are in the Irregular World, is impossible; much less, to express them: for, it is more difficult to express Irregularities, than Regularities: and what Human Creature of this World, can express a particular Confusion, much less a World of Confusions?

Which I will, however, endeavour to declare, according to the Philosophical Opinions of the Parts of my Mind.

CHAP. II. Of the Productions and Dissolutions of the Creatures of the Irregular World.

According to the Actions of Nature, all Creatures are produced by the Associations of Parts, into particular Societies, which we name, *Particular Creatures*: but, the Productions of the Parts of the Irregular World, are so Irregular, that all Creatures of that World are Monstrous: neither can there be any orderly or distinct kinds and sorts; by reason that Order and Distinction, are Regularities. Wherefore, every particular Creature of that World, hath a monstrous and different Form; insomuch, that all the several Particulars are affrighted at the Perception of each other: yet, being Parts of Nature, they must associate; but, their Associations are after a confused and perturbed manner, much after the manner of Whirlwinds, or *Aethereal Globes*, wherein can neither be Order, nor Method: and, after the same

manner as they are produced, so are they dissolved: so that, their *Births* and *Deaths* are *Storms*, and their *Lives* are *Torments*.

CHAP. III. Of Animals, and of Humans, in the Irregular World.

It has been declared in the former Chapter, *That there was not any perfect Kind or Sort of Creatures in the Irregular World*: for, though there be such Creatures as we name Animals; and amongst Animals, Humans: yet, they are so Monstrous, that, being of confused Shapes, or Forms, none of those Animal Creatures can be said to be of such, or such a sort; because they are of different disordered Forms. Also, they cannot be said to be of a perfect Animal-kind, or any Kind; by reason of the variety of their Forms: for, those that are of the nature of Animals, especially of Humans, are the most miserable and unhappy of all the Creatures of that World; and the Misery is, That Death doth not help them: for, Nature being a perpetual Motion, there is no rest either alive or dead. In this World, it's true, some Societies (*viz.* some Creatures) may, sometimes, after their Dissolutions, be united into more Happy Societies, or Forms; which, in the Irregular World, is impossible; because all Forms, Creatures, or Societies, are miserable: so that, after dissolution, those dispersed Parts cannot joyn to any other Society, but what is as bad as the former; and so those Creatures may dissolve out of one Misery, and unite into another; but cannot be released from Misery. Page 284

CHAP. IV. Of Objects, and Perceptions.

The Opinions amongst the Parts of my Mind, were, That in the Unhappy, or Miserable World, all the actions of that World, being irregular, it must needs be, that all sorts of Perceptions of that World, must also be irregular: not only because the Objects are all irregular; but, the perceptive actions are so too; in such manner, that, what with the irregularity of the Objects, and the

irregularity of the Perceptions, it must, of necessity, cause a horrid confusion, both of the Sensitive and Rational Parts of all Creatures of that World, in so much, that not only several Creatures may appear as several Devils to each other; but, one and the same Creature may appear, both to the Sense and Reason, like several Devils, at several times.

CHAP. V. The Description of the Globe of the Irregular World.

The Opinion of my Mind was, That the Globe of the Irregular World was so irregular, that it was a Horrid World: for though, being a World, it might be somewhat like other Worlds, both Globous, and a Society of it self, by its own Parts; and therefore

might have that which we name *Earth*, *Air*, *Water*, and *Fire*: but, for Sun-light, Moon-light, Starr-light, and the like, they are not parts of the World they appear to; and are Worlds of themselves. But, there can be no such Appearances in the Irregular World: for, the Irregularities do obstruct all such Appearances; and the Elemental Parts (if I may name them so) are as irregular, and therefore as horrid as can be: so that it is probable, that the Elemental Fire is not a bright shining Fire, but a dull, dead Fire, which hath the Effects of a strong Corrosive Fire, which never actually Heats, but actually Burns; so that some Creatures may both freeze and burn at once. As for the Earth of that World, it is probable that it is like corrupted Sores, by reason all Corruptions are produced by Irregular Motions; from which Corruptions, may proceed such stinking Foggs, as may be as far beyond the scent of Brimstone, or any the worst of Scents that are in this World, as *Spanish* or *Roman* Perfumes, or Essences, are beyond the scent of Carion, or *Assafoetida*; which causes all Creatures (of Airy Substances) that breathe, to be so infected, as to appear like Poysoned Bodies. As for their Elemental Water, 'tis probable, that it is as black as Ink, as bitter as Gaul, as sharp as *Aquafortis*, and as Salt as Brine, mixt irregularly together, by reason the Waters there, must needs be very troubled Waters. As for the Elemental Air, I shall declare the Opinion of my Rational Parts, in the following Chapter.

CHAP. VI. Of the Elemental Air, and Light of the Irregular WORLD.

'Tis probable, that the Elemental Air of the Irregular World, is neither perfectly Dark, nor perfectly Light; for, either would be, in some part or kind, a Perfection or Regularity: but, being irregular, it must be a perturbed Air; and, being perturbed, it is probable it produces several Colours. But, mistake me not, I do not mean such Colours as are made by perturbed Light; but, such as are made by perturbed Air: and, through the Excess of Irregularities, may be Horrid Colours; and, by reason of the *Aethereal* whirling Motions, which are Circular Motions, the Air may be of the colour of Blood, a very horrid Colour to some sorts of Creatures: but 'tis probable, this Bloody Colour is not of a pure Bloody Colour, but of a corrupted Bloody Colour: and so the Light of the Irregular World, may, probably, be of a corrupt Bloody Colour: but, by the several Irregular Motions, it may be, at several times, of several corrupted Bloody Colours: and by reason there are no intermissions of *Air*, there can be no intermissions of this *Light*, in the Irregular World. Page 287

CHAP. VII. Of Storms, and Tempests, in the Irregular World.

As for *Storms*, and *Tempests*, and such irregular Weather, 'tis probable there are continual Winds and Thunders, caused by the disturbance of the Air; and those Storms and Tempests, being irregular, must needs be violent, and therefore very horrid. There may also be Lightnings, but they are not such as those that are of a fiery colour; but such as are like the colour of Fire and Blood mixt together. As for Rain, being occasioned by the Vapours from the Earth and Waters, it is according as those Vapours gather into Clouds: but, when there is Thunder, it must needs be violent.

CHAP. VIII. Of the several Seasons, or rather, of the several Tempers in the Irregular World.

As for *several Seasons*; there can be no constant Season, because there is no Regularity; but rather, a great Irregularity, and Violence, in all Tempers and Seasons; for there is no mean Degree: and surely, their Freezing is as sharp and corroding, as their Corrosive-Burnings; and it is probable, that the Ice and Snow in that world, are not as in this world, viz. the Ice to be clear, and the Snow white; because there the water is a troubled, and black water; so that the Snow is black, and the Ice also black; not clear, or like black polished Marble; but 'tis probable, that the Snow is like black Wool; and the Ice, like unpolished black Stone; not for Solidity, but for Colour and Roughness.

CHAP. IX. The Conclusion of the Irregular and Unhappy or Cursed World.

I have declared in my former Chapter, concerning the *Irregular World*, That there could not be any exact, or perfect kind or sort, because of the Irregularities; not that there is not Animal, Vegetable, Mineral, and Elemental Actions, and so not such Creatures; but, by reason of the Irregularities, they are strangely mixt and disordered, so that every Particular seems to be of a different Kind, or sort, being not any ways like each other; and yet, may have the nature of such Kinds, and Sorts, by reason they are Natural Creatures, although irregularly Natural: but, those irregular Natural Creatures, cannot chuse, by the former Descriptions, but be Unhappy, having, in no sort or kind, Pleasure, or Ease: and for such Creatures that have such Perceptions as are any way like ours, they are most Miserable: for, by the Sense of

Touch, they freeze and burn: by the sense of Tast, they have Nauseousness, and Hunger, being not satisfied: by the sense of Scent, they are suffocated, by reason of irregular Respiration: by the sense of Hearing, and sense of Seeing, they have all the horrid Sounds and Sights, that can be in Nature: the Rational Parts are, as if they were all distracted or mad; and the

Sensitive Parts tormented with Pains, Aversions, Sickesses, and Deformities; all which is caused through the Irregular Actions of the Parts of the Irregular World; so that the Actions of all sorts of Creatures, are Violent, and Irregular.

But, to conclude: As all the Creatures of our World, were made for the Benefit of Human Creatures; so, 'tis probable, all the Creatures of the Irregular World, were produced for the Torment and Confusion of Human Creatures in that World.

The Fifth Part, Being divided into FIFTEEN SECTIONS.

Concerning Restoring-Beds, or Wombs.

I.

At the latter end of my *Philosophical Conceptions*, the Parts of my Mind grew sad, to think of the dissolving of their Society: for, the Parts of my Mind are so friendly, that although they do often Dispute and Argue for Recreation and Delight-sake; yet, they were never so irregular, as to divide into Parties, like Factious Fellows, or Unnatural Brethren: which was the reason that they were sad, to think their kind Society should dissolve, and that their Parts should be dispersed and united to other Societies, which might not be so friendly as they were. And, after many several Thoughts, (which are several Rational Discourses: for, Thoughts are the Language of the Mind) they fell into a Discourse of *Restoring Beds*, or *Wombs*, viz. *Whether there might not be Restoring Beds, as well as Producing Beds, or Breeding Beds*. And, to argue the case, they agreed to divide into Minor and Major Parts.

II.

The Major Parts of my Mind were of opinion, That there are Beds, or Wombs, of Restoration, as well as Beds of Production: for, if Nature's

actions be poysed, there must be one, as well as the other.

The Minor Part's Opinion, was, That, as all Creatures were produced, so all Creatures were subject to dissolve: so that, the poyse of Nature's Productions, was Nature's Dissolutions, and not Restorations.

The Major Part's Opinion, was, That there are Restorations in Nature: for, as some dissolved, others united in every kind and sort of Creature, which was a Restoration to the kinds and sorts of Creatures.

The Minor Part's Opinion, was, That though every sort and kind of Creatures, continued as the Species of each sort and kind; yet, they did not continue by such Restorations as they were arguing about: for though, when some Creatures dye, others of the same Sort or *Species*, are born or bred; yet, they are Produced, not restored: for, they conceived, that Restoration was a reviving and re-uniting the Parts of a Dissolved Society or Creature; which Restoration was not natural, at least, not usual.

The Major Part's Opinion, was, That Restoration was natural, and usual: for, there were many things, or Creatures, restored, in some sort, after they were dead.

The Minor Part's Opinion, was, That some Creatures might be restored from some Infirmities, or Decays; but, they could not be restored after they were dissolved, and their Parts dispersed.

The Major Part's Opinion, was, That if the Roots, Seeds, or Springs of a Society, or Creature, were not dissolved and dispersed, those Creatures might be restored to their former condition of Life, if they were put, or received, into the Restoring Beds: As for example, A dry and withered Rood of some Vegetable, although the Parts of that Vegetable be, as we say, dead; yet, they are often restored by the means of some Arts: also, dead Sprigs will, by Art, receive new Life.

The Minor Part's Opinion, was, That if there were such actions of Nature, as Restoring actions; yet, they could not be the Poysing Actions, nor the Artificial Actions: for, not any dead Creature can be restored by Art.

III.

Some of the Gravest Parts of my Mind, made this following Discourse to some other Parts of my Mind.

Dear Associates, There hath been many Human Societies, that have perswaded themselves, That there are such Restoring Actions of Nature, which will restore, not only a Dead, but a Dispersed Society; by reason they have observed, That Vegetables seem to dye in one Season, and to revive in another: as also, that the Artificial Actions of Human Creatures, can produce several Artificial Effects, that resemble those we name *Natural*; which hath occasioned many Human Creatures to wast their Time and Estates, with Fire and Furnace, cruelly torturing the Productions of Nature, to make their Experiments. Also, they trouble themselves with poring and peeping through Telescopes, Microscopes, and the like Toyish Arts, which neither get Profit, nor improve their Understanding: for, all such Arts prove rather ignorant Follies, than wise Considerations; Art being so weak and defective, that it cannot so much assist, as it doth hinder Nature: but, there is as much difference between Art and Nature, as between a Statue and a Man; and yet Artists believe they can perfect what by Nature is defective; so that they can rectifie Nature's Irregularities; and do excuse some of their Artificial actions, saying, they only endeavour to hasten the actions of Nature: as if Nature were slower than Art, because a Carver can cut a Figure or Statue of a Man, having all his Materials ready at hand, before a Child can be finished in the Breeding-Bed. But, Art being the sporting and toyish actions of Nature, we will not consider them at this time.

But, *Dear Associates*, if there be any such things in Nature, as *Restoring-Beds*, which most of our Society are willing to believe; yet, those Beds cannot possibly be *Artificial*, but must be *Natural Beds*. Nor can any one particular sort of Bed, be a general Restorer: for, every several Sort or Kind, requires a Bed, or Womb, that is proper for their Sorts or Kinds: so that, there must be as many sorts, at least, and kinds of Beds, as there are kinds of Creatures: but, what those Wombs or Beds are, we Human Creatures do not know; nor do we know whether there be any such things in this World: but, if there be such things in this World, we cannot conceive where they are.

IV.

After the former Discourse, the Parts of my Mind were a little sad: but, after many and frequent Disputes and Arguments, they all agreed, That there are Restoring Beds, or Wombs, in Nature: but that to describe their Conceptions of those Restoring Beds, was only to describe Opinions, but not known Truths: and their Opinions were, That those Beds are as lasting as Gold, or Quick-silver: for, though they may be occasioned to alter their Exterior Form; yet, not their Interior or Innate Nature. But, mistake not my Mind's Opinion: for, their Opinion is not, That those Beds are Gold, or Quicksilver: for, their Opinion was, That neither Gold, or Quicksilver, were Restorers of Life: but, if they were Restorers, they could restore no other Creatures, but only dead Metals, by reason several Creatures require several Restoring Beds proper to their Sorts or Kinds: so that a Mineral Kind or Sort, could not restore an Animal Kind or Sort; because there was no such thing in Nature, as the Elixir, or Philosophers-Stone, which the Chymists believe to be some Deity, that can restore all Sorts and Kinds.

V.

As it has formerly been declared, The Parts of my Mind were generally of opinion, That it was, at least, probable, there were such things in Nature as *Restoring-Beds*, or *Wombs*. The next Opinion was, That these Beds were of several Kinds or Sorts, viz. Animal, Vegetable, Mineral, and Elemental: so that every Kind or Sort, is a general Restorer of the Lives of their Kind or Sort. As for example, An Animal *Restoring-Bed*, may restore any dead Animal, to his former Animal Life, in case the Animal Roots or Seeds, (which we name, the *Vital Parts*) were not divided and dispersed, but inclosed, or inurned, so that no other Animal could come to feed on those Roots and Seeds of the dead Animal Body; and in case the Body was so closely kept, though dead many years, if it was put into a *Restoring-Bed*, that Animal Creature would reunite to the former Animal Life and Form.

But then there arose this Argument, *That if the Bodies of the dead Animals, did corrupt and dissolve of themselves, as most dead Animal Bodies do; Whether, after their Dissolution, they could be Restored?*

The Minor Part's Opinion was, That those dissolved Bodies, being dissolved, or divided, and their Parts out of their places, could not be restored.

The Major Part's Opinion, was, They might be restored; first, Because, though the Parts may be divided; yet, they were not annihilated. The next, That those divided Parts were not so separated and dispersed, as to be united to other Societies: Wherefore, if all those dead Animal Parts were put into a *Restoring-Womb*, or *Bed*; the Bed would occasion those Parts to place themselves into their proper Order and Form.

VI.

After the former Discourse, some of the Parts of my Mind were sad, to think, that those that had been embowelled, were made incapable of ever being restored; and, that it was a greater cruelty to murder a dead man, and to rob him of his Interior Parts; than to murder a living man, and yet suffer his whole Body to lye peaceably in the Urn, or Grave.

But, the other Parts endeavouring to comfort those sad Parts, made this Argument, viz. *Whether it might not probably be, that the Bones or Carcase of a Human Creature, were the Root of Human Life? and if so, then if all the Parts were dissolved, and none were left undissolved, but the bare Carcase; they might be restored to life.*

The sad Part's Opinion, was, That it was impossible they could be restored, by reason the Roots of Human Life, were those we name the *Vital Parts*; and those being divided from the Carcase, and dispersed,

and united unto other Societies, could not meet and joyn into their former state of Life, or Society, so as to be the same Man.

The Comforting Parts were of opinion, It was not probable that the Fleshy and Spungy Parts, being the Branches of Human Life, could also be the Roots. Wherefore, in all probability, the Bones were the Roots; and the Bones being the Roots, if the bare Carcase of a Man should be put into a Restoring Bed, all the Fleshy and Spungy Parts, both those that were the Exterior, and those that were Interior, would spring and encrease to their full Maturity.

The sad Part's Opinion, was, That if the Bones were the Roots; and that, from the Roots, all the Exterior and Interior Parts, belonging to a Human Creature, should spring, and so encrease to full Maturity; yet, those Branches would not be the same they were, viz. the same Parts of the same

Man; and besides, those Branches would rather be new Productions, than Restorations.

The Comforting Part's Opinion, was, That though the Branches were new, the Carcase, as the Root, being the same, the Man would be the same: for, though the Spungy and Fleshy Parts, divide and unite from Home, and to Forrein Parts; yet, the Man is the same: and to prove that the Bony Parts are the Roots of Human Life, doth it not happen, That if the Flesh be cut from the Bone, and the Bone be left bare; yet, in time, the bone produces new flesh: but, if any bone be separated from the Body, that Bone cannot be restored; nor can a new bone spring forth, nor can the divided bone be joyned or knit to the body, as it was before: for, although a broken bone may be set; yet, a divided bone cannot be rejoyned: All which Arguments, were a sufficient proof, That the Bones were the Roots of Life.

The Sad Part's Argument, was, That it was well known, that if any of the Vital Parts of a Human Creature, as the Liver, Lungs, Heart, Kidneys, and the like, were decayed, pierced, or wounded, the Human Creature dyed, by reason those Parts are incurable.

The Comforting Parts were of opinion, That there were many less Causes which did often occasion Human Death; yet, those Causes were not the Roots of Life: nor were those Parts the Roots of Life, although those Parts which we name *Vital*, were the chief Branches of Human Life.

But, at last, they all agreed in this opinion, That the *Bones*, were the Roots; the *Marrow*, the *Sapp*, and the *Vitals*, the chief branches of Life. Also, they agreed, That when an Human Life was restored, the bones did first fill with some Oylie Juyces; and from the bones, and the sap or juyce of the bones, did all the Parts belonging to a Human Creature, spring forth, and grow up to Maturity: and certainly, *Not to disturb the Bones of the dead*, was a Holy and Religious Charge to Human Creatures.

VII.

After the pacifying the Sad Parts of my Mind, their Argument was, *That, supposing Creatures could be restored; whether they should be restored as when they were first produced; or, as when they were at the perfection of their Age; or, as when they were at old Age?*

But, after many Disputes, they all agreed, That those that should be restored, should be restored to that degree of Age and Strength, which is the most perfect: and, as all Productions arrived towards Perfection by degrees; so those that were restored, should return to Perfection by degrees, if they were past the perfect time of their age: and those that were not arrived to their Perfection, before they dyed, should arrive to it, however, as those that had it: so that, both *Youth* and *Age*, shall meet in Perfection: for, as the one encreases, as it were, forward; so the other return to their Strength and Perfection of their past Age.

VIII.

After the former Opinions, the Parts of my Mind were somewhat puzzled in their *Arguments* concerning the degrees of the Restoring Times; as, *Whether Restoration was done by a General Act, or by Degrees?*

The most Doubting Part's Opinion, was, That it was not natural to Restore, although it was natural to Produce; and, that all Natural Productions, were by degrees: but, for Restorations, (being not Natural Productions) they could not be done by degrees: and therefore the Action of Restoration, was but as one Action, although of many Parts.

The Believing Parts of my Mind were of opinion, That all Nature's Actions, being by degrees, all Restorations were also by degrees.

The Doubting Part's Opinion was, That there were some actions that had no degrees: for, One action might signifie a Thousand.

The other Part's Opinion was, That a Thousand actions, or degrees, were in the figure of One.

The Doubting Parts were of opinion, That it was impossible. But, at last, they agreed, That the Restoring actions were by degrees.

IX.

The Parts of my Mind were divided into Minor and Major Parts, about the Time or Degrees of Restoration of Human Creatures.

The Minor's Opinion was, That the Restoring actions of Nature, were so much quicker than the Producing actions, that a Human Creature might be restored in a Months time; whereas the production of a Human Creature was in ten Months: for, though a Human Creature may Quicken at Three Months time; yet, it was not fully Ripe for Birth, before the time of Ten Months.

The Major Part's Opinion was, That Restoration was according as the Creature was Dissolved: for, a Man that was newly dead; or not so long dead, that his Parts were not yet divided; that Man might be restored to Life in an Hour's time, or less: but, if all the Parts, excepting the bare Carcase, were dissolved, there would require as long a time in Restoring, as in Producing.

The Minor's Opinion, was, That the Restoring-time, was no longer than the time of Quickning.

The Major Part's Opinion, was, That though the Exterior Form or Frame of a Child, might be before the Quickning; yet, it was not a perfect Animal, until it was Quick: and although it might be a perfect Animal when it was Quick; yet, not ripe, that is, not at the full Perfection of a Human Creature. As it is with Fruits: for, a Green Plumb is not like a Ripe Plumb; but, any Green Fruit, is like a Dead Fruit, in comparison of a Ripe Fruit.

At last, the Parts of my Mind did agree, That if a Human Creature was dissolved, excepting the bare Carcase; it would require Ten Months time ere it could perfectly be restored: for, the Springing Parts would require so long a time ere they could come to full Maturity.

X.

The Question being stated, *Whether the Restoring-Bed, was a Fleshy Bed*; All the Parts of my Mind, after many Disputes, agreed, That it could not be a Fleshy Bed, by reason the nature of Flesh is so corruptible, dissolvable, and easie to be dissolved, that it could not possibly be of such a lasting nature, as is required for *Restoring-Beds*. But yet, they agreed, they were like Flesh, for Softness, or Spunginess; as also, for Colour. Also, they agreed, That the Animal *Restoring-Bed*, was of such a Nature or Property, that it could dilate and contract, as it had occasion; in so much, that it could

contract to the compass of the smallest, or extend to the magnitude of the largest Animal. Also, they did agree, That it was somewhat like the Stomack of a Human Creature, or of the like Animal, that could open and shut the Orifice; and that when an Animal Creature was put into the *Restoring-Bed*, it would immediately inclose the Animal: and when it had caused a perfect Restoration, the *Restoring-Bed* would open it self, and deliver it to its own Liberty.

XI.

Another Question amongst the Parts of my Mind concerning *Restoring-Beds*, or *Wombs*, was: *That in case there were such Restoring-Beds in Nature, as in all probability there were; Where could those Restoring-Beds be? viz. Whether there were any in this World? If not in this World, in any other World?*

The Minor Parts were of opinion, There were none in this World; but, that there were some in other Worlds.

The Major Part's Opinion, was, That there were such Beds; but, that Human Creatures would not know them, though they could perceive them: nor, if they could perceive them, could they tell how to make use of them.

At last they all agreed, That those *Restoring-Beds* were in the Center of the World: but, where the Center is, no Human Creature, no, not the most Subtile and *Learned Mathematicians, Geometricians, or Astrologers*, could, with their most Laborious Arts, and Subtile Observations, know; and therefore, unless by a special Decree from God, no such Restoration can be made.

XII.

The Parts of my Mind were very studious to conceive where the Center of the World was: Some of the Parts of my Mind was of opinion, That there were four Centers, *viz.* A Center in the Earth, a Center in the Air, a Center in the Sea, and a Center in the Element of Fire.

Upon which Opinion, the Parts of my Mind divided into Minor and Major Parts.

The Minor Parts were of opinion, That there were Centers in all the Four Elemental Parts; and that the *Restoring-Beds*, were only of four Kinds: but yet, there might be many several sorts of each particular Kind; and that each particular Kind, with all the several Sorts, was produced in each particular Elemental Center.

The Major Part was of opinion, That there might be infinite Centers, if there were infinite Worlds: also, there might be many Centers in this World; for, every round Globe hath a Center. But, their Opinion concerning the *Restoring-Beds*, was, That they were in the Center of the Globe of our whole World, and not of any of the Parts of the World: for, the *Air* could have but an uncertain Center; neither could the *Water* have a very solid Center; and the *Earth* was too solid to have a Center, consisting of the Four kinds of Elements: neither could the Elemental *Fire* have such a Center, as to breed such different kinds and sorts of *Beds*, as the *Restoring-Beds* are, because many of them are quite of a different nature from the nature of Elemental Fire: wherefore, it must be the Center of the World, which must consist of all the Elemental kinds.

XIII.

After the former *Argument*, the Parts of my Mind were very studious in conceiving, where the Center of the whole Universe of this our World, might be: at last they all agreed, It was the *Sea*, which is the Watry Element: for, the *Sea* is inclosed with the *Airy*, *Fiery*, and *Earthy* Parts of the Universe, and therefore must be the Center. And, though the *Sea* was the Center of the World; yet, there was a Center of the *Sea*: so that, there was a Center in a Center; in which Center, were the *Restoring-Beds*.

XIV.

After the former Conceptions, the Parts of my Mind were very studious, to conceive where the Center's Center might be. But, they could not possibly conceive it, by reason they could not possibly imagine how large, and of what compass the *Sea* may be of: for they did verily believe, that the utmost extension of the *Sea*, is not, as yet, known to Human-Kind: for, that Circle about which the Ships of *Cavendishe*, and *Drake*, did swim, might be, in

comparison to the whole Body of the Sea, but such a Circle as a Boy may occasion, with throwing a small Stone, or such like thing, into a Pond of Water.

XV.

The last Conception of my Mind, concerning *Restoring-Beds*, was, That the Parts of my Mind did conceive, That the Center of the whole Universe, was the Sea; and in the Center of the Sea, was a small Island; and in the Center of the Island, was a Creature, like (in the outward Form) to a great and high Rock: Not that this Rock was Stone; but, it was of such a nature, (by the natural Compositions of Parts) that it was compounded of Parts of all the principal Kinds and Sorts of the Creatures of this World, viz. Of *Elemental*, *Animal*, *Mineral*, and *Vegetable* kinds: and, being of such a nature, did produce, out of it self, all kinds and sorts of *Restoring-Beds*; whereof, some sorts were so loose, that they only hung by Strings, or Nerves: others stuck close. Some were produced at the top, or upper parts: others were produced out of the middle parts; and some were produced from the lower parts, or at the bottom. In short, the Opinion of the Parts of my Mind, was, That this Rocky Creature was all covered with its own Productions; which Productions were of all Kinds and Sorts: not that they were numerous; but, various Productions: also, that these various Productions, were *Restoring-Beds*: for, the nature of this Rocky Creature, is as lasting as the Sun, or other Planets; which was the reason that those Productions are not subject to decay, as other Productions are: nor can they produce new Creatures; but only restore former Creatures; as, those that had been Produced, and were partly Dissolved.

THE CONCLUSION.

After the Wisest Parts of my Mind had ended their *Arguments*, there being some of the Dullest, and the most Unbelieving, or rather, Strange Parts of my Mind, that had retired into the *Glandula* of my Brain, which is a kind of

a Kernel; which they made use of, instead of a Pulpit: out of which, they declared their Opinions, thus:

Dear Associates, We, that were not Parties of your Disputations, or Argumentations, concerning *Restoring-Beds*; being retired into the *Glandula* of the Brain, where we have been informed by the Nerves, and Sensitive Spirits, of your wise Opinions, and subtile Arguments,

Considering that your Conclusion was as improbable, if not as impossible, as the Chymical *Philosophers-Stone*, or *Elixir*; We desire you (being Parts of one and the same Society) not to trouble the whole Society, in the search of that, which, if it was in Nature, will never be found. But to prevent, that your painful Studies, and witty Arguments, be not buried in Oblivion; We advise you, To perswade the Sensitive Parts of our Society, to record them, so that they may be divulged to all the Societies of our own Kind or Sort of Creatures; as *Chymists* do, who, after they have wasted their Times and Estates, to gain the *Philosophers-Stone*, or *Elixir*; write Books to teach it to the Sons of Art: which is impossible, at least, very improbable, ever to be learn'd, there being no such Art in Nature: but, were it possible such an Art was to be obtained; yet, when obtained, the Artist would never divulge it in Print. But, those great Practitioners, finding, after much Loss and Pains, nothing but Despair, write Books of that Art; which, instead of the *Elixir*, did produce *Despair*; which again, though produced by Art, did produce, naturally, that Vice, named *Malice*; and *Malice*, being a Pregnant Seed, sowed upon the Fertile Ground of their Writings, produces so much Mischief, that many men of good Estates, have been undone, in following their Rules in *Chymistry*: And if your Books should be as succesful as *Chymistry* hath been (I dare not say, among *Fools*; but) amongst Credulous Men; your Books will cause as much Mischief as theirs have done; not by the ways of *Fire*, but by the ways of *Water*: for, your Books send men to Sea, a much Cooler Element than *Fire*; but, more Dangerous than *Chymical Fire*, unless *Chymical Fire* be *Hell-Fire*.

Upon which Discourse, the rest of my Thoughts were very angry, and pull'd them out of their Pulpit, the *Glandula*; and not only so, but put them out of their Society, believing they were a Factious Party, which, in time, might cause the Society's Dissolution.

FINIS.

*** END OF THE PROJECT GUTENBERG EBOOK GROUNDS OF
NATURAL PHILOSOPHY: DIVIDED INTO THIRTEEN PARTS ***

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